MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
4000 PROCESS PLANT-MOSQUE & ABLUTION
DESIGN OF BUILDING STRUCTURE

Document Number **MD-216-4000-EG-CV-CAL-1070**
Revision: **D01**
Revision Date: **30.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AN03	CED01	40004	OU100682976	2
L&T Document Number:	O19045	900	C	K	24031	001	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	12.05.2020	FIRST ISSUE	MF	AVG	SJM	
C01	11.06.2020	ISSUED FOR CONSTRUCTION	MF	AVG	SJM	
D01	30.10.2023	AS BUILT	MF	AVG	SJM	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of a Mosque Ablution Building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Mosque & Ablution Building only.

4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab) and a Walkway was proposed with Steel Sections Connecting Masjid and Ablution Area along with separate Analysis and Design Sections for Dome and Minaret of Concrete Structure.

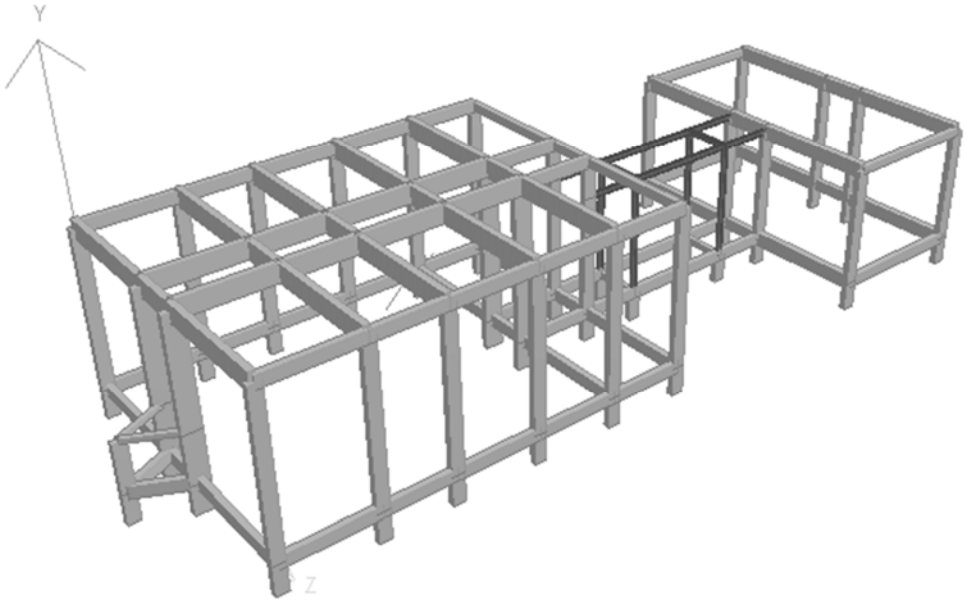
The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ and Steel Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column

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Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007- Structural Loading & Forces.

[2] SBC-303-2007- Structural Soil & Foundations.

[3] SBC-304-2007- Structural Concrete Structures.

[4] SASO SSA 2 - Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 - Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 - Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 - Building Code Requirements.

[8] PCI –MNL-120-04 - Design Handbook, 6th edition.

[9] SBC-306-2007- Structural – Steel Structures.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load(DL)

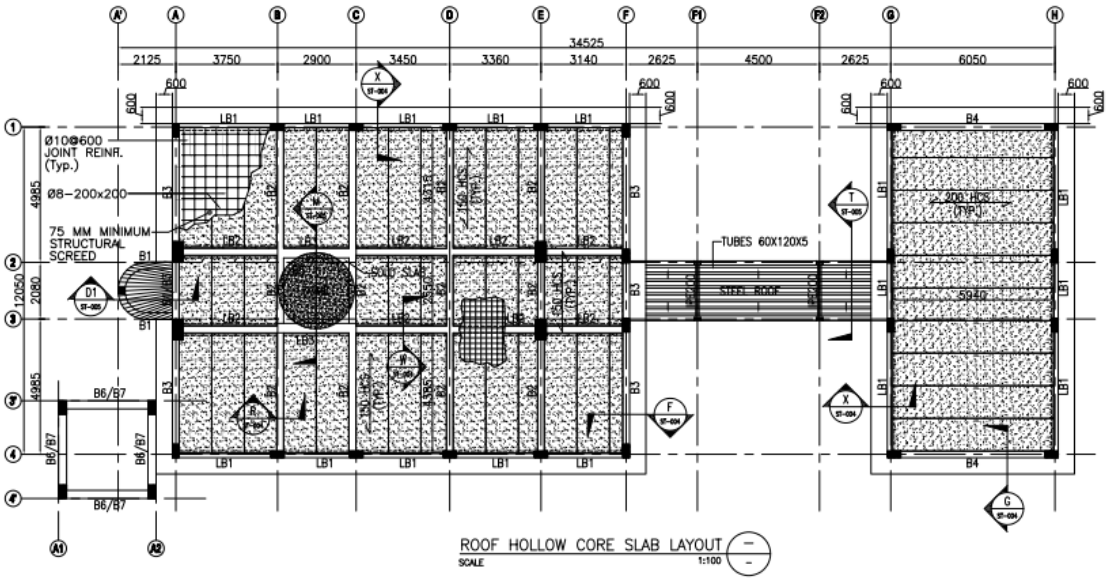
5.1.1. Roof Dead Loads

• Provision for Electrical Lighting		= 0.05 kN/m²
Ceiling		
Suspended steel channel system		= 0.10 kN/m²
Acoustical Fiber Tile (16mm)		= 0.05 kN/m²
	Sub-total	= **0.2 kN/m²**

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- Hollow Core Slab

  (HCS-200)(To be confirmed by Vendor) b/w Grid Line's (1 & 4), (G & H) = **3.10 KN/m2**

  (HCS-150)(To be confirmed by Vendor) b/w Grid Line's (1 & 2),(A & F) = **2.52 KN/m2**

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	**Self Wt-KN/M2**	**No. of Cores in Unit**	**Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2**							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

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- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load b/w Grid's (1 & 4) =3.1+0.2+2.52= 5.82 **kN/m²**

Total Dead Load b/w Grid's (G & H) =3.10+0.2+3.1= 6.40 **kN/m²**

TOTAL DL as UDL ON BEAMS

- UDL on LB's Along Grid Line-1 & 4= 4.985 x 5.82/2 = **14.50 Kn/m**
- UDL on LB's Along Grid Line-2 & 3=14.5+(5.82x 2.08)/2 =**20.55 Kn/m**
- UDL on LB's Along Grid Line's-G, H =6.40 x 6.00/2 =**19.20 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20) *0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 5.8m clear height = **23.2 kN/m**

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 4.5m clear height = **18 kN/m**

5.1.3. Dome Load on Beams

Thickness of Dome = 150mm

Diameter of the Dome = 2.4m

Height of the Dome = 1m

Weight of the Dome = 2.7 x 24 = 65 Kn

Udl of dome on the Supporting Beams = 65/10 = 6.5 Kn/m

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5.1.4. Roof Live Loads (Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on LB's Along Grid's (1 & 4) = 0.50 x 4.985/2 = **1.25 Kn/m**
- UDL on LB's Along Grid's (2 & 3) = 1.25+(0.50 x 2.08/2) =**1.77 Kn/m**
- UDL on Ledger Beams Along Grid Line's-G,H =0.50 x 6.00/2 =**1.5 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

5.2. Wind Load (WL)

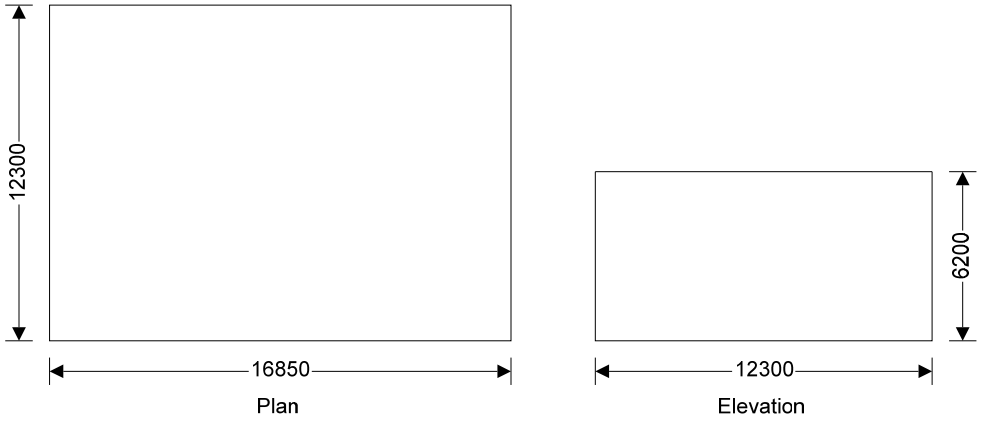
Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON-PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

5.3. Wind Load (Analytical Verification)

5.3.1. WIND LOADING (SBC-301)-MOSQUE AREA

Using the directional design method



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Building data

Type of roof;	Flat
Length of building;	b = 16850 mm
Width of building;	d = 12300 mm
Height to eaves;	H = 6200 mm
Mean height;	h = 6200 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant;	K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)
Velocity pressure equation;	$q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$ V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
6200	1.08	1.35

Pressures and forces

Net pressure;	$p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;
Net force;	$F_w = p \times A_{ref}$;

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Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	6.20	0.80	1.35	0.18	104	14.83
B	6.20	-0.50	1.35	-1.32	104	-137.97
C	6.20	-0.70	1.35	-1.55	76	-118.28
D	6.20	-0.70	1.35	-1.55	76	-118.28

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	6.20	0.80	1.35	1.67	104	170
B	6.20	-0.50	1.35	0.17	104	17.69
C	6.20	-0.70	1.35	-0.06	76	-4.65
D	6.20	-0.70	1.35	-0.06	76	-4.65

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	6.20	0.80	1.35	0.18	76	11
B	6.20	-0.43	1.35	-1.24	76	-94.5
C	6.20	-0.70	1.35	-1.55	104	-162.03
D	6.20	-0.70	1.35	-1.55	104	-162.03

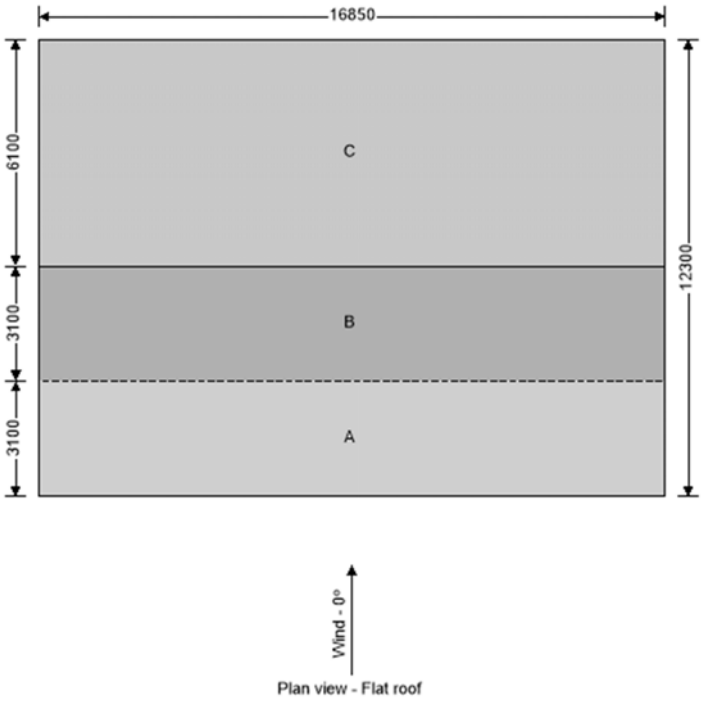
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

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Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	6.20	0.80	1.35	1.67	76.26	124
B	6.20	-0.43	1.35	0.25	76.26	19.41
C	6.20	-0.70	1.35	-0.06	104.50	-6.30
D	6.20	-0.70	1.35	-0.06	104.50	-6.30

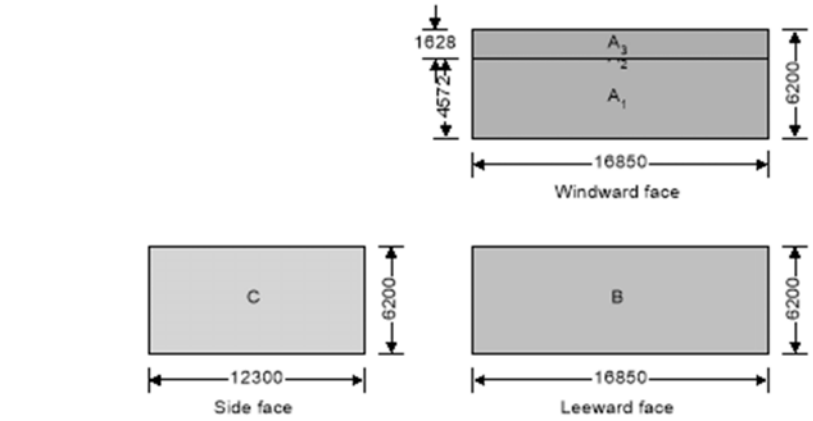


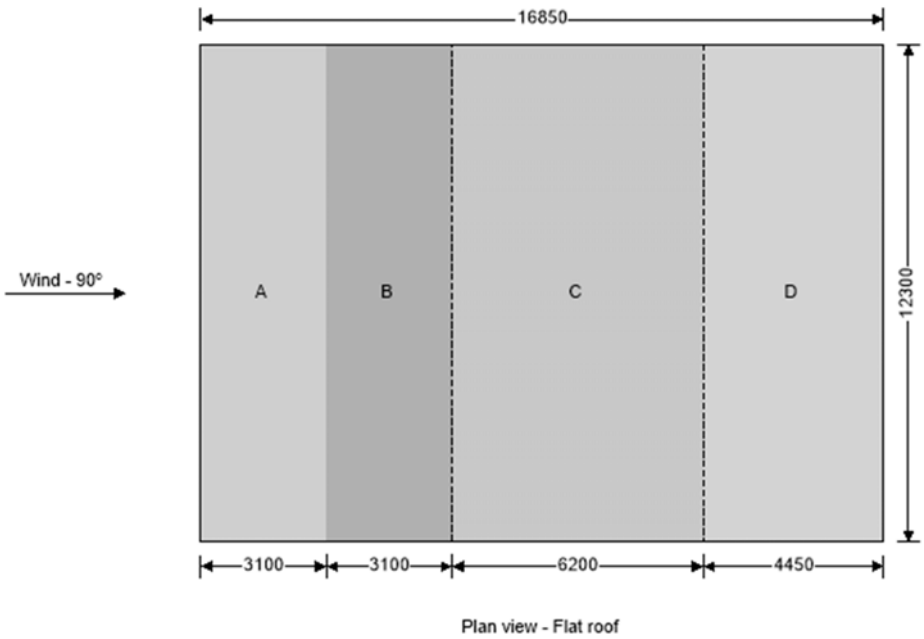
Plan view - Flat roof

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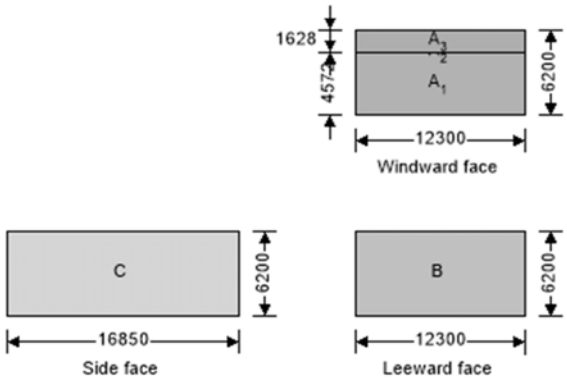


Plan view - Flat roof

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

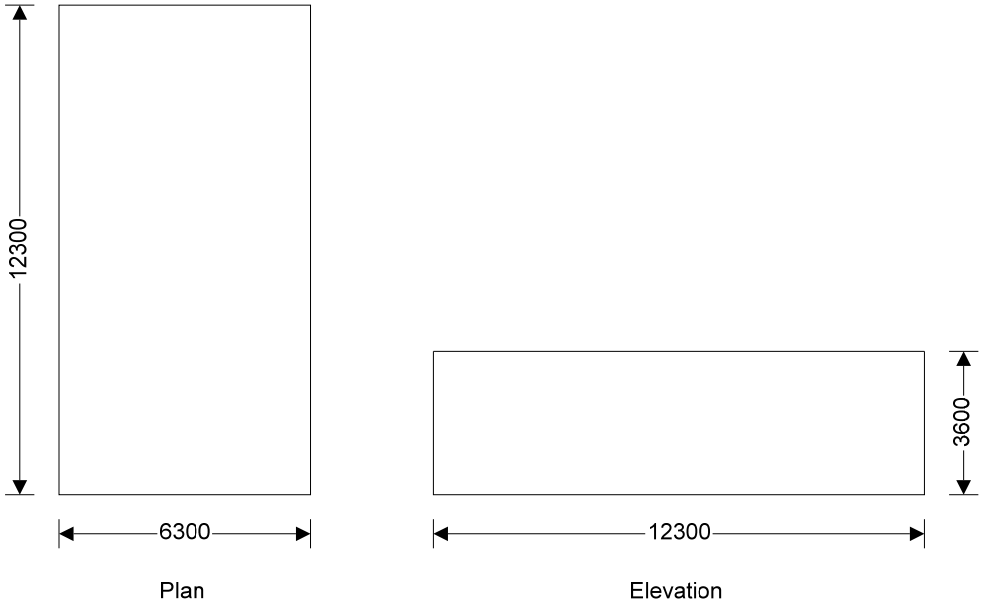
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5.3.2. WIND LOADING (SBC-301)-ABLUTION AREA

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 6300 mm
Width of building;	d = 12300 mm
Height to eaves;	H = 3600 mm
Mean height;	h = 3600 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)

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Internal pressure coef +ve; GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)

Internal pressure coef –ve; GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)

Gust effect factor; G_f = **0.85**(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3600	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	0.17	22.68	3.80
B	3.60	-0.31	1.29	-1.05	22.68	-23.76
C	3.60	-0.70	1.29	-1.48	44.28	-65.33
D	3.60	-0.70	1.29	-1.48	44.28	-65.33

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	1.58	22.68	35.94
B	3.60	-0.31	1.29	0.37	22.68	8.38

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Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
C	3.60	-0.70	1.29	-0.06	44.28	-2.57
D	3.60	-0.70	1.29	-0.06	44.28	-2.57

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	0.17	44.28	7.50
B	3.60	-0.50	1.29	-1.26	44.28	-55.63
C	3.60	-0.70	1.29	-1.48	22.68	-33.46
D	3.60	-0.70	1.29	-1.48	22.68	-33.46

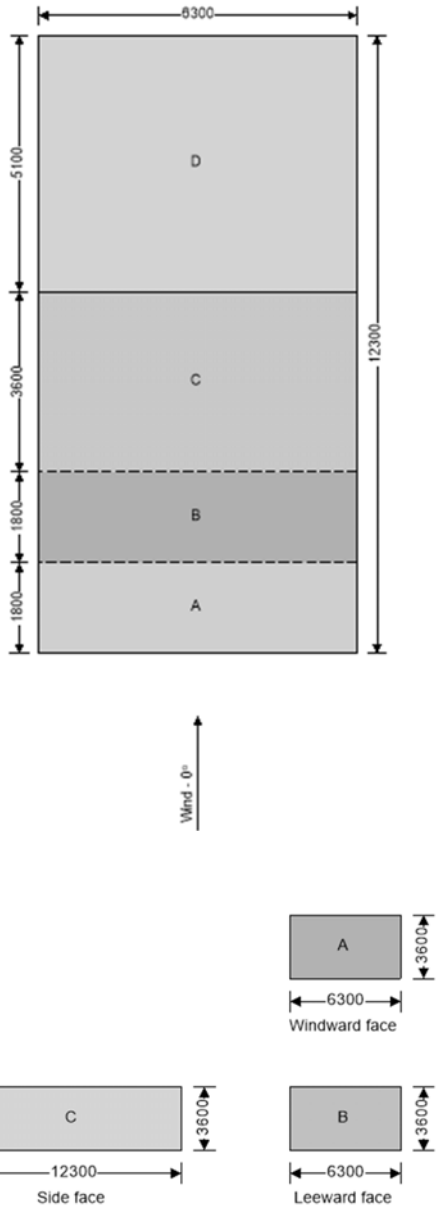
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	1.58	44.28	70.10
B	3.60	-0.50	1.29	0.16	44.28	7.13
C	3.60	-0.70	1.29	-0.06	22.68	-1.32
D	3.60	-0.70	1.29	-0.06	22.68	-1.32

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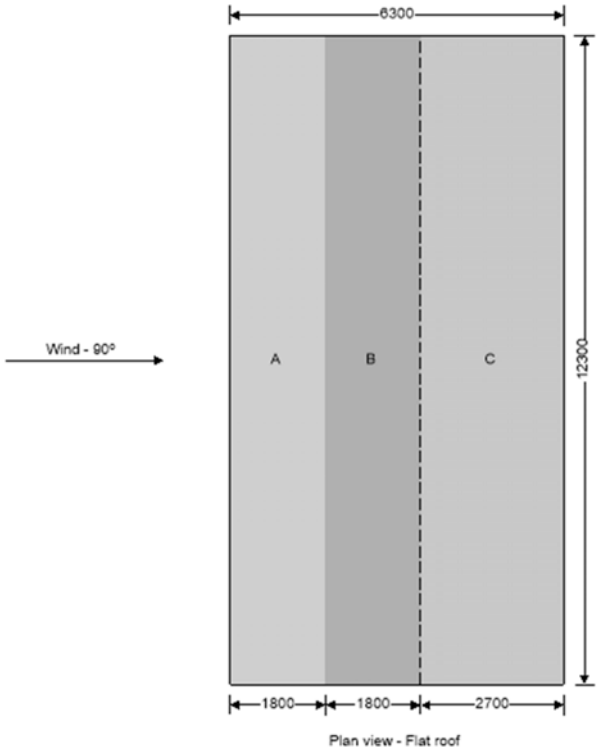
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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.4. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.5. Thermal/Temperature Loads(Self-restraining ,Thermal Expansion & Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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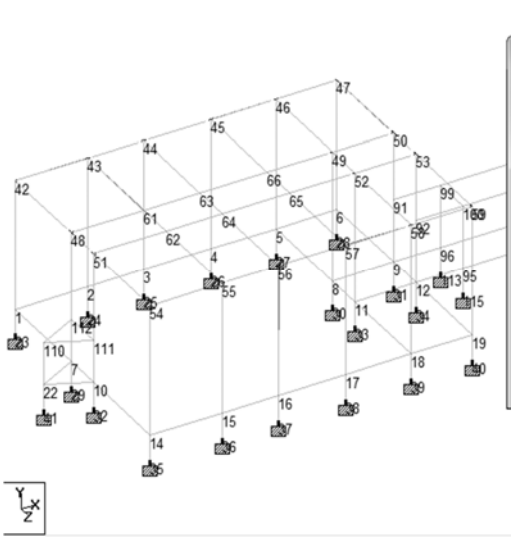
8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

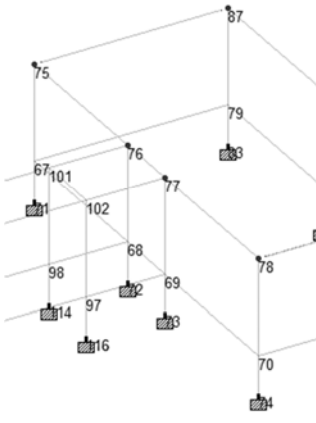
L/600: for members supporting masonry under total loads

Where, L = span of member between support



Masjid L&T 03 Mar 2020 EW & EQ R.std - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm
63	17	3.801	-29.263	5.994	30.111
64	17	3.697	-28.691	5.991	29.542
56	17	1.431	-0.447	5.989	6.174
57	40	1.407	-0.345	-5.344	5.537
55	40	1.405	-0.395	-5.333	5.529
45	40	1.290	-0.355	-5.330	5.495
65	40	3.775	-16.274	-5.329	17.535
66	40	3.742	-16.649	-5.327	17.876
62	40	3.833	-22.705	-5.315	23.631
43	40	1.273	-0.406	-5.312	5.477
61	40	3.745	-23.184	-5.311	24.077
45	17	1.296	-0.345	5.233	5.402
66	17	3.770	-16.585	5.217	17.790
57	17	1.442	-0.355	5.217	5.424
65	17	3.695	-16.311	5.214	17.518
43	17	1.292	-0.396	5.092	5.268
61	17	3.828	-23.119	5.072	23.977



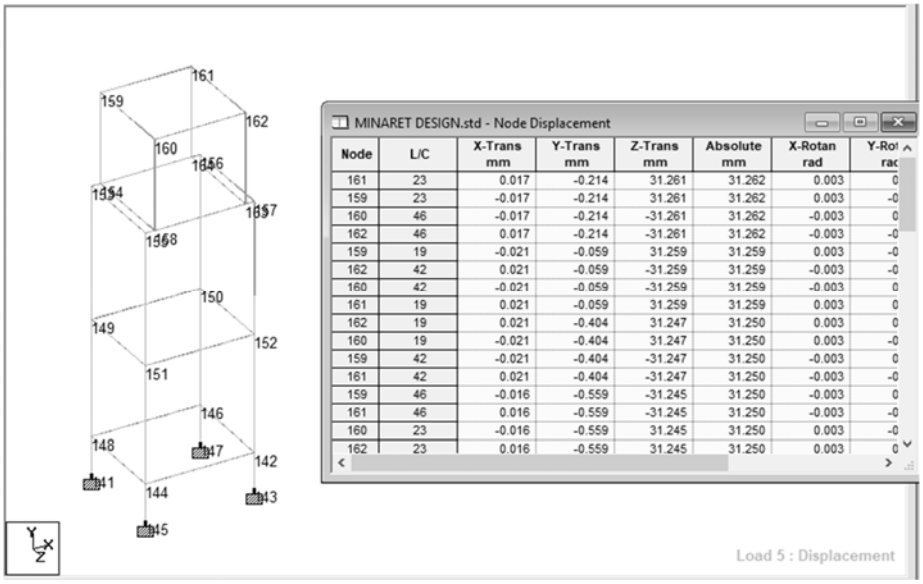
Masjid L&T 03 Mar 2020 EW & EQ R.std - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm
88	46	10.694	-0.173	-0.907	10.734
89	23	10.694	-0.173	0.907	10.734
88	42	10.318	-0.108	-0.907	10.359
89	19	10.318	-0.108	0.907	10.359
89	42	10.316	-0.090	-0.906	10.356
88	19	10.316	-0.090	0.906	10.356
89	46	10.691	-0.155	-0.906	10.731
88	23	10.691	-0.155	0.906	10.731
87	42	0.735	-0.091	-0.895	1.162
90	19	0.734	-0.091	0.895	1.161
75	23	0.780	-0.141	0.882	1.185
78	46	0.780	-0.141	-0.881	1.186
87	46	0.780	-0.149	-0.880	1.186
90	23	0.779	-0.149	0.880	1.185
75	19	0.727	-0.083	0.867	1.134

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Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rot rad
161	23	0.017	-0.214	31.261	31.262	0.003	0
159	23	-0.017	-0.214	31.261	31.262	0.003	-0
160	46	-0.017	-0.214	-31.261	31.262	-0.003	0
162	46	0.017	-0.214	-31.261	31.262	-0.003	-0
159	19	-0.021	-0.059	31.259	31.259	0.003	-0
162	42	0.021	-0.059	-31.259	31.259	-0.003	-0
160	42	-0.021	-0.059	-31.259	31.259	-0.003	0
161	19	0.021	-0.059	31.259	31.259	0.003	0
162	19	0.021	-0.404	31.247	31.250	0.003	0
160	19	-0.021	-0.404	31.247	31.250	0.003	-0
159	42	-0.021	-0.404	-31.247	31.250	-0.003	0
161	42	0.021	-0.404	-31.247	31.250	-0.003	-0
159	46	-0.016	-0.559	-31.245	31.250	-0.003	0
161	46	0.016	-0.559	-31.245	31.250	-0.003	-0
160	23	-0.016	-0.559	31.245	31.250	0.003	-0
162	23	0.016	-0.559	31.245	31.250	0.003	0

Maximum attained Drift as per Analysis from above figure's = **29 mm** (Mosque) & **10mm** (Ablution Building) & 32mm (Minarete)

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 6500/100 = **65mm** (Mosque), 3500/100= **35mm** (Ablution Building) & 11300/100=**113mm** (Minarete)

Hence the Structure is safe considering drift check…

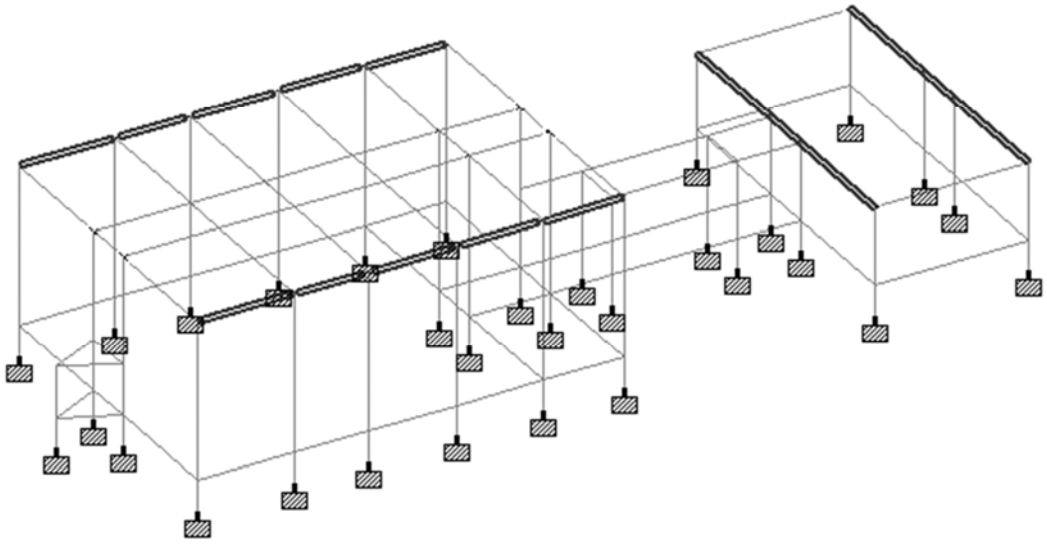
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9. CONCRETE ROOF BEAM DESIGN

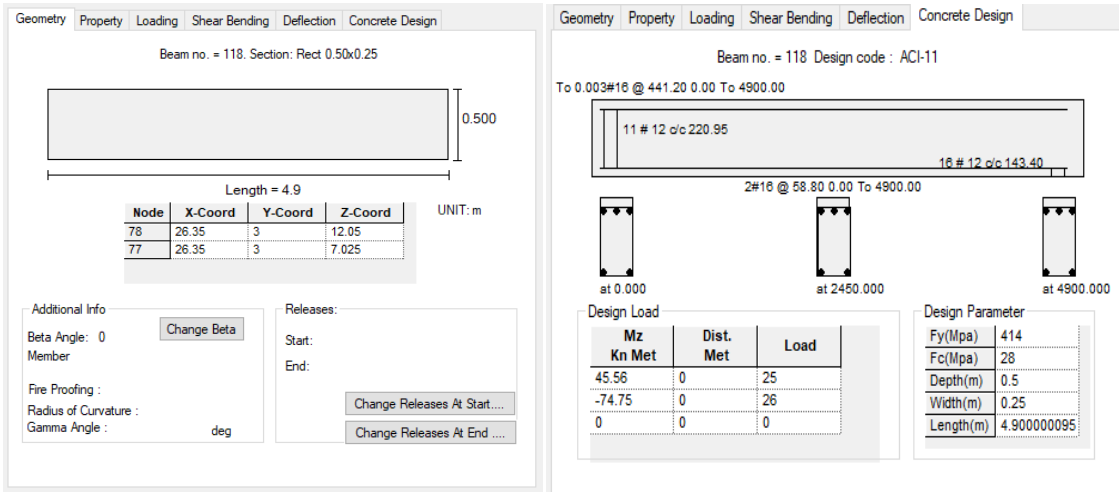
9.1. Design of Ledger Beam(LB1)



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Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 74 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ =0.9

Ast.req = **475 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **45 Knm (Sagging)**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

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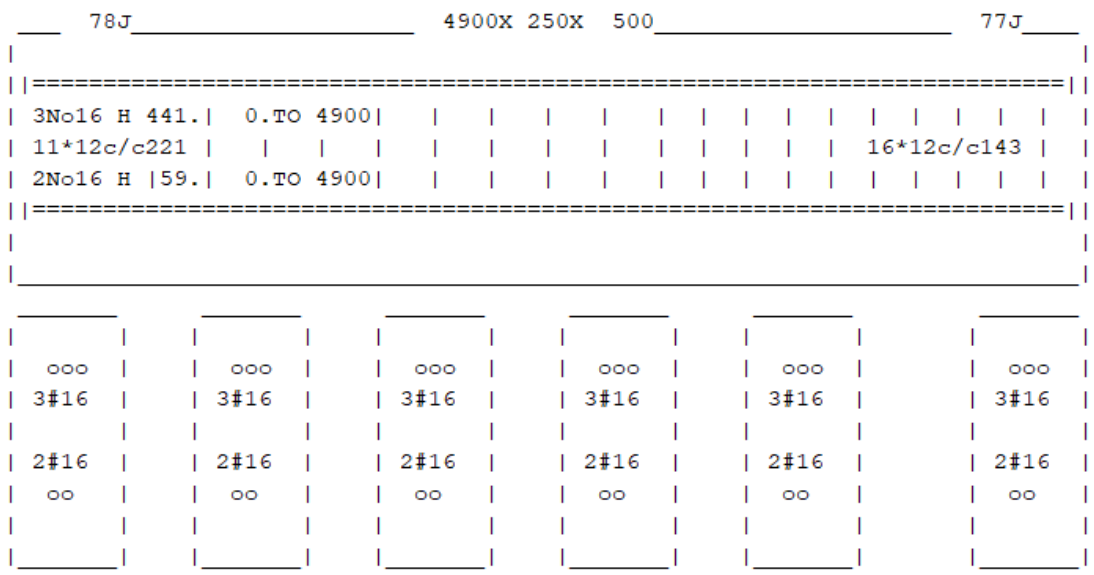
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9.1.1. Staad Output for LB1

```
        B E A M  N O.   118 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   50.95 KNS  Vc=  123.66 KNS  Vs=    0.00 KNS
Tu=    2.84 KN-MET  Tc=    3.5 KN-MET  Ts=    0.0 KN-MET  LOAD    28
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2013. MM

AT END   SUPPORT - Vu=   64.27 KNS  Vc=   96.99 KNS  Vs=    0.00 KNS
Tu=    4.64 KN-MET  Tc=    3.4 KN-MET  Ts=    6.2 KN-MET  LOAD    49
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2013. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.52 SQ.CM.
```

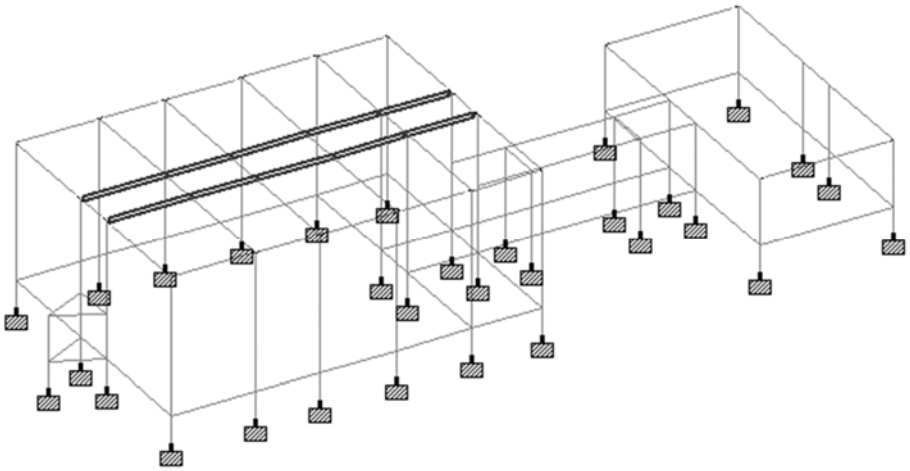


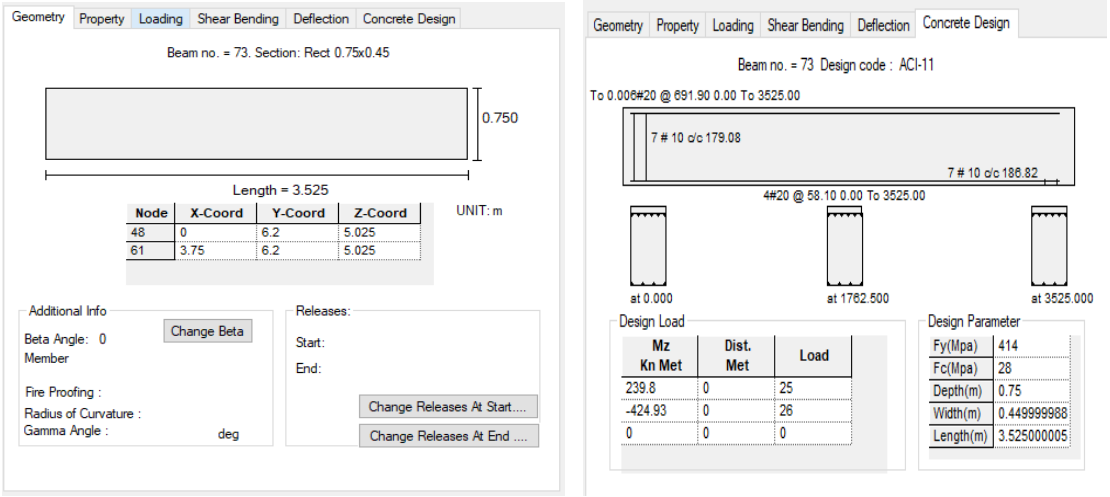
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9.2. Design of Ledger Beam(LB2 & LB3)





Design of Beam (LB2)

Maximum Moment required from Analysis = 425 **Knm (Hogging)**

Therefore $Ast._{req}$ = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=450mm, Fy=414 Mpa, d=750-40-10-20/2=690mm, Fc'=35 Mpa & ϕ=0.9

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Ast.req = **1715 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **1200mm²**

Maximum Moment required from Analysis = **239 Knm (Sagging)**

However Provide 8-20 dia-Full Straight Bars @ Top &
6-20 dia-Full Straight Bars, 4-20 dia- Additional@ Bottom considering Long term deflection check as shown below in section 8.2.1

Ast.pro**>>>>>>(Ast.min, Ast.req)**

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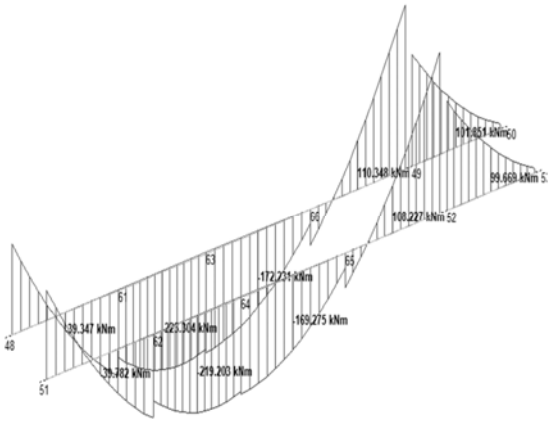
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9.2.1. Long Term Deflection Check for LB2 & LB3 Beam

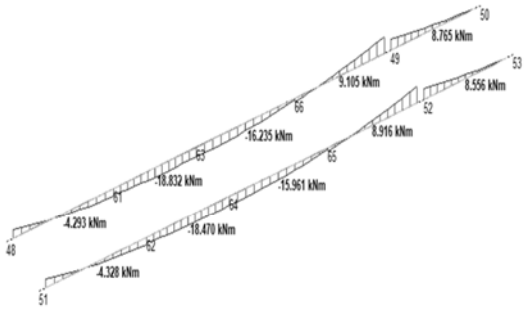
INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m
Unit wt. of steel	=	77.00	kN/m
Depth of beam, h	=	750	mm
width of beam, b	=	450	mm
Length of beam, L	=	13.50	m
Bottom bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of bottom bars	=	10.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	10.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	690.00	mm
FROM STAAD RESULTS Mdead	=	223.00	kN-m
Mlive	=	18.00	kN-m



CHECK DEFLECTION: $M_{d+\ell}$ = 241.00 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 232.00 kN-m



modulus of rupture, modulus of elasticity, modular ra

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa ←*ACI(318M-08) Sec. 9.5.2.3*

$n_s = \frac{Es}{Ec}$ = 7.19

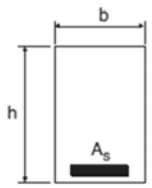
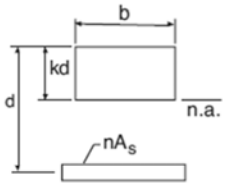
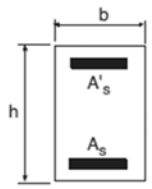
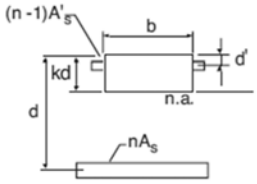
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 1.58E+10 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.02 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 217.77 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 6.59E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 2.40$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 154.74 \text{ kN-m} \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

under dead load

$$\frac{M_{cr}}{M_d} = 0.69$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 9.67E+09 \text{ mm}^4 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.67$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 9.33E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.64$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 9.03E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 15.740 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 16.982 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 18.218 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 2.477 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^\ddagger$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^\S$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\lambda \le \frac{\lambda}{180}$ = 75.00 mm > 2.48 mm *ok!*

$(\Delta_i)_\lambda \le \frac{\lambda}{360}$ = 37.50 mm > 2.48 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$\rho = \frac{As}{bd}$ = 0.0101

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.33 ←*ACI(318M-08) Sec. 9.5.2.5*

Page 4

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 22.55 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda$ = 25.03 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480}$ = 28.13 mm > 25.03 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240}$ = 56.25 mm > 25.03 mm *ok!*

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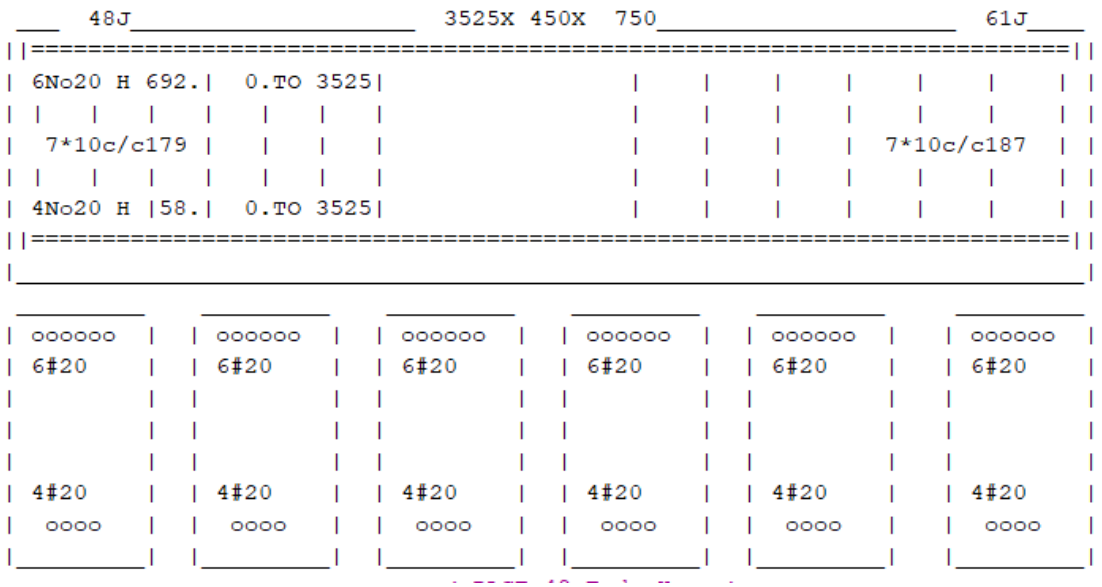
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9.2.2. Staad Output for LB2

```
            B E A M  N O.    73 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=  233.20 KNS  Vc=  270.39 KNS  Vs=   40.54 KNS
   Tu=   46.85 KN-MET  Tc=   16.5 KN-MET  Ts=   62.5 KN-MET  LOAD    25
   STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 179. MM  C/C FOR 1071. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   7.56 SQ.CM.

   AT END   SUPPORT - Vu=  122.85 KNS  Vc=  269.73 KNS  Vs=    0.00 KNS
   Tu=   53.55 KN-MET  Tc=   15.5 KN-MET  Ts=   71.4 KN-MET  LOAD    48
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 187. MM  C/C FOR 1071. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   8.64 SQ.CM.
```



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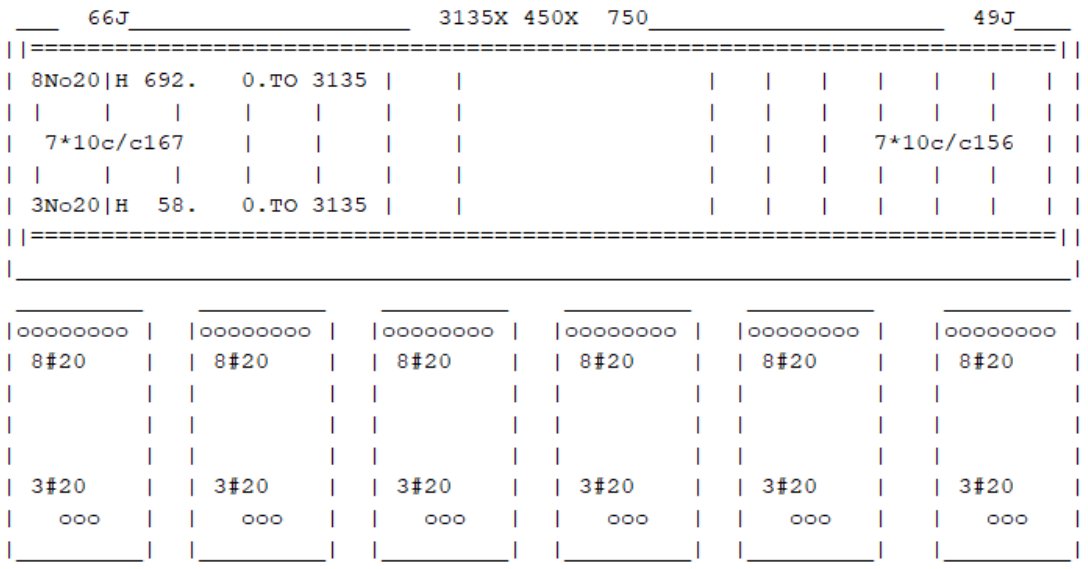
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9.2.3. Staad Output for LB3

```
        B E A M  N O.   100 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=  148.89 KNS  Vc=  428.23 KNS  Vs=    0.00 KNS
Tu=   59.96 KN-MET  Tc=   15.5 KN-MET  Ts=   79.9 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM  C/C FOR  876. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   9.67 SQ.CM.

AT END   SUPPORT - Vu=  211.89 KNS  Vc=  262.87 KNS  Vs=   19.65 KNS
Tu=   59.96 KN-MET  Tc=   15.5 KN-MET  Ts=   79.9 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 156. MM  C/C FOR  876. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   9.67 SQ.CM.
```

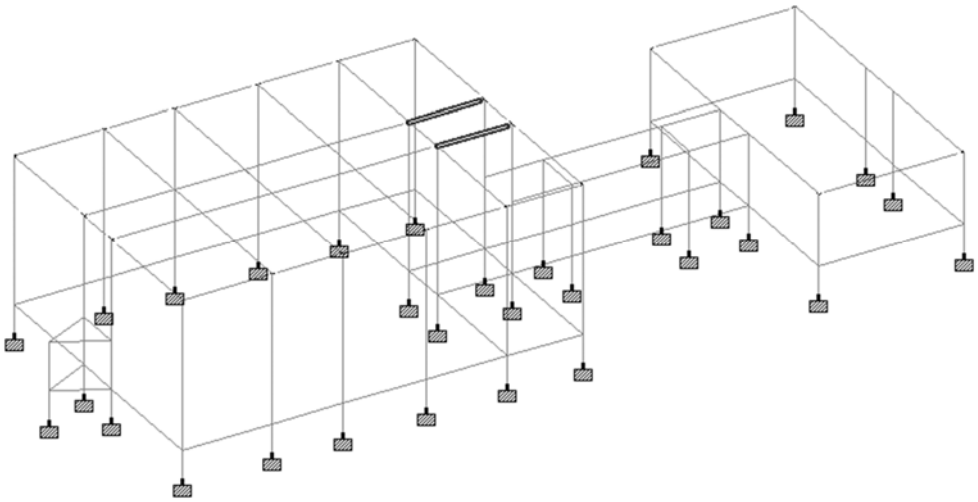


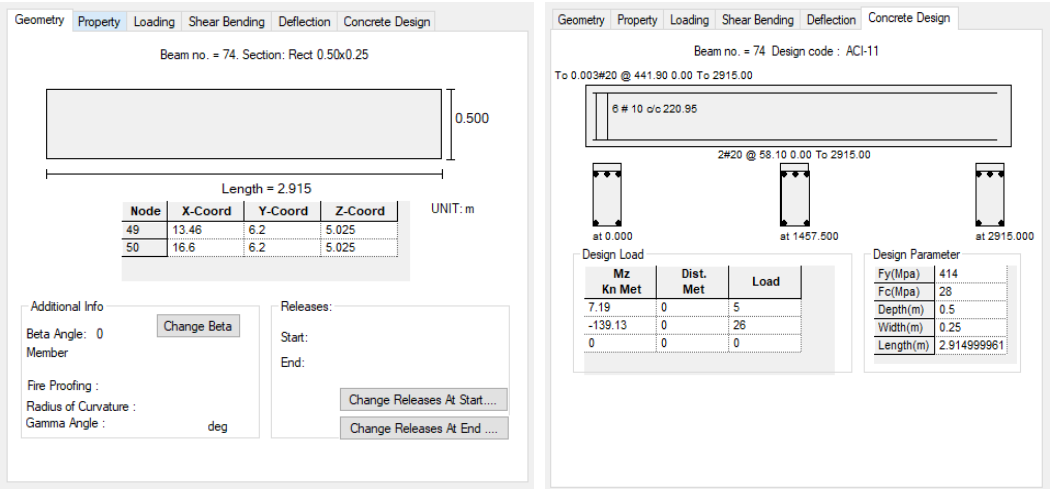
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9.3. Design of Ledger Beam(LB4)





Design of Beam (LB4)

Maximum Moment required from Analysis = 139 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **896 mm²**

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Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **392mm²**

Maximum Moment required from Analysis = **7 Knm (Sagging)**

However Provide 3-20 dia-Full Straight Bars @ Top & Bottom

Ast.pro**>>>>>>(Ast.min, Ast.req)**

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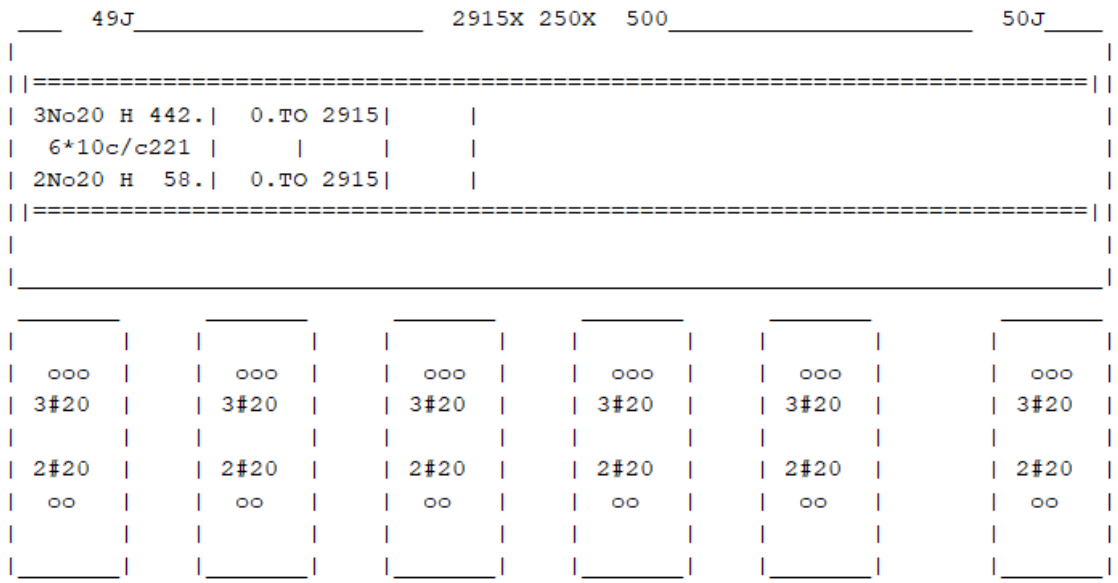
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9.3.1. Staad Output for LB4

```
        B E A M  N O.    74 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   57.45 KNS  Vc=   91.91 KNS  Vs=    0.00 KNS
 Tu=    0.82 KN-MET  Tc=    3.4 KN-MET  Ts=    0.0 KN-MET  LOAD     3
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1016. MM

 AT END   SUPPORT - Vu=    9.71 KNS  Vc=  168.92 KNS  Vs=    0.00 KNS
 Tu=    0.82 KN-MET  Tc=    3.4 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                  STIRRUPS ARE NOT REQUIRED.
```

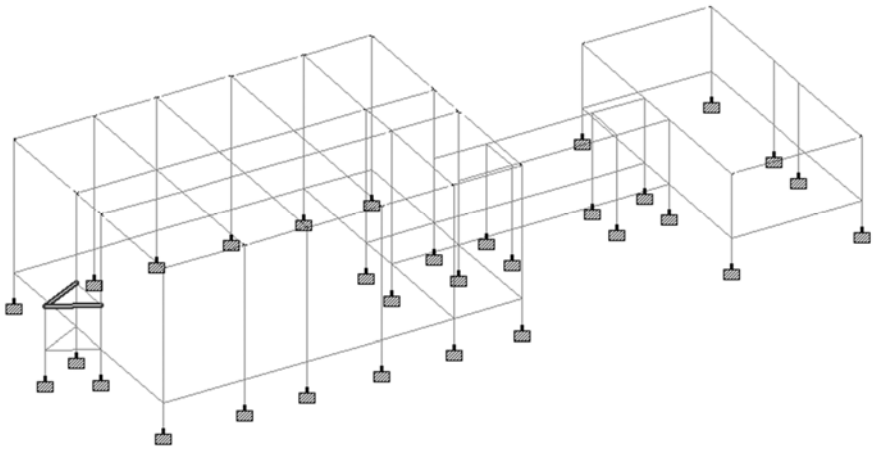


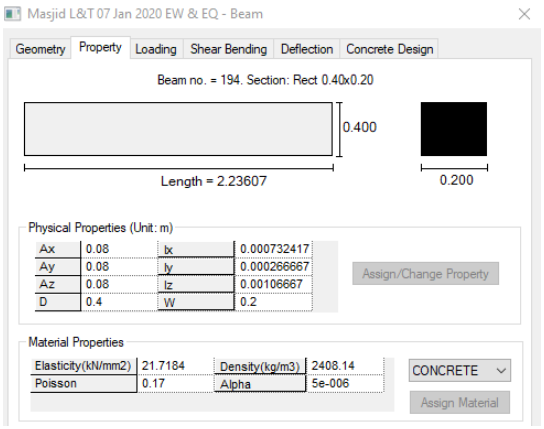
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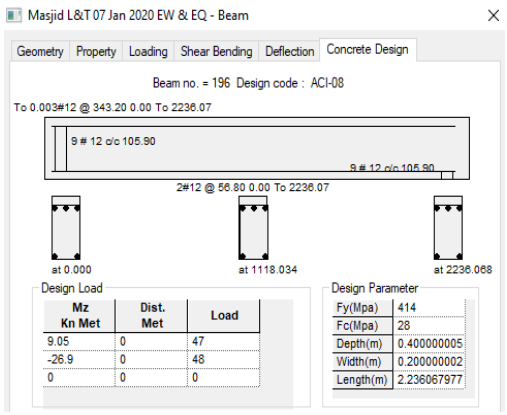
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9.4. Design of Beam(B1)







Hence Provide 3-12 dia-Full Straight Bars Top & Bottom

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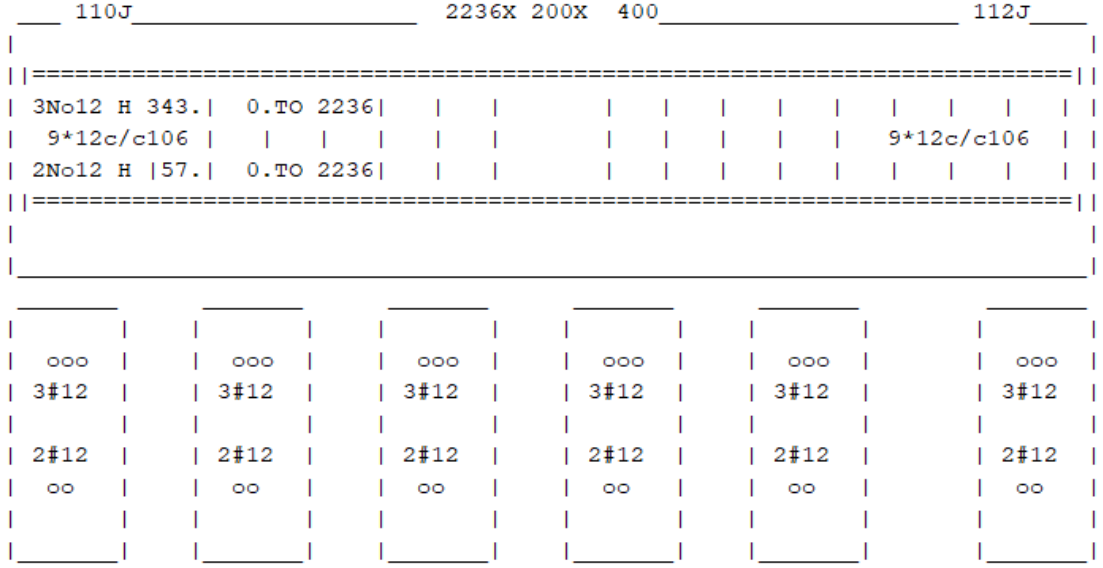
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9.4.1. Staad Output for B1

```
            B E A M   N O.   196 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    19.95 KNS   Vc=    61.22 KNS   Vs=     0.00 KNS
 Tu=     3.10 KN-MET   Tc=     1.8 KN-MET   Ts=     4.1 KN-MET   LOAD     28
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 12 MM 2-LEGGED STIRRUPS AT 106. MM  C/C FOR  782. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.43 SQ.CM.

 AT END    SUPPORT - Vu=     3.28 KNS   Vc=    57.35 KNS   Vs=     0.00 KNS
 Tu=     3.10 KN-MET   Tc=     1.8 KN-MET   Ts=     4.1 KN-MET   LOAD     28
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 12 MM 2-LEGGED STIRRUPS AT 106. MM  C/C FOR  782. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.43 SQ.CM.
```

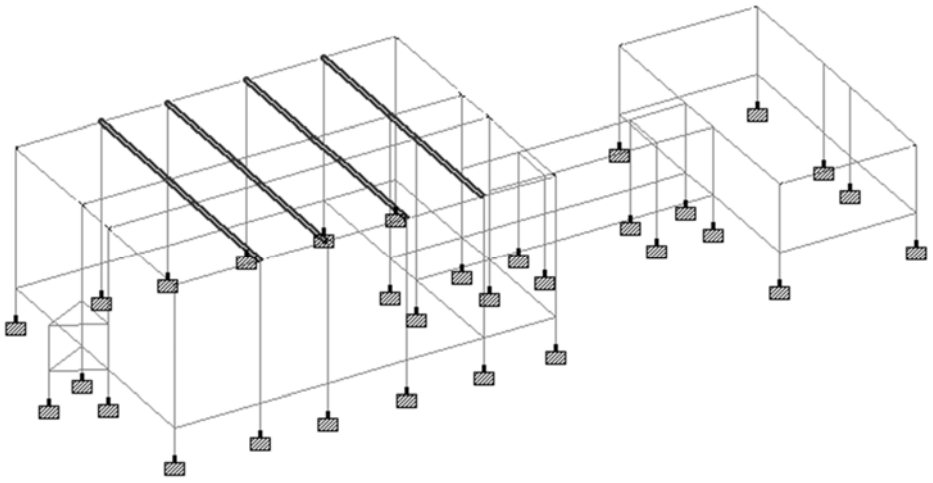


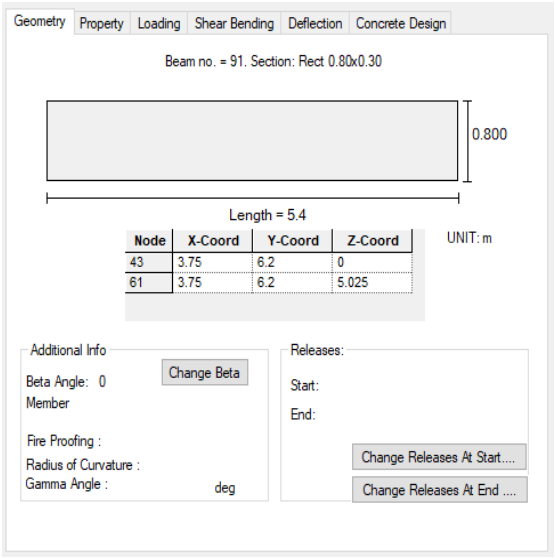
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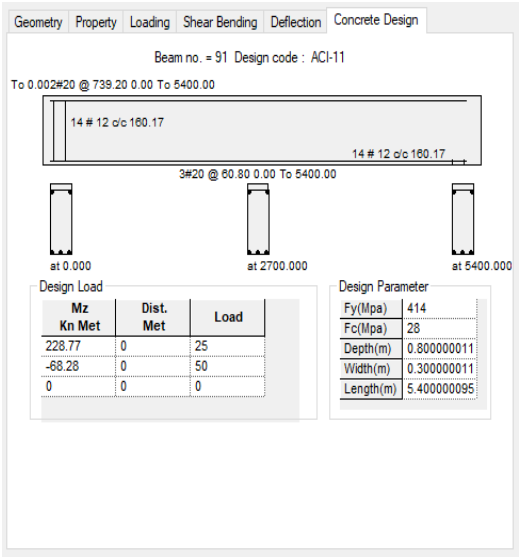
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9.5. Design of Beam-(B2)







Design of Beam (B2)

Maximum Moment required from Analysis = 228 **Knm (Sagging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=300mm, Fy=414 Mpa, d=800-40-10-20/2=740mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **847 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **791mm²**

Maximum Moment required from Analysis = **68 Knm (Hogging)**

Ast.pro**>>>>>>(Ast.min, Ast.req)**

Hence Provide 3-25 dia-Full Straight Bars Top & 6-20 dia-Full Straight Bars Bottom considering Long term deflection check as shown below in section 8.4.1.

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9.5.1. Long Term Deflection Check for B2 Beam

INPUT DATA:

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	750	mm
width of beam, b	=	250	mm
Length of beam, L	=	5.00	m
Bottom bar diam., db	=	16.00	mm
area of 1 bar Ab	=	201.06	mm^2
no. of bottom bars	=	6.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	20.00	mm
area of 1 bar Av	=	314.16	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	682.00	mm
Mdead	=	182.00 kN-m	(FROM STAAD RESULTS)
M live	=	20.00 kN-m	

CHECK DEFLECTION: $M_{d+\ell}$ = 202.00 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 192.00 kN-m

modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa ←*ACI(318M-08) Sec. 9.5.2.3*

$n_s = \frac{Es}{Ec}$ = 7.19

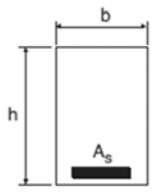
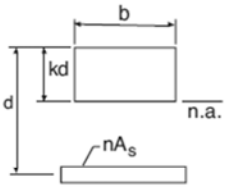
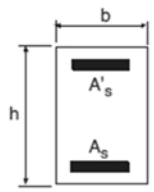
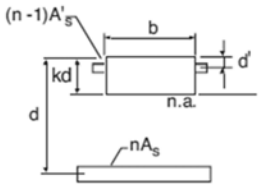
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 8.79E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.03 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 185.63 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 2.67E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 3.29$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 85.97 \text{ kN-m}$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.47$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 3.32E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.45$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 3.22E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.43$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 3.14E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 5.141 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 5.584 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 6.020 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 0.879 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\ell \le \frac{\ell}{180} = 27.78 \text{ mm} > 0.88 \text{ mm} \quad ok!$$

$$(\Delta_i)_\ell \le \frac{\ell}{360} = 13.89 \text{ mm} > 0.88 \text{ mm} \quad ok!$$

additional long-term deflection

$\xi = 2.00$

5 years or more 2.0
12 months..... 1.4
6 months..... 1.2
3 months..... 1.0

$$\rho = \frac{As}{bd} = 0.0071$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.48$$ ←*ACI(318M-08) Sec. 9.5.2.5*

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 8.25 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell = 9.13 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480} = 10.42 \text{ mm} > 9.13 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240} = 20.83 \text{ mm} > 9.13 \text{ mm} \quad ok!$$

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9.5.2. Staad Output for B2

ACI 318-11 BEAM NO. 91 DESIGN RESULTS

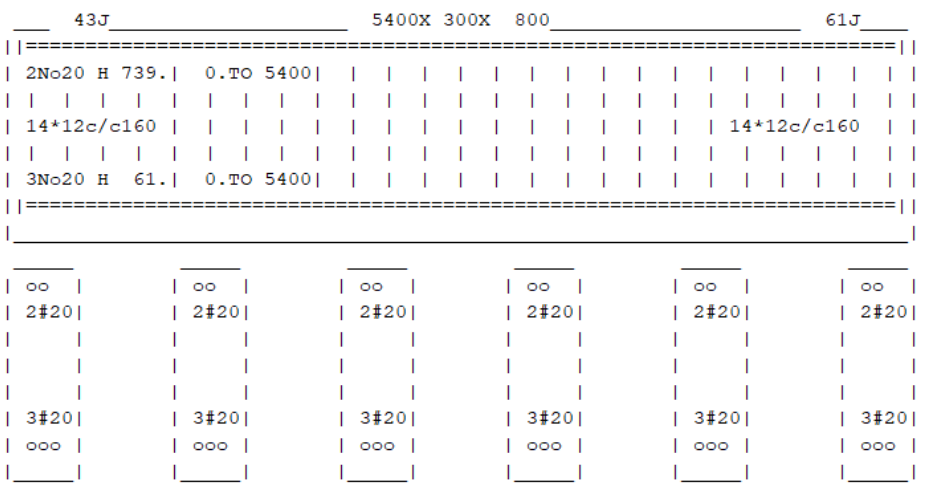
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 800. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	61.	3 - 20MM	0.	5400.	YES	YES
2	739.	2 - 20MM	0.	5400.	YES	YES

B E A M N O. 91 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.96 KNS Vc= 277.99 KNS Vs= 0.00 KNS
Tu= 56.15 KN-MET Tc= 8.7 KN-MET Ts= 74.9 KN-MET LOAD 25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 160. MM C/C FOR 1963. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 13.04 SQ.CM.

AT END SUPPORT - Vu= 38.80 KNS Vc= 186.05 KNS Vs= 0.00 KNS
Tu= 56.15 KN-MET Tc= 8.7 KN-MET Ts= 74.9 KN-MET LOAD 25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 160. MM C/C FOR 1963. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 13.04 SQ.CM.

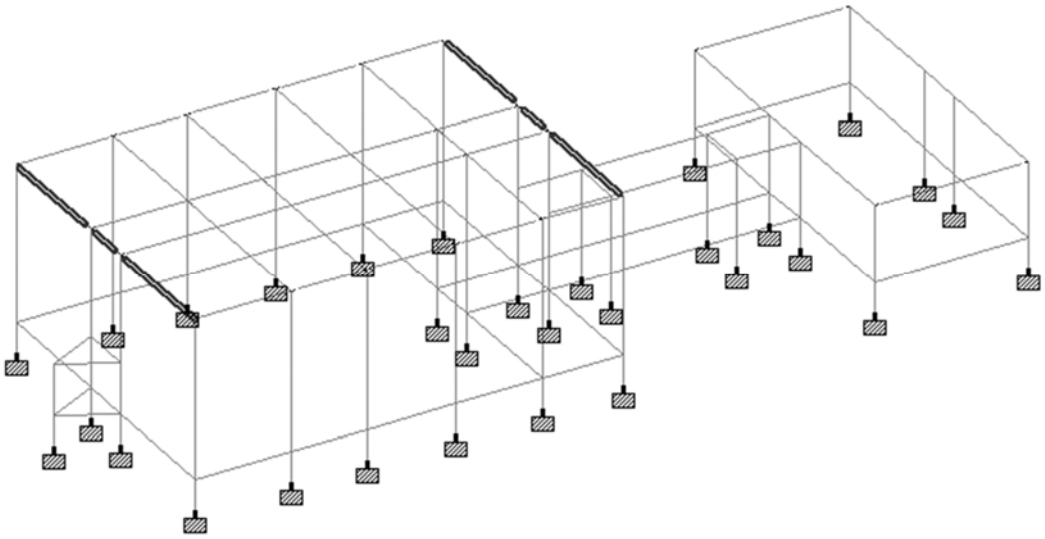


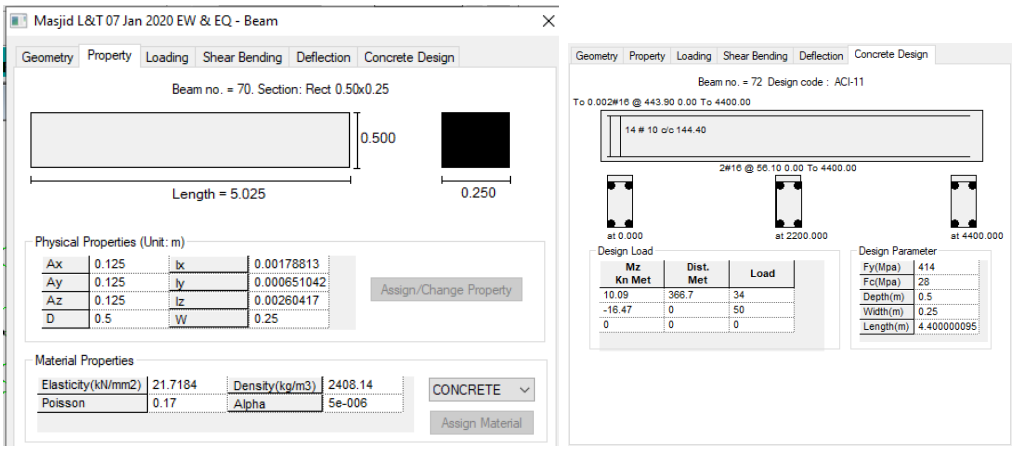
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9.6. Design of Beam-(B3)





Hence Provide minimum 3-16 dia-Full Straight Bars Top & Bottom…

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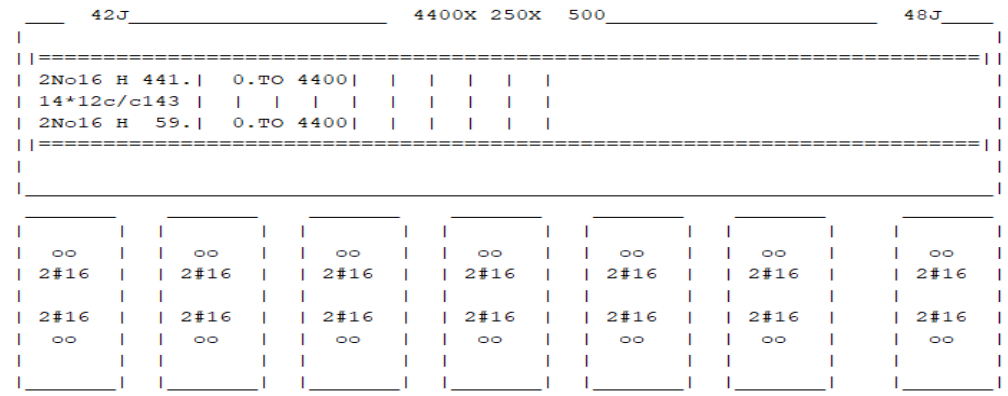
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9.6.1. Staad Output for B3

```
            B E A M  N O.    70 D E S I G N  R E S U L T S - SHEAR


  AT END   SUPPORT - Vu=   12.38 KNS  Tu=   25.2 KN-MET
                     Vc=   93.6 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD    24 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
  AT START SUPPORT - Vu=    2.04 KNS  Vc=   91.96 KNS  Vs=    0.00 KNS
  Tu=   20.74 KN-MET  Tc=    3.5 KN-MET  Ts=   27.7 KN-MET  LOAD    25
  STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1764. MM
  ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   6.77 SQ.CM.
```

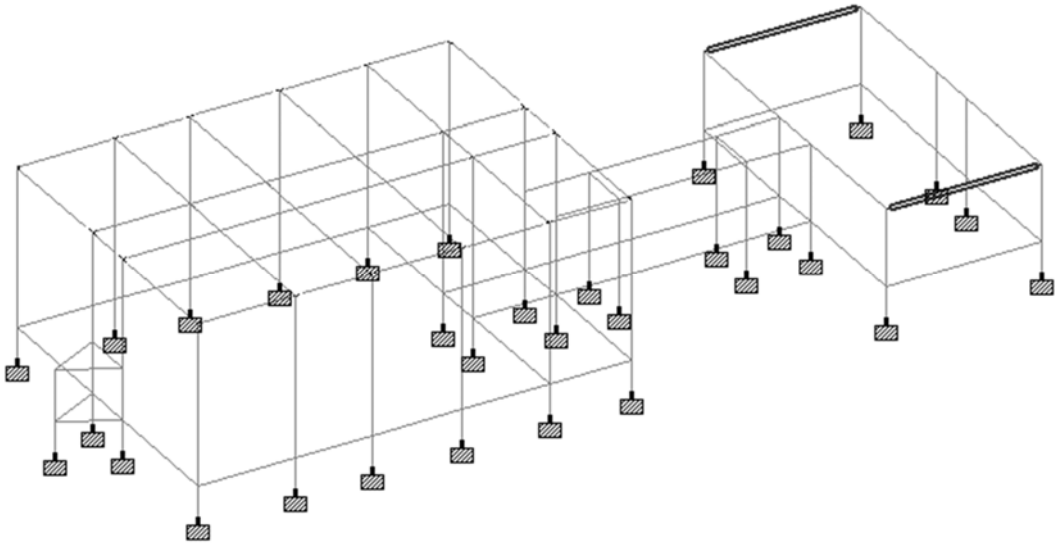


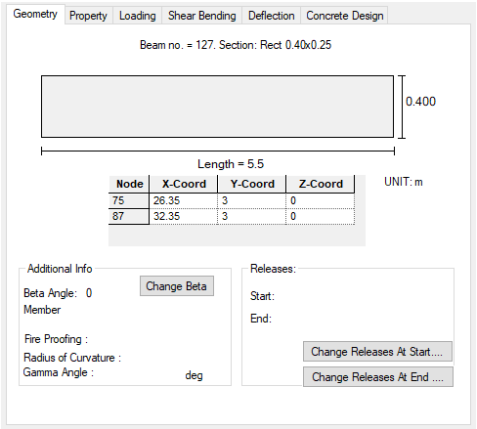
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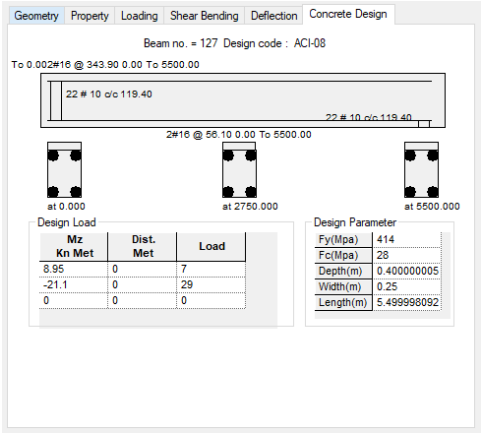
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9.7. Design of Beam-(B4)







Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

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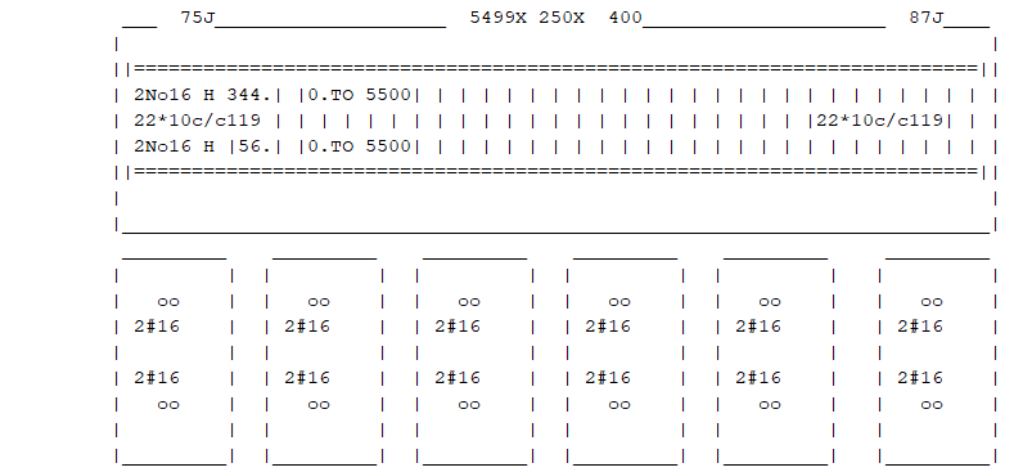
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9.7.1. Staad Output for B4

```
B E A M  N O.   127 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=    4.22 KNS  Vc=  132.91 KNS  Vs=    0.00 KNS
Tu=    3.50 KN-MET  Tc=    2.6 KN-MET  Ts=    4.7 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 119. MM  C/C FOR 2408. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.23 SQ.CM.

AT END   SUPPORT - Vu=    8.29 KNS  Vc=   72.13 KNS  Vs=    0.00 KNS
Tu=    3.67 KN-MET  Tc=    2.6 KN-MET  Ts=    4.9 KN-MET  LOAD    25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 119. MM  C/C FOR 2408. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.29 SQ.CM.
```



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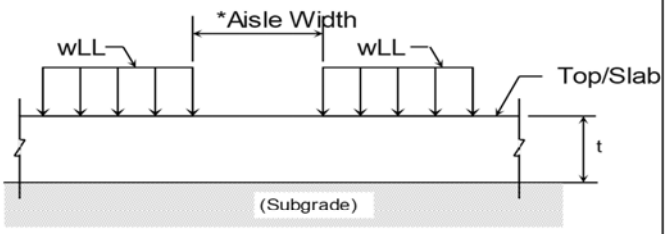
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

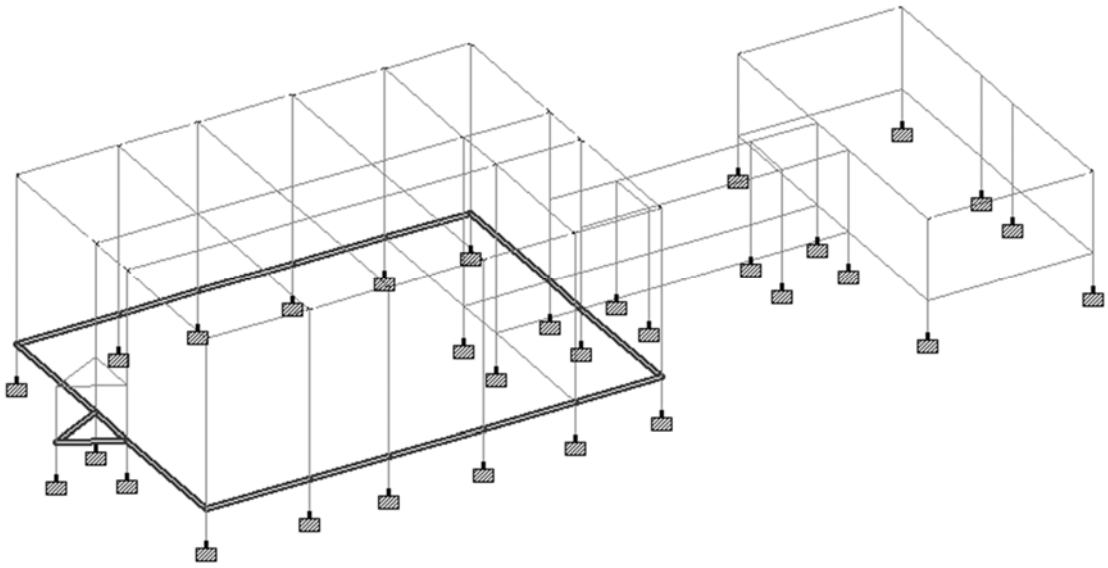
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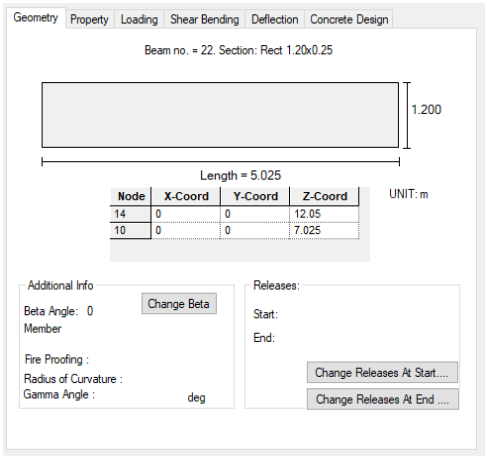
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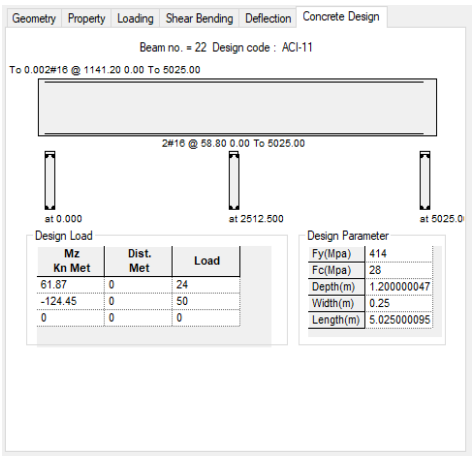
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11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1/GB4)







Maximum Moment required from Analysis = 124 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=1200-50-10-16/2=1132mm, Fc'=35 Mpa & ϕ=0.9

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Ast.req = **300 mm²**

Ast.min= $$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **1008mm²**

Maximum Moment required from Analysis = **61 Knm (Sagging)**

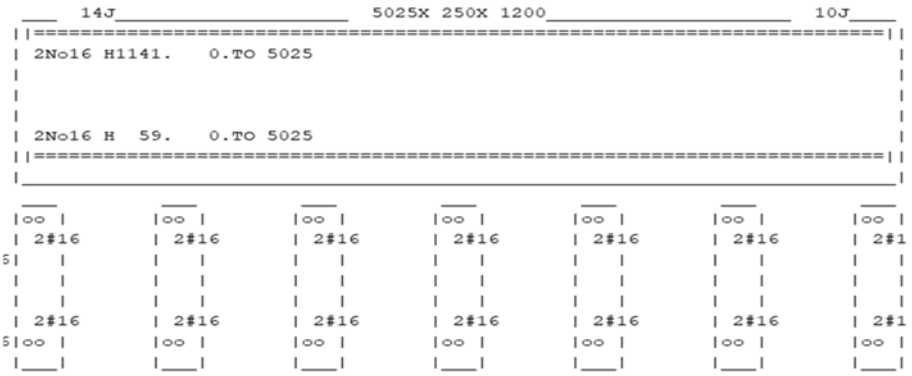
Hence Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bars Bottom. Also the stirrups shall be 12Dia@150 c-c considering retaining effects of the Grade Beam.

Ast.pro= **1470mm² >>>>>>(Ast.min, Ast.req)**

11.1.1. Staad Output for GB1/GB4

```
            B E A M  N O.    22 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=    36.19 KNS   Vc=  244.72 KNS   Vs=     0.00 KNS
   Tu=     4.97 KN-MET   Tc=    10.3 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=    47.16 KNS   Vc=  335.82 KNS   Vs=     0.00 KNS
   Tu=     4.97 KN-MET   Tc=    10.3 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
```

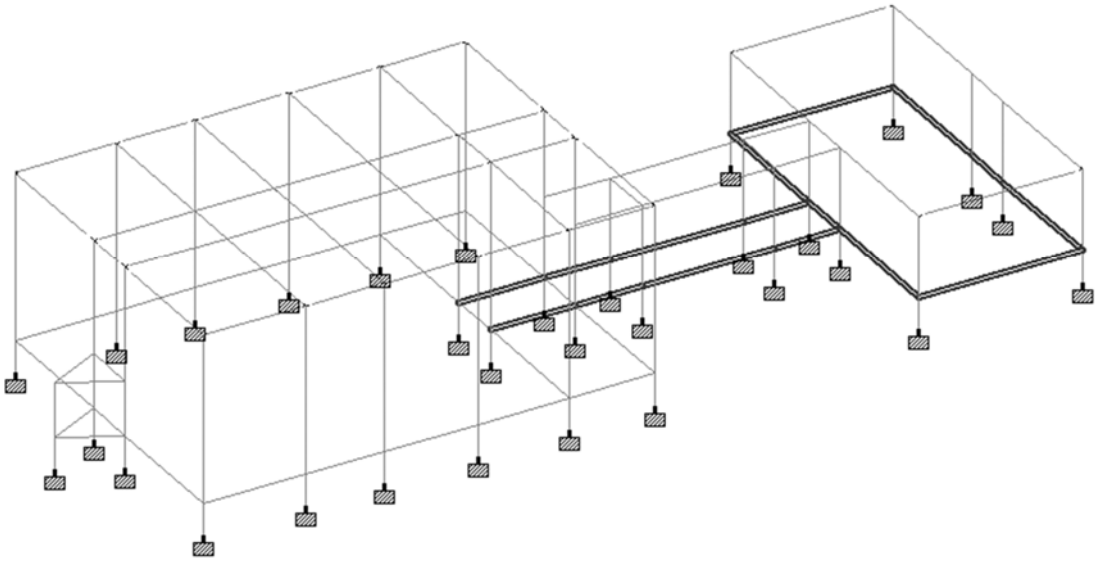


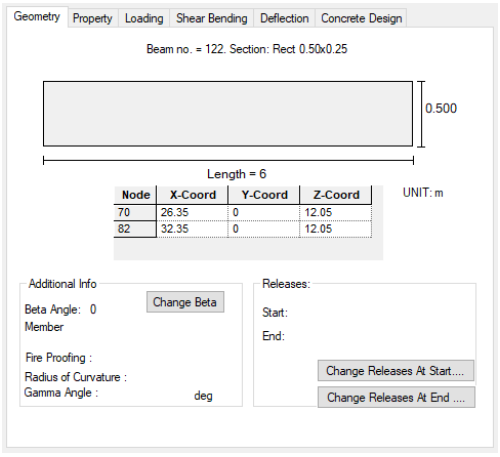
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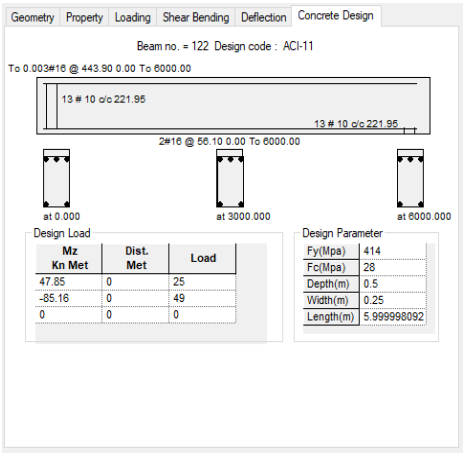
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11.2. Grade Beam(GB2)







Hence Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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11.2.1. Staad Output for GB2

```
        B E A M  N O.   122 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=    53.46 KNS  Vc=    97.46 KNS  Vs=     0.00 KNS
Tu=     0.00 KN-MET  Tc=     3.7 KN-MET  Ts=     0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 2558. MM

AT END   SUPPORT - Vu=    53.73 KNS  Vc=    97.35 KNS  Vs=     0.00 KNS
Tu=     0.00 KN-MET  Tc=     3.7 KN-MET  Ts=     0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 2558. MM
```

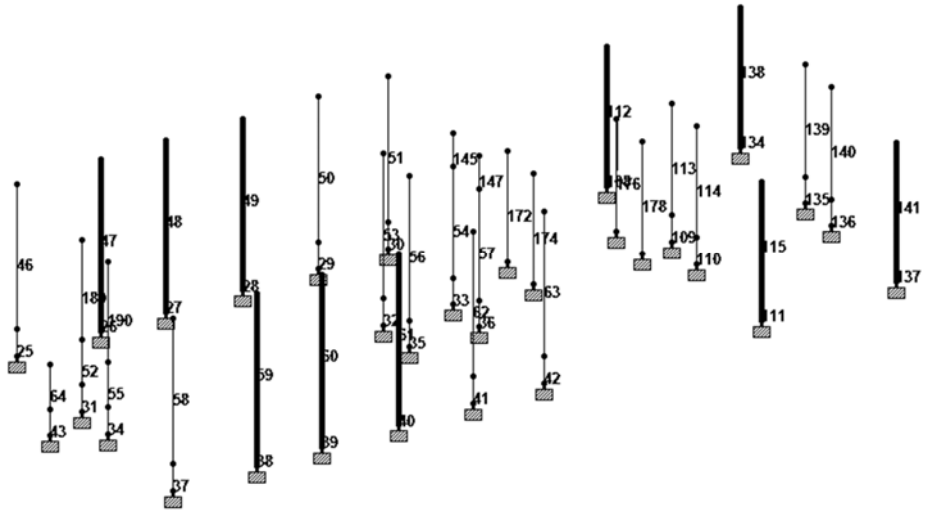

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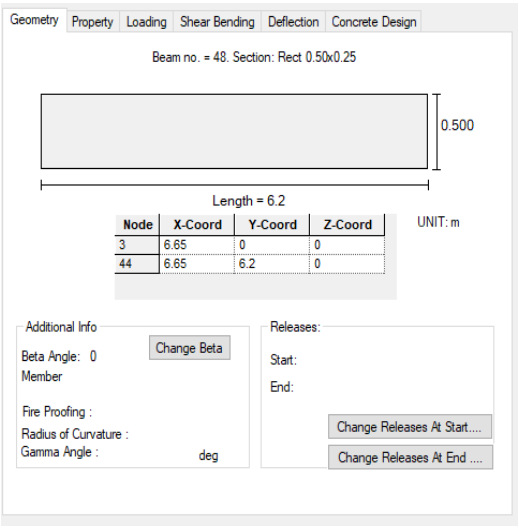
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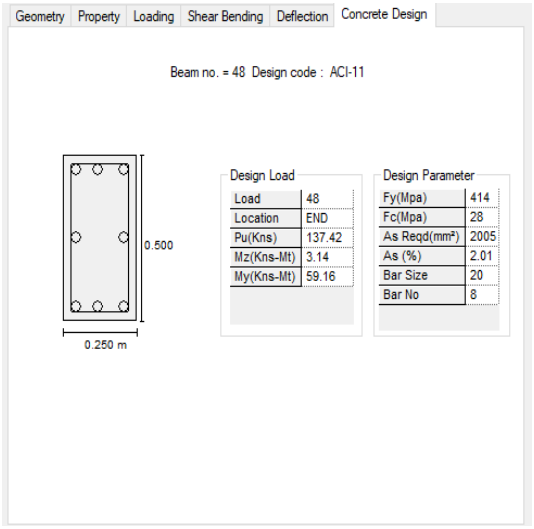
12. CONCRETE COLUMN DESIGN

12.1. Design of Column-(C1)



Column numbers C1





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12.1.1. Staad Output for Column C1

```
    ACI 318-11   COLUMN  NO.    48   DESIGN RESULTS
    ==============================================


-------------------------------< PAGE 97 Ends Here >--------------------------
 STAAD SPACE                                                 -- PAGE NO.   98

FY - 413.7  FC -  27.6 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
       AREA OF STEEL REQUIRED =  2005.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

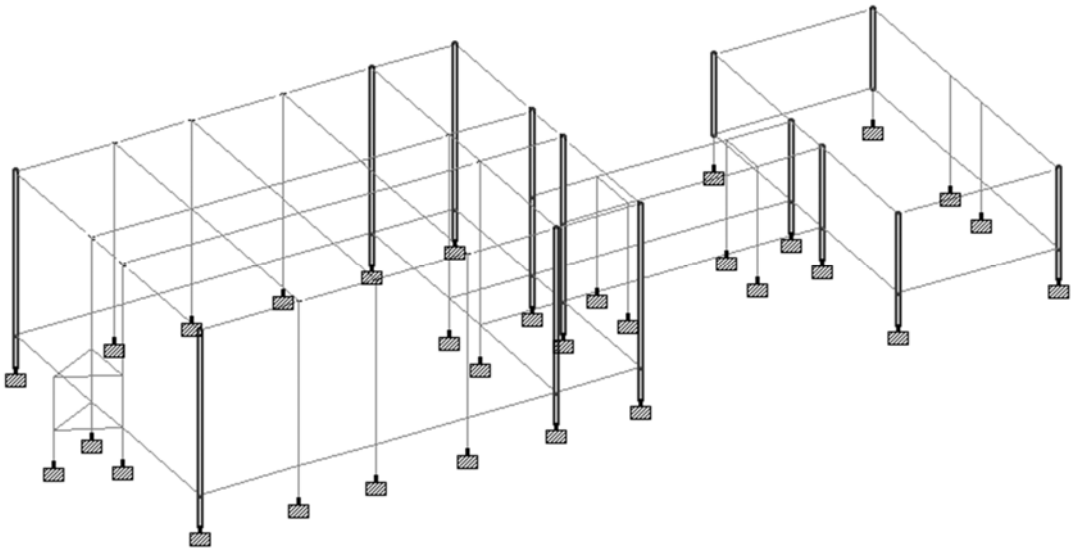
  8 - 20 MM               2.011       48      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   12 SPACING 250.00 MM
```


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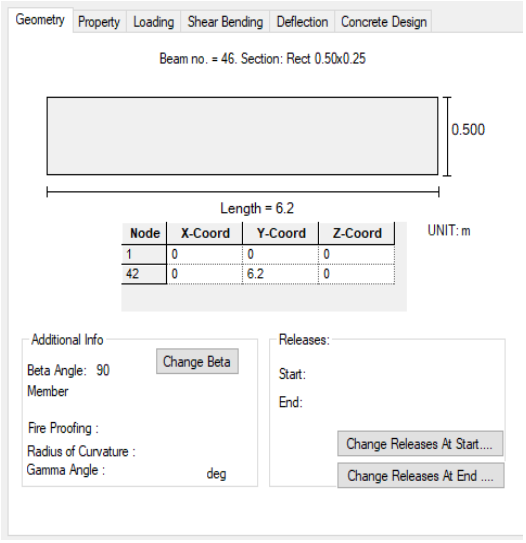
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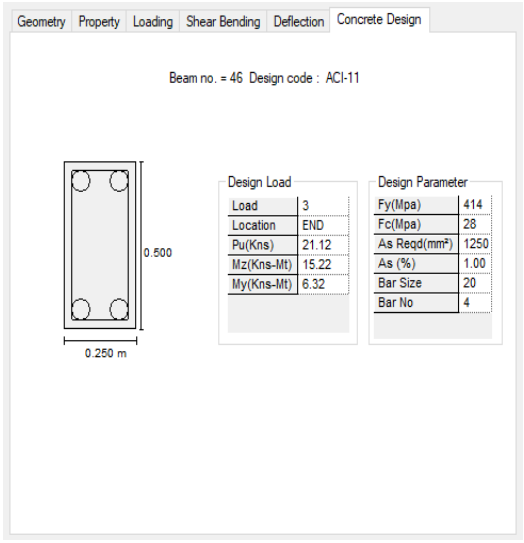
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12.2. Design of Column-(C2)



Column numbers C2





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12.2.1. Staad Output for Column C2

```
ACI 318-11   COLUMN  NO.    48   DESIGN RESULTS
==================================================

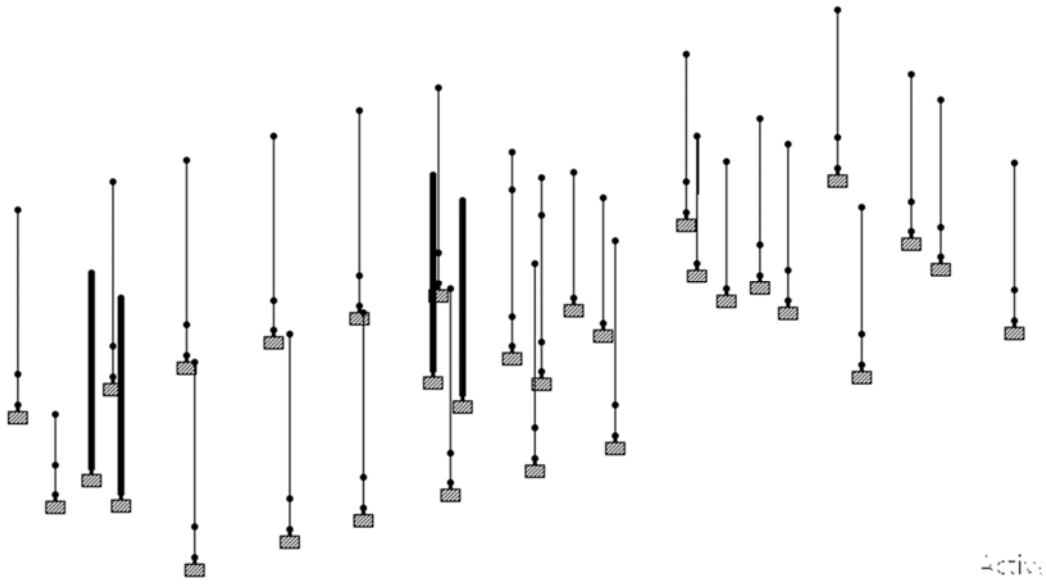
-------------------------------< PAGE 97 Ends Here >--------------------------
 STAAD SPACE                                                  -- PAGE NO.   98

FY - 413.7  FC -  27.6 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
       AREA OF STEEL REQUIRED =  2005.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

  8 - 20 MM               2.011       48      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   12 SPACING 250.00 MM
```

12.3. Design of Column-(C4)



Column numbers C4

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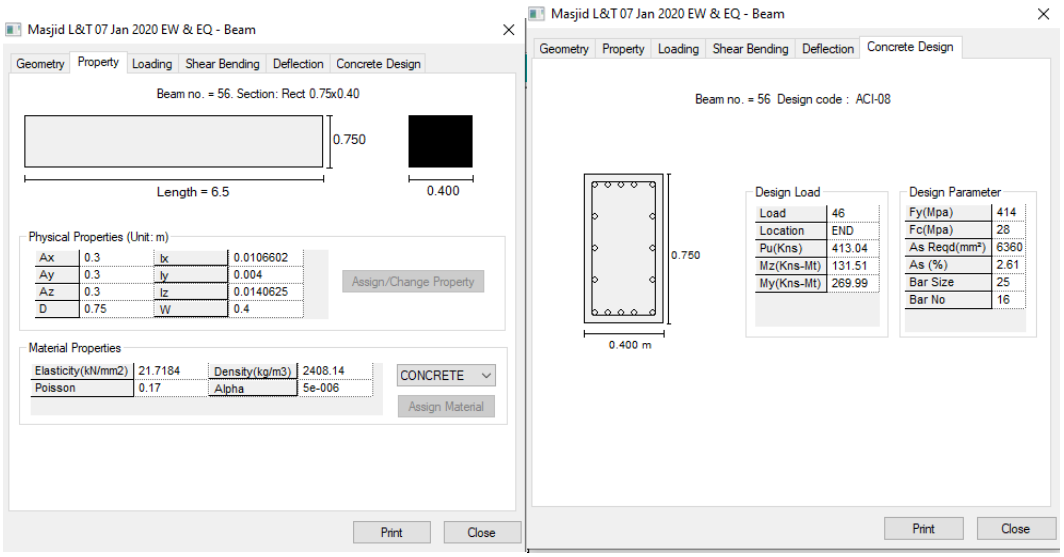
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Beam End Force Summary

The signs of the forces at end B of each beam have been reversed. For example: this means that the Min Fx entry gives the large tension value for an beam.

				Axial	Shear		Torsion	Bending	
	Beam	Node	L/C	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kNm)	My (kNm)	Mz (kNm)
Max Fx	35	27	27:GENERATE	**267.766**	31.686	-1.987	0.290	-6.825	-26.132
Min Fx	32	2	5:WL+X	**-16.820**	18.206	-6.743	0.181	2.613	11.730
Max Fy	2	5	23:GENERATE	14.757	**74.525**	-0.000	4.179	0.001	77.262
Min Fy	2	6	23:GENERATE	14.757	**-60.162**	-0.000	-5.017	0.000	33.814
Max Fz	36	9	27:GENERATE	168.833	20.726	**33.468**	-0.564	-17.631	18.676
Min Fz	33	3	23:GENERATE	153.453	30.158	**-24.278**	0.005	19.769	24.819
Max Mx	7	12	27:GENERATE	11.594	-57.245	1.017	**5.284**	2.642	36.032
Min Mx	6	10	27:GENERATE	7.480	35.251	-2.059	**-5.224**	4.180	9.636
Max My	36	28	27:GENERATE	173.084	20.726	33.468	-0.564	**32.571**	-12.413
Min My	36	9	23:GENERATE	153.453	30.158	24.278	-0.005	**-19.769**	24.819
Max Mz	2	5	23:GENERATE	14.757	74.525	-0.000	4.179	0.001	**77.262**
Min Mz	477	2	31:GENERATE	4.046	-15.647	0.254	-0.126	-0.532	**-43.277**



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12.3.1. Staad Output for Column C4

```
=========================================================================

     COLUMN  NO.    56  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  27.6 MPA,  RECT SIZE - 400.0 X 750.0 MMS, TIED
        AREA OF STEEL REQUIRED =  6360.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

 16 - 25 MM              2.618       46      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   12 SPACING 256.00 MM
```


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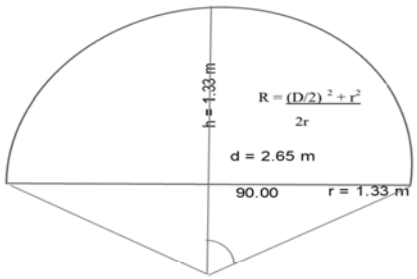
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13. DOME & RING BEAM(B5) DESIGN

Dome & Ring Beam Design as per ACI-318-08(PCA Notes)

Dimensions of Dome

Diameter	d =	2650 mm
Height	h =	1325 mm
Thickness	t =	120 mm
Radius of Sphere	R =	1325 mm
	Φ =	90.00
	Θ =	0 to 90.00



Loading

Dead Load	DL =	2.88 KN/m
Live Load	LL =	0.10 KN/m
Wind Load	WL =	0.10 KN/m
Total Load	W =	3.08 KN/m
Factored Load	Wu =	4.62 KN/m

Meridional Stress =WR/t(1/1+cosθ)
Hoop Stresses =WR/tX((cos^2(θ)+cosθ-1)/(1+cosθ))

Meridional Stress		Hoop Stress	
Θ	Mt	Θ	Mt
90.00	0.051 MPa	90.00	-0.032 MPa
45.00	0.030 MPa	45.00	0.004 MPa
40.00	0.029 MPa	40.00	0.006 MPa
35.00	0.028 MPa	35.00	0.009 MPa
30.00	0.027 MPa	30.00	0.010 MPa
25.00	0.027 MPa	25.00	0.012 MPa
20.00	0.026 MPa	20.00	0.013 MPa
15.00	0.026 MPa	15.00	0.015 MPa
5.00	0.026 MPa	5.00	0.016 MPa
0.00	0.026 MPa	0.00	0.016 MPa
Maximum Meridional Stress	0.051 MPa	Maximum Hoop Stress	0.032 MPa

fck	35
Fy	415 MPa
δst	230.000 MPa

Area of steel	**27 sqmm**	**Area of steel**	**17 sqmm**
Bar Dia	10 mm	Bar Dia	10 mm
Spacing	**2951 c/c**	Spacing	**4734 c/c**

However Provide Minimum Reinf both Circumferentially & meridionally in the Dome as per Clause 19.4.9

Astmin= 0.35%*t*h = 557 sqmm

Spacing of Reinf. 1000*(Abar)/Astmin
= 140 mm
Both Ways

19.4.8 Concentration of Reinforcement

In the case of long barrel shells (or domes) it is often desirable to concentrate tensile reinforcement near the edges rather than distribute the reinforcement over the entire tensile zone. When this is done, a minimum amount of reinforcement equal to 0.0035bh must be distributed over the remaining portion of the tensile zone, as shown in Fig. 27-3. This amount in practical terms is twice the minimum steel requirement for shrinkage and temperature stresses.

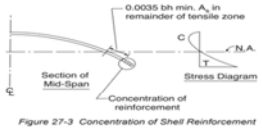


Figure 27-3 Concentration of Shell Reinforcement

19.4.10 Spacing of Reinforcement

Maximum permissible spacing of reinforcement is the smaller of 5 times the shell thickness and 18 in. Therefore, for shells less than 3.6 in. thick, 5 times the thickness controls. For thicker shells, the spacing of bars must not exceed 18 in.

Ring Beam Design

The Meredian thrust@support will have a horizontal component which will cause supporting walls to burst outwards, causing its failure. In order to bear this horizontal component, a Ring Beam is provided @Base of Dome.

Meridional Thrust @ Base(T1) **11 KN/m** Wr/(1+cos Θ)

Hoop Tension on Ring Beam(Th) = **6 KN**
Area of steel **15 sqmm** T1*(cos Θ)* D/2

However Provide minimum 3-16mm Dia T&B-Full Straight Bars

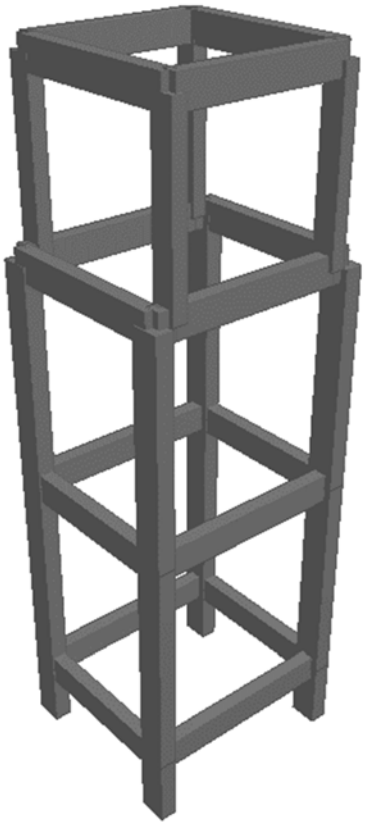
Bar Dia	16 mm
No of Bars	**3 nos**

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14. MINARETE DESIGN

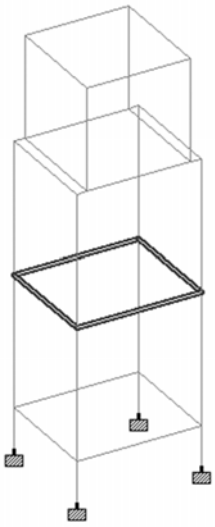


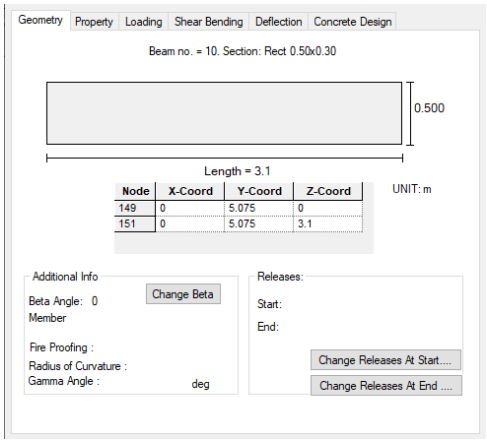
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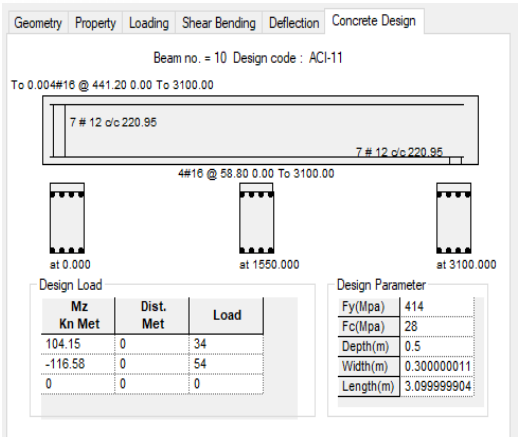
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14.1. Design of Beam-(B6)







Maximum Moment required from Analysis = 116 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=300mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **736mm^2**

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Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **472mm²**

Maximum Moment required from Analysis = **104 Knm (Sagging)**

Hence Provide 4-16 dia-Full Straight Bars Top & Bottom

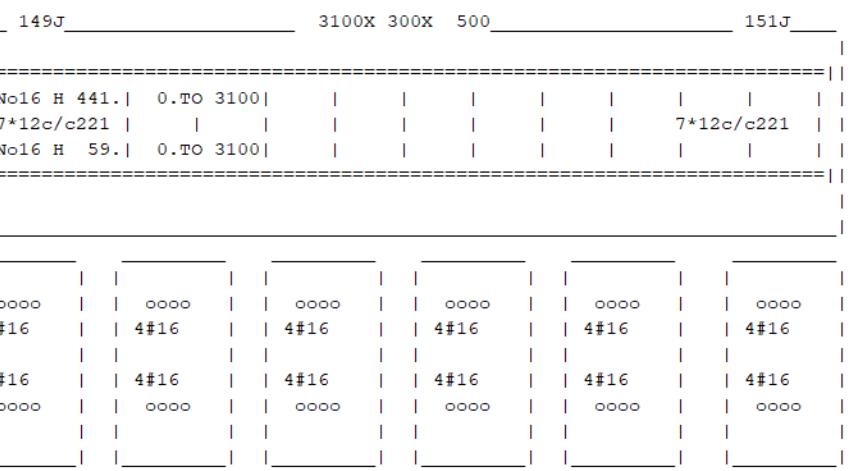
Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

14.1.1. Staad Output for B6

```
            B E A M   N O.     10 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=    44.50 KNS   Vc=   114.39 KNS   Vs=     0.00 KNS
Tu=     0.00 KN-MET   Tc=     4.6 KN-MET   Ts=     0.0 KN-MET   LOAD      6
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1114. MM

AT END    SUPPORT - Vu=    44.50 KNS   Vc=   114.39 KNS   Vs=     0.00 KNS
Tu=     0.00 KN-MET   Tc=     4.6 KN-MET   Ts=     0.0 KN-MET   LOAD      6
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1114. MM
```

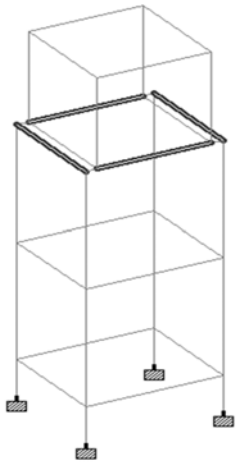


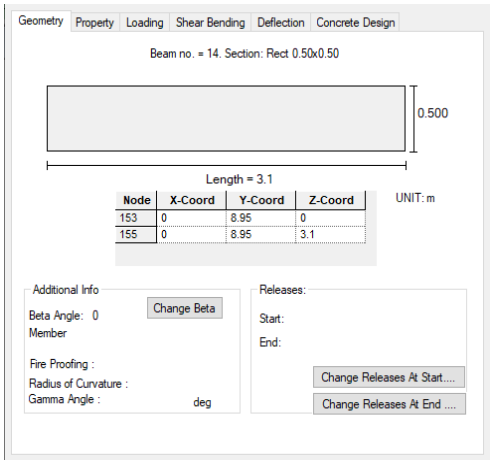
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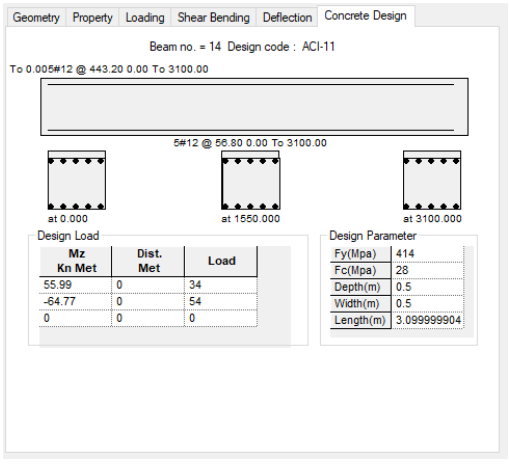
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14.2. Design of Beam-(B7)







Maximum Moment required from Analysis = 64 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **784mm²**

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Maximum Moment required from Analysis = **55 Knm (Sagging)**

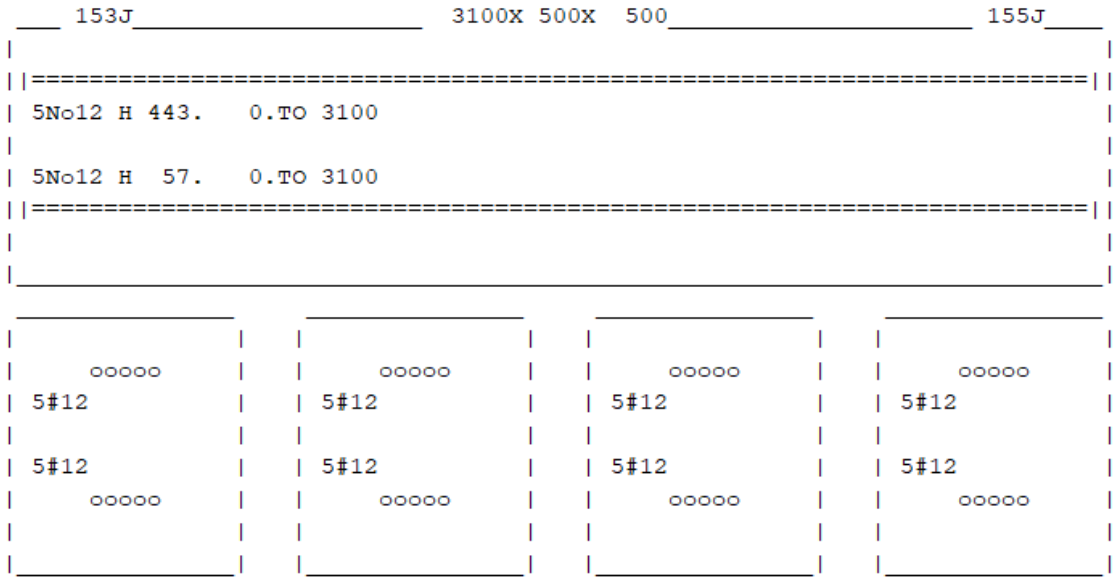
Hence Provide 5-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **1000mm^2 >>>>>>(Ast.min, Ast.req)**

14.2.1. Staad Output for B7

```
        B E A M  N O.    14 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   17.71 KNS  Vc=  199.51 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=   10.2 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   17.71 KNS  Vc=  199.51 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=   10.2 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                   STIRRUPS ARE NOT REQUIRED.
```

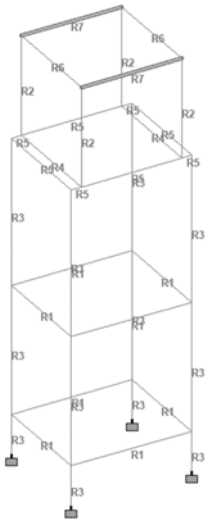


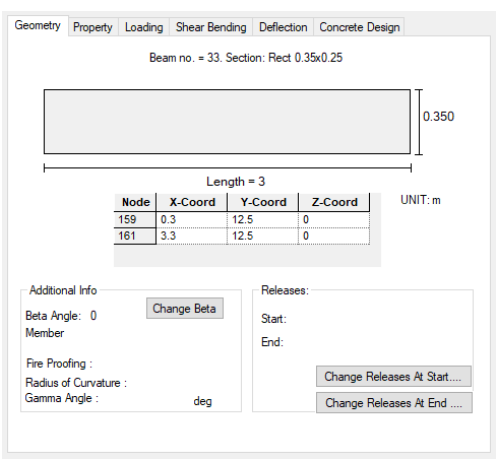
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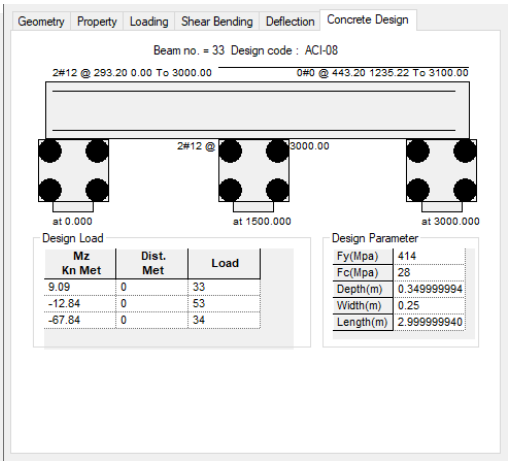
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14.3. Design of Beam-(LB5)







Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

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14.3.1. Staad Output for LB5

BEAM NO. 33 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

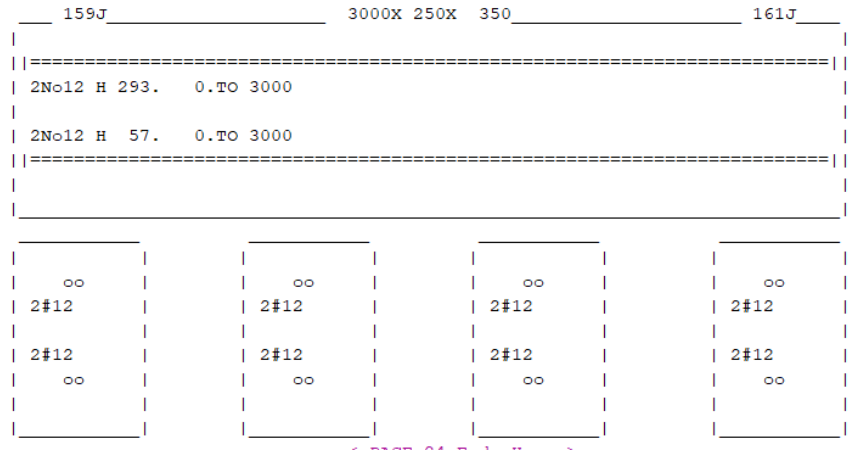
LEN - 3000. MM FY - 414. FC - 28. MPA, SIZE - 250. X 350. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	3000.	YES	YES
2	293.	2 - 12MM	0.	3000.	YES	YES

B E A M N O. 33 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.51 KNS Vc= 62.42 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.1 KN-MET Ts= 0.0 KN-MET LOAD 3
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.51 KNS Vc= 62.42 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.1 KN-MET Ts= 0.0 KN-MET LOAD 3
STIRRUPS ARE NOT REQUIRED.

AT START SUPPORT - Vu= 2.51 KNS Vc= 62.42 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.1 KN-MET Ts= 0.0 KN-MET LOAD 3
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.51 KNS Vc= 62.42 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.1 KN-MET Ts= 0.0 KN-MET LOAD 3
STIRRUPS ARE NOT REQUIRED.

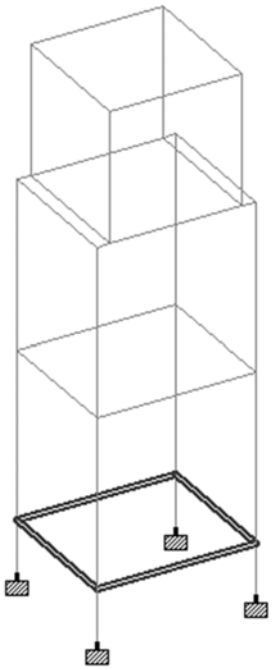


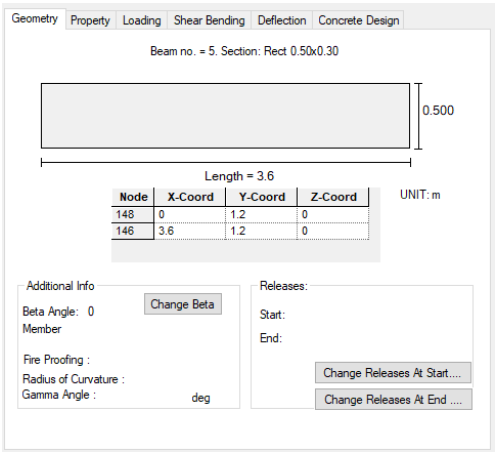
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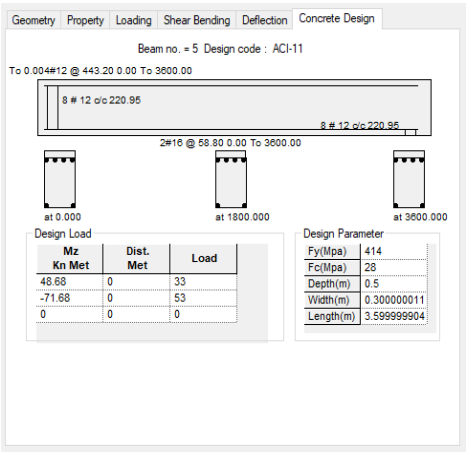
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14.4. Grade Beam(GB3)







Hence Provide Astmin 4-16 dia-Full Straight Bars Top & 4-16 dia-Full Straight Bars Bottom

Ast.pro= **800mm² >>>>>>(Ast.min, Ast.req)**

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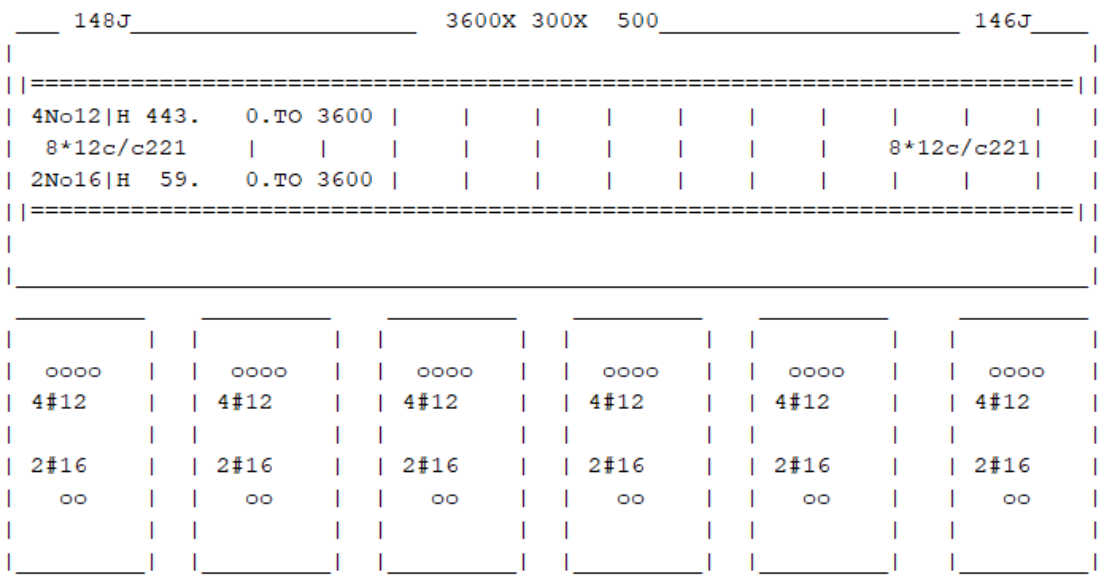
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14.4.1. Staad Output for GB3

```
        B E A M  N O.     5 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   50.05 KNS  Vc=  113.05 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    4.7 KN-MET  Ts=    0.0 KN-MET  LOAD    53
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1364. MM

AT END   SUPPORT - Vu=   50.05 KNS  Vc=  113.05 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    4.7 KN-MET  Ts=    0.0 KN-MET  LOAD    33
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1364. MM
```

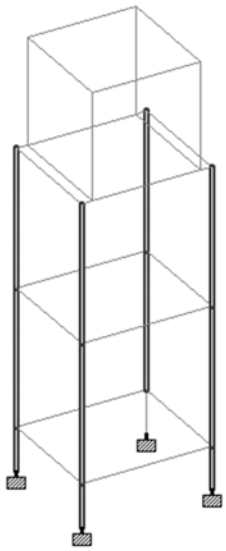


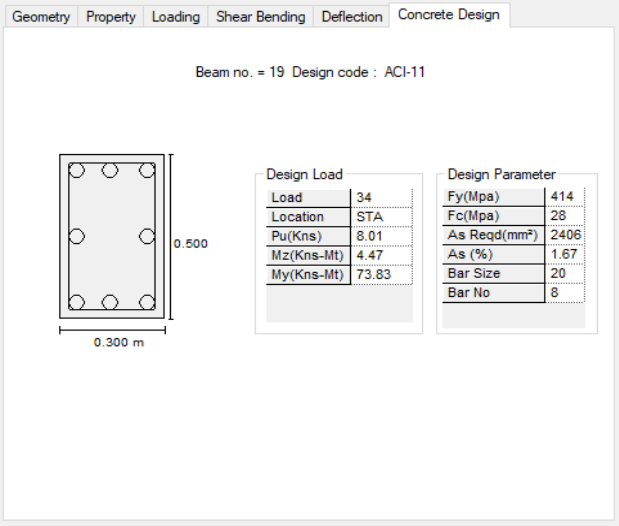
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14.5. Design of Column-(C3)





14.5.1. Staad Output for Column-(C3)

```
        ACI 318-11   COLUMN  NO.    19   DESIGN RESULTS
        ================================================


FY - 413.7  FC -  27.6 MPA,  RECT SIZE - 300.0 X 500.0 MMS, TIED
        AREA OF STEEL REQUIRED =  2406.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

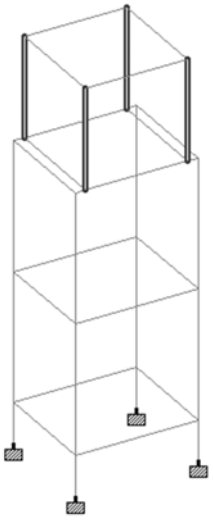
  8 - 20 MM               1.676       34      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   12 SPACING 300.00 MM
```

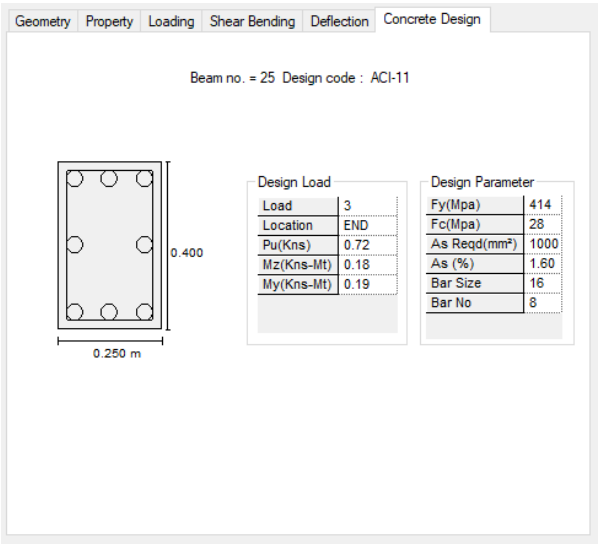

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14.6. Design of Column-(C6)





14.6.1. Staad Output for Column-(C6)

```
      ACI 318-11    COLUMN  NO.     25    DESIGN RESULTS
      ================================================


FY - 413.7   FC -   27.6 MPA,   RECT SIZE - 250.0 X 400.0 MMS, TIED
         ONLY MINIMUM STEEL IS REQUIRED.
         AREA OF STEEL REQUIRED =   1000.0   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  8 - 16 MM                 1.609         3       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    12 SPACING 250.00 MM
```


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15. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		
Calculations					
Shear force Vu = U*le	**8.096**	**kN**			
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**			

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b l_e/360 **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = $M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage
$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000
Where:-
ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318
ρ' = $A_{s\ b}$/(b*d) 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

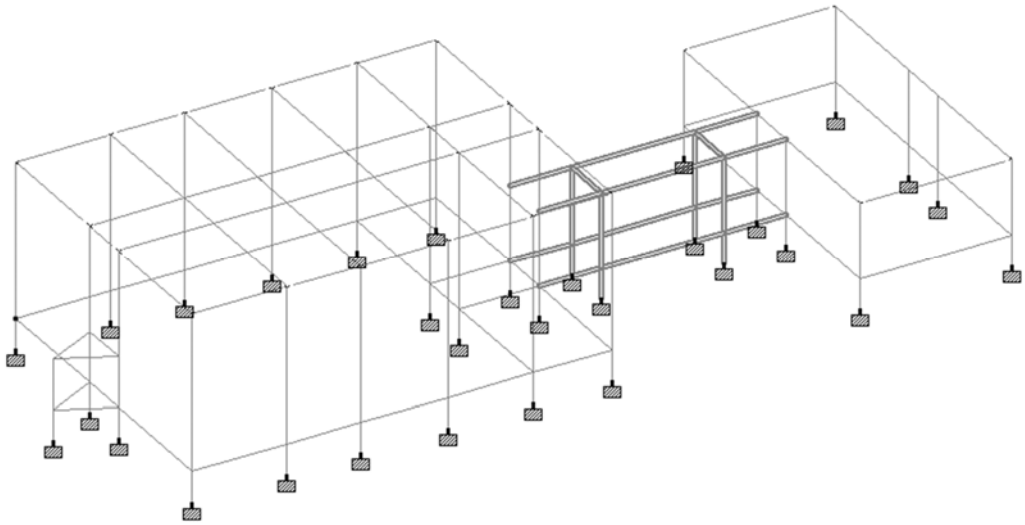
Allowable deflection = l_e/240 3.333 mm > 0.079 mm O.K

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16. STEEL WALKWAY DESIGN

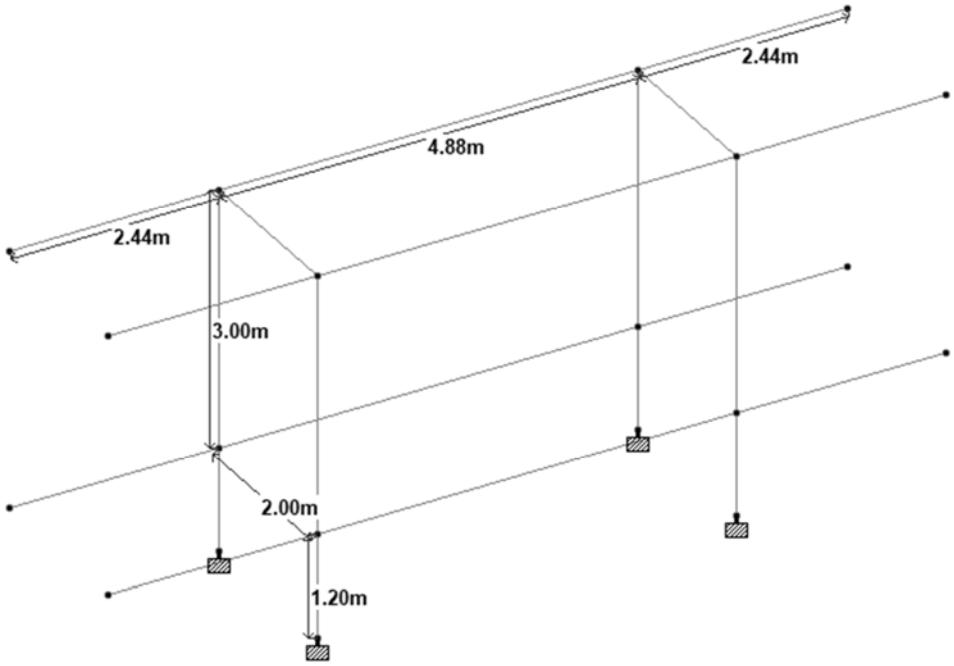


WALKWAY AREA HIGHLIGHTED

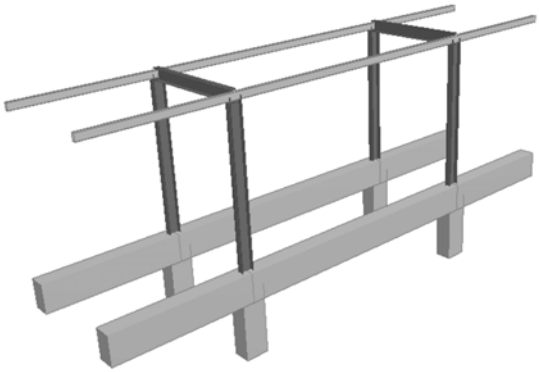
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WALKWAY DIMENSIONS

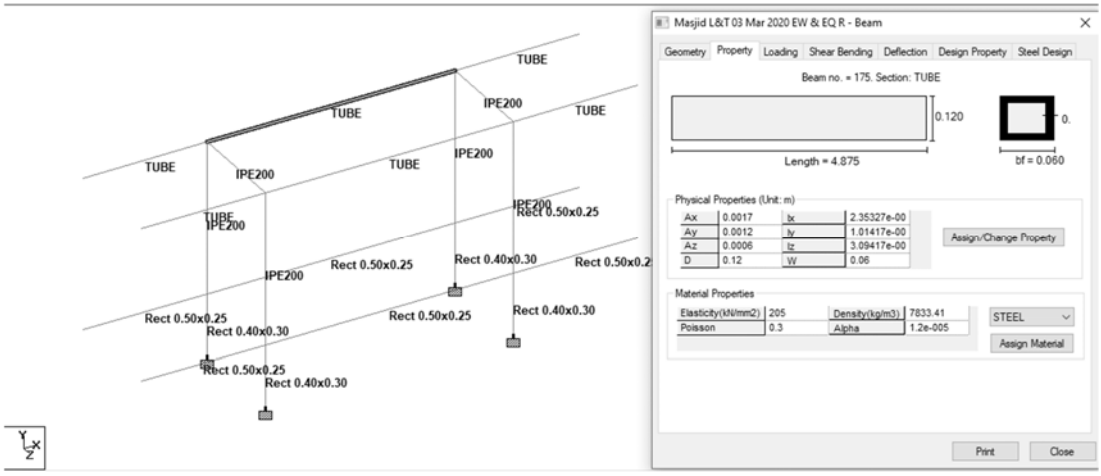


WALKWAY- 3D ISOMETRIC VIEW

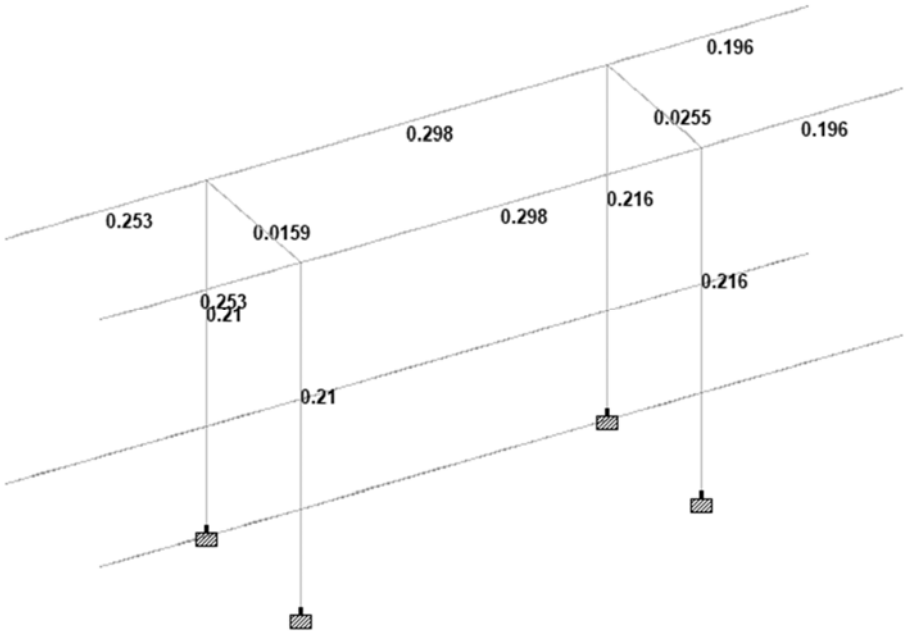
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MEMBER PROPERTIES OF WALKWAY



WALKWAY-UTILIZATION RATIO

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17. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-4000-EG-AR-GAD-3017
2.)	GENERAL PROCESS PLANT LAYOUT	MD-216-1000-EG-ME-PPD-1002
3.)	LAYOUT OF FDN, COL & GRADE BEAM	MD-216-4000-EG-CV-GAD-1067
4.)	LAYOUT OF ROOF & GRADE SLAB	MD-216-4000-EG-CV-GAD-1068
5.)	DETAILS OF FDN, COL & GRADE BEAM	MD-216-4000-EG-CV-DED-1069
6.)	DETAILS OF ROOF BEAMS & GRADE SLAB	MD-216-4000-EG-CV-DED-1070
7.)	DETAILS OF WALKWAY SHED & ROOF DOME	MD-216-4000-EG-CV-GAD-1071
8.)	DETAILS OF MINARETE & MIHRAB	MD-216-4000-EG-CV-DED-1073
9.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
10.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001
11.)	BUILDING ELEVATIONS & SECTION	MD-216-4000-EG-AR-GAD-3018

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD INPUT FOR MOSQUE & ABLUTION AREA:

STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 3.75 0 0; 3 6.65 0 0; 4 10.1 0 0; 5 13.46 0 0; 6 16.6 0 0;
7 0 0 5.025; 8 13.46 0 5.025; 9 16.6 0 5.025; 10 0 0 7.025; 11 13.46 0 7.025;
12 16.6 0 7.025; 14 0 0 12.05; 15 3.75 0 12.05; 16 6.65 0 12.05;
17 10.1 0 12.05; 18 13.46 0 12.05; 19 16.6 0 12.05; 22 -2 0 6.025; 23 0 -1.2 0;
24 3.75 -1.2 0; 25 6.65 -1.2 0; 26 10.1 -1.2 0; 27 13.46 -1.2 0;
28 16.6 -1.2 0; 29 0 -1.2 5.025; 30 13.46 -1.2 5.025; 31 16.6 -1.2 5.025;
32 0 -1.2 7.025; 33 13.46 -1.2 7.025; 34 16.6 -1.2 7.025; 35 0 -1.2 12.05;
36 3.75 -1.2 12.05; 37 6.65 -1.2 12.05; 38 10.1 -1.2 12.05;
39 13.46 -1.2 12.05; 40 16.6 -1.2 12.05; 41 -2 -1.2 6.025; 42 0 6.2 0;
43 3.75 6.2 0; 44 6.65 6.2 0; 45 10.1 6.2 0; 46 13.46 6.2 0; 47 16.6 6.2 0;
48 0 6.2 5.025; 49 13.46 6.2 5.025; 50 16.6 6.2 5.025; 51 0 6.2 7.025;
52 13.46 6.2 7.025; 53 16.6 6.2 7.025; 54 0 6.2 12.05; 55 3.75 6.2 12.05;
56 6.65 6.2 12.05; 57 10.1 6.2 12.05; 58 13.46 6.2 12.05; 59 16.6 6.2 12.05;
61 3.75 6.2 5.025; 62 3.75 6.2 7.025; 63 6.65 6.2 5.025; 64 6.65 6.2 7.025;
65 10.1 6.2 7.025; 66 10.1 6.2 5.025; 67 26.35 0 0; 68 26.35 0 5.025;
69 26.35 0 7.025; 70 26.35 0 12.05; 71 26.35 -1.2 0; 72 26.35 -1.2 5.025;
73 26.35 -1.2 7.025; 74 26.35 -1.2 12.05; 75 26.35 3 0; 76 26.35 3 5.025;
77 26.35 3 7.025; 78 26.35 3 12.05; 79 32.35 0 0; 80 32.35 0 5.025;
81 32.35 0 7.025; 82 32.35 0 12.05; 83 32.35 -1.2 0; 84 32.35 -1.2 5.025;
85 32.35 -1.2 7.025; 86 32.35 -1.2 12.05; 87 32.35 3 0; 88 32.35 3 5.025;
89 32.35 3 7.025; 90 32.35 3 12.05; 91 16.6 3 5.025; 92 16.6 3 7.025;
95 19.0375 0 7.025; 96 19.0375 0 5.025; 97 23.9125 0 7.025; 98 23.9125 0 5.025;
99 19.0375 3 5.025; 100 19.0375 3 7.025; 101 23.9125 3 5.025;
102 23.9125 3 7.025; 110 -2 2 6.025; 111 0 2 7.025; 112 0 2 5.025;
113 19.0375 -1.2 5.025; 114 23.9125 -1.2 5.025; 115 19.0375 -1.2 7.025;
116 23.9125 -1.2 7.025;
MEMBER INCIDENCES
1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 1 7; 7 5 8; 8 6 9; 10 8 9; 11 7 10;
12 8 11; 13 9 12; 15 11 12; 17 14 15; 18 15 16; 19 16 17; 20 17 18; 21 18 19;
22 14 10; 23 18 11; 24 19 12; 25 1 23; 26 2 24; 27 3 25; 28 4 26; 29 5 27;
30 6 28; 31 7 29; 32 8 30; 33 9 31; 34 10 32; 35 11 33; 36 12 34; 37 14 35;
38 15 36; 39 16 37; 40 17 38; 41 18 39; 42 19 40; 43 22 41; 44 22 7; 45 22 10;
46 1 42; 47 2 43; 48 3 44; 49 4 45; 50 5 46; 51 6 47; 52 7 112; 53 8 49;
54 9 91; 55 10 111; 56 11 52; 57 12 92; 58 14 54; 59 15 55; 60 16 56; 61 17 57;
62 18 58; 63 19 59; 64 22 110; 65 42 43; 66 43 44; 67 44 45; 68 45 46;
69 46 47; 70 42 48; 71 46 49; 72 47 50; 73 48 61; 74 49 50; 75 48 51; 76 49 52;
77 50 53; 78 51 62; 79 52 53; 80 54 55; 81 55 56; 82 56 57; 83 57 58; 84 58 59;
85 54 51; 86 58 52; 87 59 53; 90 61 63; 91 43 61; 92 62 64; 93 55 62; 94 63 66;
95 44 63; 96 64 65; 97 56 64; 98 65 52; 99 57 65; 100 66 49; 101 45 66;
102 61 62; 103 63 64; 104 66 65; 105 67 68; 106 68 69; 107 70 69; 108 67 71;
109 68 72; 110 69 73; 111 70 74; 112 67 75; 113 68 76; 114 69 77; 115 70 78;
116 75 76; 117 76 77; 118 78 77; 119 67 79; 122 70 82; 127 75 87; 130 78 90;
131 79 80; 132 80 81; 133 82 81; 134 79 83; 135 80 84; 136 81 85; 137 82 86;
138 79 87; 139 80 88; 140 81 89; 141 82 90; 142 87 88; 143 88 89; 144 90 89;
145 91 50; 146 76 101; 147 92 53; 148 92 100; 171 99 91; 172 96 99;
173 100 102; 174 95 100; 175 101 99; 176 98 101; 177 102 77; 178 97 102;
185 99 100; 186 101 102; 189 112 48; 190 111 51; 194 110 111; 196 110 112;
197 112 111; 198 9 96; 199 96 98; 200 98 68; 201 12 95; 202 95 97; 203 97 69;
204 96 113; 205 98 114; 206 95 115; 207 97 116;
START GROUP DEFINITION

```
MEMBER
_LB1 65 TO 69 80 TO 84 116 TO 118 142 TO 144
_B2 71 76 86 91 93 95 97 99 101 TO 104
_GRADEBEAMS 1 TO 8 10 TO 13 15 17 TO 24 44 45 105 TO 107 119 122 131 TO 133
_B3 70 72 75 77 85 87
_LB2 73 74 78 79 90 92 94 96 98 100
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
25 TO 29 33 36 TO 42 46 TO 51 54 57 TO 63 108 TO 115 134 TO 141 145 -
147 PRIS YD 0.5 ZD 0.25
31 32 34 35 52 53 55 56 189 190 PRIS YD 0.75 ZD 0.45
MEMBER PROPERTY EUROPEAN
172 174 176 178 185 186 TABLE ST IPE200
MEMBER PROPERTY EUROPEAN
70 72 74 75 77 79 85 87 PRIS YD 0.5 ZD 0.25
65 TO 69 80 TO 84 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY EUROPEAN
194 196 PRIS YD 0.4 ZD 0.2
MEMBER PROPERTY EUROPEAN
43 64 PRIS YD 0.2 ZD 0.2
MEMBER PROPERTY EUROPEAN
71 76 86 91 93 95 97 99 101 TO 104 PRIS YD 0.8 ZD 0.3
MEMBER PROPERTY EUROPEAN
116 TO 118 127 130 142 TO 144 PRIS YD 0.4 ZD 0.25
MEMBER PROPERTY EUROPEAN
73 78 90 92 94 96 98 100 PRIS YD 0.75 ZD 0.45
MEMBER PROPERTY EUROPEAN
10 15 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY EUROPEAN
197 PRIS YD 0.2 ZD 0.2
MEMBER PROPERTY AMERICAN
146 148 171 173 175 177 TABLE ST TUBE TH 0.005 WT 0.06 DT 0.12
MEMBER PROPERTY AMERICAN
7 12 23 105 TO 107 119 122 131 TO 133 198 TO 203 PRIS YD 0.5 ZD 0.25
204 TO 207 PRIS YD 0.4 ZD 0.3
MEMBER PROPERTY AMERICAN
1 TO 6 8 11 13 17 TO 22 24 30 44 45 PRIS YD 1.2 ZD 0.25
CONSTANTS
BETA 90 MEMB 25 30 TO 37 42 46 51 TO 58 63 109 110 113 114 135 136 139 140 -
145 147 189 190
MATERIAL CONCRETE MEMB 1 TO 8 10 TO 13 15 17 TO 87 90 TO 119 122 127 -
```

```
130 TO 145 147 189 190 194 196 TO 207
MATERIAL STEEL MEMB 146 148 171 TO 178 185 186
*********************************************************************************
SUPPORTS
23 TO 41 71 TO 74 83 TO 86 113 TO 116 FIXED
MEMBER CRACKED
25 TO 43 46 TO 53 55 56 58 TO 64 108 TO 112 115 134 TO 141 145 147 189 -
190 REDUCTION RIY 0.700000 RIZ 0.700000
1 TO 8 10 TO 13 15 17 TO 24 44 45 65 TO 87 90 TO 107 116 TO 119 122 127 130 -
131 TO 133 142 TO 144 REDUCTION RIY 0.350000 RIZ 0.350000
MEMBER OFFSET
66 TO 68 81 TO 83 127 130 START 0.25 0 0
66 TO 68 81 TO 83 127 130 END -0.25 0 0
65 69 80 84 START 0.125 0 0
65 69 80 84 END -0.125 0 0
71 85 87 91 95 101 END 0 0 0.375
116 142 START 0 0 0.125
70 72 START 0 0 0.25
87 START 0 0 -0.25
70 72 75 77 END 0 0 -0.375
75 77 START 0 0 0.375
73 78 START 0.225 0 0
74 79 END -0.225 0 0
86 93 97 99 118 144 START 0 0 -0.125
98 100 END -0.225 0 0
DEFINE IBC 2003
SDS 0.1 SD1 1.36 S1 0.02 IE 1 RX 3 RZ 3 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
65 TO 69 80 TO 84 UNI 14.5
73 74 78 79 90 92 94 96 98 100 UNI 20.55
116 TO 118 142 TO 144 UNI 19.2
1 TO 6 8 17 TO 22 24 44 45 UNI 23.2
90 92 102 103 194 196 UNI 7
105 107 119 122 131 133 UNI 18
7 12 23 UNI 14
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 40.000 M 20.000 M 2.000 0.010 0 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
0 0 0 0.878 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.67105 1.67105 1.67612 1.68107 1.68589 1.6906 1.6952 1.69969 1.70408 -
 1.70838 1.71259 1.71671 1.72074 1.7247 1.72859 HEIG  0  4.572  4.72031 -
 4.86862  5.01692  5.16523  5.31354  5.46185  5.61015  5.75846  5.90677 -
 6.05508  6.20338  6.35169  6.5
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 40.000 M 20.000 M 2.000 0.010 1 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
0 0 0 0.878 -0.300 0.550
!> END GENERATED DATA BLOCK
INT -1.1227 -1.1227 HEIG  0  6.5
TYPE 3 WIND X SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
```

```
1 6.500 M 40.000 M 20.000 M 2.000 0.010 2 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.58044 -1.58044 HEIG  0  6.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 40.000 M 2.000 0.010 0 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.64206 1.64206 1.64697 1.65176 1.65643 1.66099 1.66544 1.66979 1.67404 -
 1.6782 1.68228 1.68627 1.69017 1.69401 1.69777 HEIG  0  4.572  4.72031 -
 4.86862  5.01692  5.16523  5.31354  5.46185  5.61015  5.75846  5.90677 -
 6.05508  6.20338  6.35169  6.5
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 40.000 M 2.000 0.010 1 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
0 0 0 0.861 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.35341 -1.35341 HEIG  0  6.5
TYPE 6 WIND Z SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 40.000 M 2.000 0.010 2 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.58044 -1.58044 HEIG  0  6.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
65 TO 69 80 TO 84 UNI GY -14.5
73 74 78 79 90 92 94 96 98 100 UNI GY -20.55
116 TO 118 142 TO 144 UNI GY -19.2
1 TO 6 8 13 17 TO 22 24 44 45 UNI GY -23.2
70 75 85 116 TO 118 UMOM GZ 1
72 77 87 142 TO 144 UMOM GZ -1
80 TO 84 130 UMOM GX 1
65 TO 69 127 UMOM GX -1
90 92 102 103 194 196 UNI GY -7
105 TO 107 119 122 131 TO 133 UNI GY -18
7 12 23 UNI GY -14
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
65 TO 69 80 TO 84 UNI GY -2.5
```

73 74 78 79 90 92 94 96 98 100 UNI GY -3.5
116 TO 118 142 TO 144 UNI GY -3
65 TO 69 127 UMOM GX -0.18
80 TO 84 130 UMOM GX 0.18
70 75 85 116 TO 118 UMOM GZ 0.18
72 77 87 142 TO 144 UMOM GZ -0.18
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 6.5
WIND LOAD X 1 TYPE 1 XR 26.35 26.35 YR 0 3
WIND LOAD -X 1 TYPE 2 XR 16.6 16.6 YR 0 6.5
WIND LOAD -X 1 TYPE 2 XR 32.35 32.35 YR 0 3
WIND LOAD -Z 1 TYPE 3 XR 0 16.6 YR 0 6.5
WIND LOAD -Z -1 TYPE 3 XR 0 16.6 YR 0 6.5
WIND LOAD -Z 1 TYPE 3 XR 26.35 32.35 YR 0 6.5
WIND LOAD -Z -1 TYPE 3 XR 26.35 32.35 YR 0 6.5
WIND LOAD -Z 1 TYPE 3 XR 16.6 26.35 YR 0 6.5 OPEN
WIND LOAD -Z -1 TYPE 3 XR 16.6 26.35 YR 0 6.5 OPEN
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 XR 0 16.6 YR 0 6.5
WIND LOAD Z 1 TYPE 4 XR 26.35 32.35 YR 0 6.5
WIND LOAD Z 1 TYPE 4 XR 16.7 26.3 YR 0 6.5 OPEN
WIND LOAD -Z 1 TYPE 5 XR 0 16.6 YR 0 6.5
WIND LOAD -Z 1 TYPE 5 XR 26.35 32.35 YR 0 6.5
WIND LOAD -Z 1 TYPE 5 XR 16.7 26.3 YR 0 6.5 OPEN
WIND LOAD -X 1 TYPE 6 XR 16.6 16.6 YR 0 6.5
WIND LOAD -X 1 TYPE 6 XR 32.35 32.35 YR 0 3
WIND LOAD -X -1 TYPE 6 YR 0 6.5
WIND LOAD -X -1 TYPE 6 XR 26.35 26.35 YR 0 3
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 XR 16.6 16.6 YR 0 6.5
WIND LOAD X -1 TYPE 1 XR 32.35 32.35 YR 0 3
WIND LOAD -X -1 TYPE 2 XR 26.35 26.35 YR 0 3
WIND LOAD -X -1 TYPE 2 XR -5 45 YR 0 14
WIND LOAD -Z 1 TYPE 3 XR -5 16.6 YR 0 13
WIND LOAD -Z 1 TYPE 3 XR 26.35 32.35 YR 0 13
WIND LOAD -Z 1 TYPE 3 XR 16.7 26.3 YR 0 13 OPEN
WIND LOAD -Z -1 TYPE 3 XR -5 16.6 YR 0 13
WIND LOAD -Z -1 TYPE 3 XR 26.35 32.35 YR 0 13
WIND LOAD -Z -1 TYPE 3 XR 16.7 26.3 YR 0 13 OPEN
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 XR 0 16.6 YR 0 6.5
WIND LOAD Z -1 TYPE 4 XR 26.35 32.35 YR 0 6.5
WIND LOAD Z -1 TYPE 4 XR 16.7 26.3 YR 0 6.5 OPEN
WIND LOAD -Z -1 TYPE 5 XR 0 16.6 YR 0 6.5
WIND LOAD -Z -1 TYPE 5 XR 26.35 32.35 YR 0 6.5
WIND LOAD -Z -1 TYPE 5 XR 16.7 26.3 YR 0 6.5 OPEN
WIND LOAD -X 1 TYPE 6 XR 16.6 16.6 YR 0 6.5
WIND LOAD -X 1 TYPE 6 XR 32.35 32.35 YR 0 3
WIND LOAD -X -1 TYPE 6 YR 0 6.2
WIND LOAD -X -1 TYPE 6 XR 26.35 26.35 YR 0 3
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
46 TO 87 90 TO 104 112 TO 118 127 130 138 TO 148 171 TO 178 185 186 189 190 -
194 196 TEMP 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)

3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0

```
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
MINSEC 10 ALL
MINMAIN 20 MEMB 71 73 74 76 78 79 86 90 TO 104
MAXMAIN 20 MEMB 71 73 74 76 78 79 86 90 TO 104
MINMAIN 16 MEMB 1 TO 8 10 TO 13 15 17 TO 24 44 45 65 TO 70 72 75 77 -
80 TO 85 87 105 TO 107 116 TO 119 122 127 130 TO 133 142 TO 144
MAXMAIN 16 MEMB 1 TO 8 10 TO 13 15 17 TO 24 44 45 65 TO 70 72 75 77 -
80 TO 85 87 105 TO 107 116 TO 119 122 127 130 TO 133 142 TO 144
MINMAIN 12 MEMB 44 45 194 196
MAXMAIN 12 MEMB 44 45 194 196
DESIGN BEAM 1 TO 8 10 TO 13 15 17 TO 24 44 45 65 TO 87 90 TO 107 116 TO 119 -
122 127 130 TO 133 142 TO 144 194 196
DESIGN COLUMN 25 TO 43 46 TO 64 108 TO 115 134 TO 141 145 147 189 190 204 -
205 TO 207
SFACE 0.5 MEMB 65 TO 70 72 75 77 80 TO 85 87 116 TO 118 142 TO 144
SFACE 0.75 MEMB 71 73 74 76 78 79 86 90 TO 104
EFACE 0.5 MEMB 65 TO 69 80 TO 84 116 TO 118 142 TO 144
EFACE 0.75 MEMB 71 73 76 78 86 90 TO 104
END CONCRETE DESIGN
PARAMETER 2
CODE AISC UNIFIED 2005
CHECK CODE MEMB 146 148 171 TO 178 185 186
FINISH
```

STAAD INPUT FOR MINARET:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
141 0 0 0; 142 3.6 1.2 3.1; 143 3.6 0 3.1; 144 0 1.2 3.1; 145 0 0 3.1;
146 3.6 1.2 0; 147 3.6 0 0; 148 0 1.2 0; 149 0 5.075 0; 150 3.6 5.075 0;
151 0 5.075 3.1; 152 3.6 5.075 3.1; 153 0 9.5 0; 154 0.3 9.5 0; 155 0 9.5 3.1;
156 3.6 9.5 0; 157 3.6 9.5 3.1; 158 0.3 9.5 3.1; 159 0.3 12.5 0;
160 0.3 12.5 3.1; 161 3.3 12.5 0; 162 3.3 12.5 3.1; 163 3.3 9.5 3.1;
164 3.3 9.5 0;
MEMBER INCIDENCES
1 142 143; 2 144 145; 3 146 147; 4 148 141; 5 148 146; 6 148 144; 7 146 142;
8 144 142; 9 149 150; 10 149 151; 11 150 152; 12 151 152; 13 153 154;
14 153 155; 15 156 157; 16 155 158; 17 159 160; 18 161 162; 19 148 149;
20 144 151; 21 142 152; 22 149 153; 23 151 155; 24 152 157; 25 154 159;
26 158 160; 27 162 163; 28 161 164; 29 146 150; 30 150 156; 31 160 162;
32 154 164; 33 159 161; 34 158 163; 35 163 157; 36 164 156; 37 154 158;
38 164 163;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
END DEFINE MATERIAL
CONSTANTS
MATERIAL CONCRETE ALL
MEMBER PROPERTY EUROPEAN
5 TO 12 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY EUROPEAN
25 TO 28 PRIS YD 0.4 ZD 0.25
MEMBER PROPERTY EUROPEAN
1 TO 4 19 TO 24 29 30 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY EUROPEAN
37 38 PRIS YD 0.1 ZD 0.1
MEMBER PROPERTY EUROPEAN
13 TO 16 32 34 TO 36 PRIS YD 0.5 ZD 0.5
MEMBER PROPERTY
17 18 PRIS YD 0.35 ZD 0.2
31 33 PRIS YD 0.35 ZD 0.25
*********************************************************************************
SUPPORTS
```

```
141 143 145 147 FIXED
MEMBER CRACKED
1 TO 5 19 TO 30 REDUCTION RIY 0.700000 RIZ 0.700000
5 TO 16 32 34 TO 36 REDUCTION RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.1 SD1 1.36 S1 0.02 IE 1 RX 3 RZ 3 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
5 TO 12 14 15 32 34 UNI 10
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 13.000 M 40.000 M 15.000 M 2.000 0.010 0 -
0 0 1 0 1.236 1.000 1.000 0.850 0 -
0 0 0 0.876 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.76679 1.76679 1.78802 1.80719 1.82467 1.84077 1.85571 1.86966 1.88275 -
 1.8951 1.90678 1.91788 1.92845 1.93855 1.94822 HEIG  0  4.572  5.22031 -
 5.86862  6.51692  7.16523  7.81354  8.46185  9.11015  9.75846  10.4068 -
 11.0551  11.7034  12.3517  13
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 13.000 M 40.000 M 15.000 M 2.000 0.010 1 -
0 0 1 0 1.236 1.000 1.000 0.850 0 -
0 0 0 0.876 -0.200 0.550
!> END GENERATED DATA BLOCK
INT -1.12937 -1.12937 HEIG  0  13
TYPE 3 WIND X SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 13.000 M 40.000 M 15.000 M 2.000 0.010 2 -
0 0 1 0 1.236 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.78292 -1.78292 HEIG  0  13
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 13.000 M 15.000 M 40.000 M 2.000 0.010 0 -
0 0 1 0 1.236 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.73932 1.73932 1.75991 1.7785 1.79545 1.81107 1.82556 1.83909 1.85178 -
 1.86376 1.87509 1.88585 1.8961 1.9059 1.91528 HEIG  0  4.572  5.22031  5.86862 -
 6.51692  7.16523  7.81354  8.46185  9.11015  9.75846  10.4068  11.0551 -
 11.7034  12.3517  13
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 13.000 M 15.000 M 40.000 M 2.000 0.010 1 -
0 0 1 0 1.236 1.000 1.000 0.850 0 -
0 0 0 0.856 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.52324 -1.52324 HEIG  0  13
TYPE 6 WIND Z SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
```

1 13.000 M 15.000 M 35.000 M 2.000 0.010 2 -
0 0 1 0 1.236 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.78292 -1.78292 HEIG 0 13
LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead TITLE DL
MEMBER LOAD
5 TO 12 14 15 32 34 UNI GY -10
SELFWEIGHT Y -1
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 XR 0 3.6 YR 0 13 ZR 0 3.1
WIND LOAD -X 1 TYPE 2 XR 3.3 3.6 YR 0 14 ZR 0 3.1
WIND LOAD -Z 1 TYPE 3 YR 0 14
WIND LOAD -Z -1 TYPE 3 YR 0 14
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 13
WIND LOAD -Z 1 TYPE 5 YR 0 13
WIND LOAD -X 1 TYPE 6 YR 0 13
WIND LOAD -X -1 TYPE 6 YR 0 13
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 13
WIND LOAD -X -1 TYPE 2 YR 0 13
WIND LOAD -Z 1 TYPE 3 YR 0 13
WIND LOAD -Z -1 TYPE 3 YR 0 13
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 13
WIND LOAD -Z -1 TYPE 5 YR 0 13
WIND LOAD -X 1 TYPE 6 YR 0 13
WIND LOAD -X -1 TYPE 6 YR 0 13
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
9 TO 38 TEMP 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)

3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
MINMAIN 16 MEMB 25 TO 28
MAXMAIN 16 MEMB 25 TO 28
MINMAIN 20 MEMB 1 2 4 19 TO 24 29 30
MAXMAIN 20 MEMB 1 2 4 19 TO 24 29 30
DESIGN BEAM 5 TO 18 31 TO 38
DESIGN COLUMN 1 TO 4 19 TO 30
END CONCRETE DESIGN
PARAMETER 2
CODE AISC UNIFIED 2005
FINISH

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

PROPRIETARY & CONFIDENTIAL

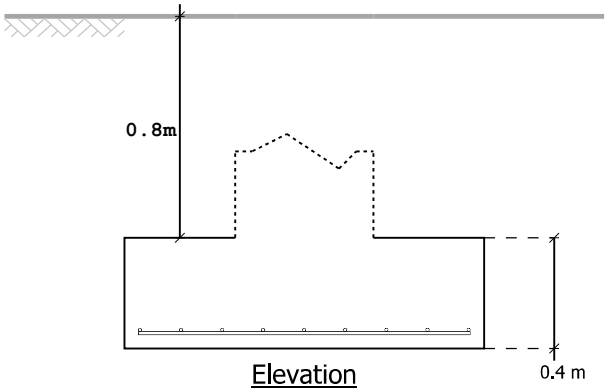
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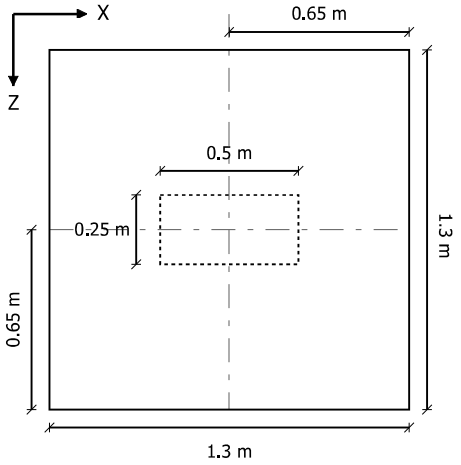
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
37	1	1.300m	1.300m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
37	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.300m

Initial Width (W_o) = 1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	210.879	-2.312	6.430	13.751	-2.940
12	222.152	-1.956	6.992	14.965	-3.372
13	219.333	-2.045	6.851	14.661	-3.264
14	212.318	-2.254	8.725	19.733	-3.051
15	214.414	2.459	6.551	14.013	-10.595
16	222.440	4.835	17.980	24.708	-12.361
17	220.944	-2.727	18.652	30.774	-2.648
18	130.670	7.785	18.696	21.647	-13.894
19	128.675	-2.297	19.592	29.735	-0.943
20	130.062	3.384	3.979	8.512	-9.419
21	127.966	-1.329	6.154	14.232	-1.875

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	215.021	6.861	21.268	27.147	-15.070
23	213.027	-3.222	22.164	35.235	-2.120
39	215.815	-8.635	17.803	24.321	5.517
40	217.457	-2.813	-6.944	-3.203	-2.470
41	121.836	-10.174	18.459	21.131	9.943
42	124.026	-2.412	-14.536	-15.568	-0.706
45	206.188	-11.099	21.031	26.631	8.767
46	208.378	-3.336	-11.964	-10.068	-1.882

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	295.231	-3.236	9.001	19.251	-4.116
25	300.867	-3.058	9.283	19.858	-4.332
26	300.867	-3.058	9.283	19.858	-4.332
27	274.404	5.134	20.486	29.160	-13.922
28	262.127	-4.052	33.172	51.482	-2.431
29	265.319	12.080	31.737	38.541	-23.151
30	262.127	-4.052	33.172	51.482	-2.431
31	258.104	4.042	7.889	16.875	-14.464
32	255.111	-2.692	10.995	25.046	-3.686
33	196.419	12.595	29.527	33.809	-22.054
34	193.227	-3.537	30.962	46.750	-1.333
35	191.847	-1.998	9.066	20.921	-2.804
36	194.841	4.735	5.960	12.750	-13.582
47	267.338	-9.234	20.297	28.747	5.147
48	254.689	-4.235	-21.434	-21.002	-2.050
49	251.185	-16.656	31.359	37.716	14.988
50	254.689	-4.235	-21.434	-21.002	-2.050
53	182.285	-16.140	29.149	32.984	16.086
54	185.789	-3.720	-23.644	-25.734	-0.953

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.690m^2

Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 1.347m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 11

Width (W_2) = 1.300 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 30

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

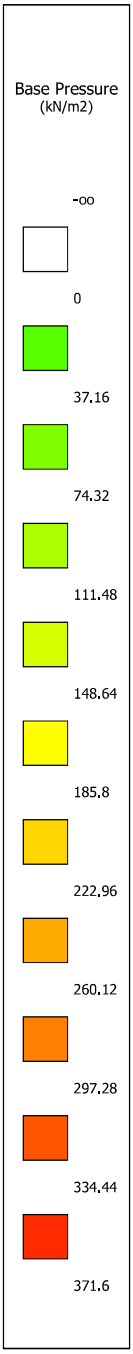
Gross Soil Bearing Capacity = 371.60 kN/m2

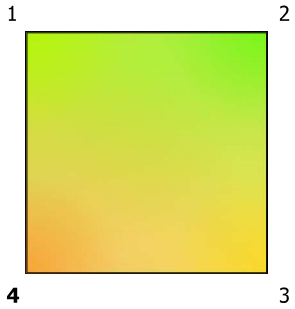
Soil Weight On Top Of Footing = 22.535 kN

Pressures at 4 Corners

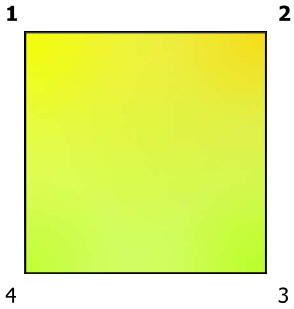
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

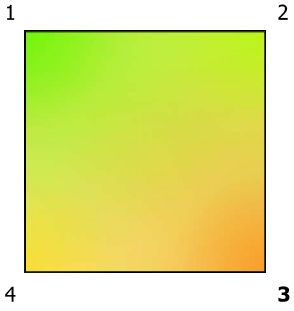




Corner 4

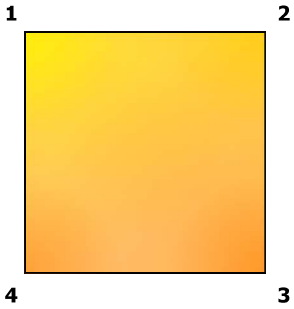


Corner 1, 2



Corner 3

Governing Base Pressure Contour



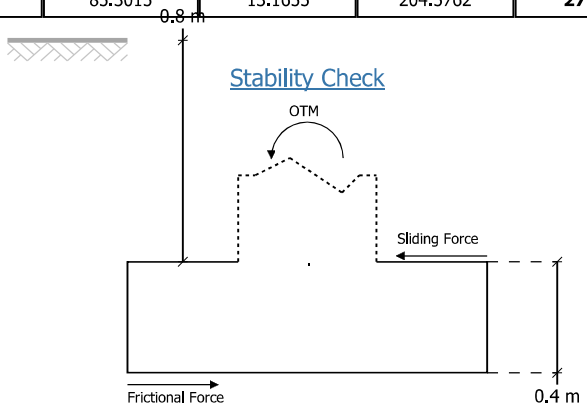
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**185.3049**	188.2938	107.1645	104.1756	0.000
46	185.3049	**188.2938**	107.1645	104.1756	0.000
22	4.1425	101.4444	**296.1888**	198.8870	0.000
45	85.3015	13.1655	204.5762	**276.7121**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**185.3049**	188.2938	107.1645	104.1756
46	185.3049	**188.2938**	107.1645	104.1756
22	4.1425	101.4444	**296.1888**	198.8870
45	85.3015	13.1655	204.5762	**276.7121**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	53.993	19.413	25.835	1.500	9.995	80.501	1.500
12	66.710	18.658	25.411	1.500	9.598	65.493	1.500
13	63.115	18.835	25.525	1.500	9.691	68.585	1.500
14	55.693	14.388	19.701	1.500	7.065	75.932	1.500
15	51.473	19.324	25.585	1.500	9.946	14.212	1.500
16	27.013	7.264	9.920	1.500	5.350	11.877	1.500
17	47.611	6.962	9.742	1.500	4.438	108.397	1.500
18	10.881	4.531	5.916	1.500	3.811	6.475	1.500
19	36.441	4.273	6.002	1.500	2.920	4449.971	1.500
20	24.944	21.214	22.854	1.500	10.947	10.186	1.500
21	62.707	13.547	18.726	1.500	6.544	80.692	1.500
22	18.496	5.966	8.030	1.500	4.651	9.260	1.500
23	39.072	5.680	7.949	1.500	3.731	197.007	1.500
39	14.741	7.150	9.098	1.500	5.291	18.445	1.500
40	45.541	18.448	24.181	1.500	27.994	123.853	1.500
41	7.892	4.350	5.388	1.500	3.692	7.449	1.500
42	33.749	5.599	7.812	1.500	4.989	408.431	1.500
45	11.035	5.823	7.283	1.500	4.568	12.055	1.500
46	37.037	10.328	14.069	1.500	10.874	293.560	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41

Governing Disturbing Force : -10.174kN

Governing Restoring Force : 80.297kN

Minimum Sliding Ratio for the Critical Load Case : 7.892

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 37.571kNm

Governing Resisting Moment : 109.707kNm

Minimum Overturning Ratio for the Critical Load Case : 2.920

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 19.592kN

Governing Restoring Force : 83.717kN

Minimum Sliding Ratio for the Critical Load Case : 4.273

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -17.008kNm

Governing Resisting Moment : 110.126kNm

Minimum Overturning Ratio for the Critical Load Case : 6.475

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 41

Governing Disturbing Force : 21.078kN

Governing Restoring Force : 113.558kN

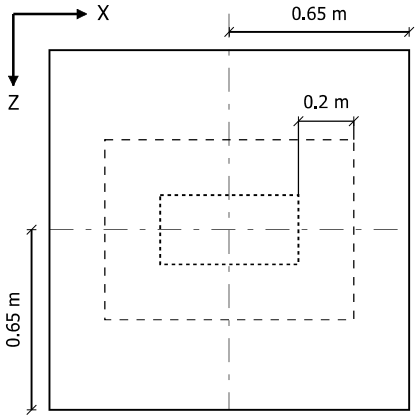
Minimum Sliding Ratio for the Critical Load Case : 5.388

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 238.876kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1944.021kN

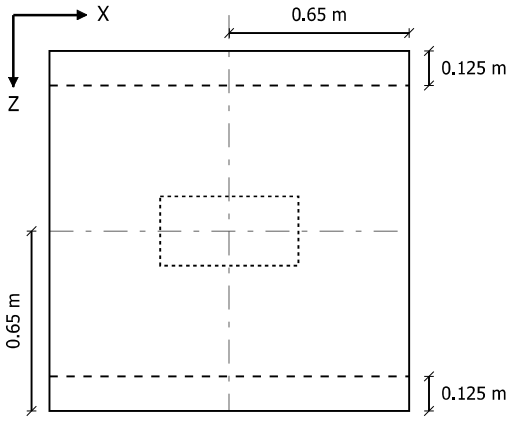
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.181m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

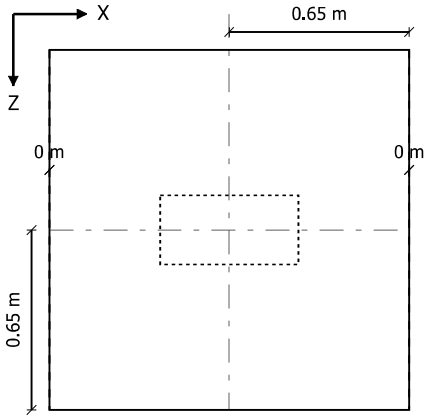
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 77.709 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.056 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

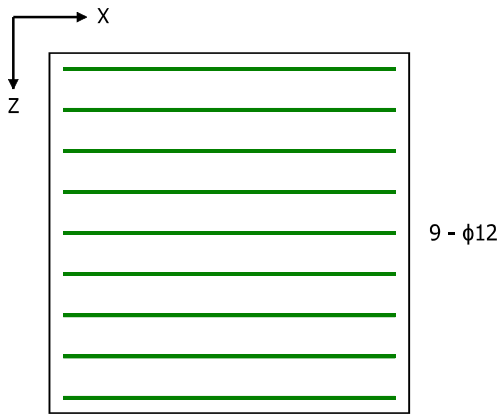
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z} =$ 18.114 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00172
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.900 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	25.300 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	28.111 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00044
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00038
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	896.155 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.350	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	9	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.869	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00228	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

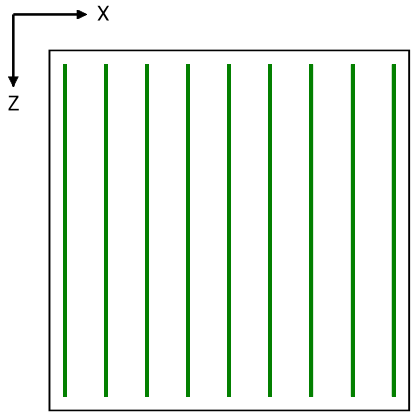
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.136

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 30

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00176

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.775 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 55.070 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 61.189 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00102

(Based on gross depth) $\rho \times d_{eff}$ / Depth = 0.00085

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 914.436 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.475 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

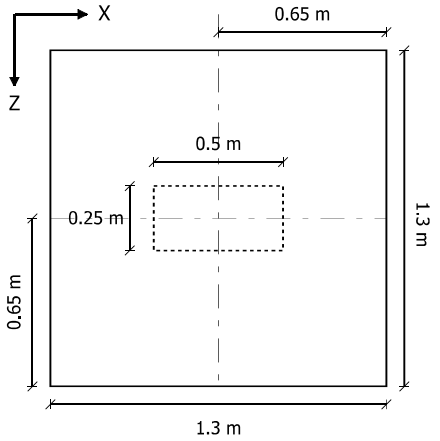
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

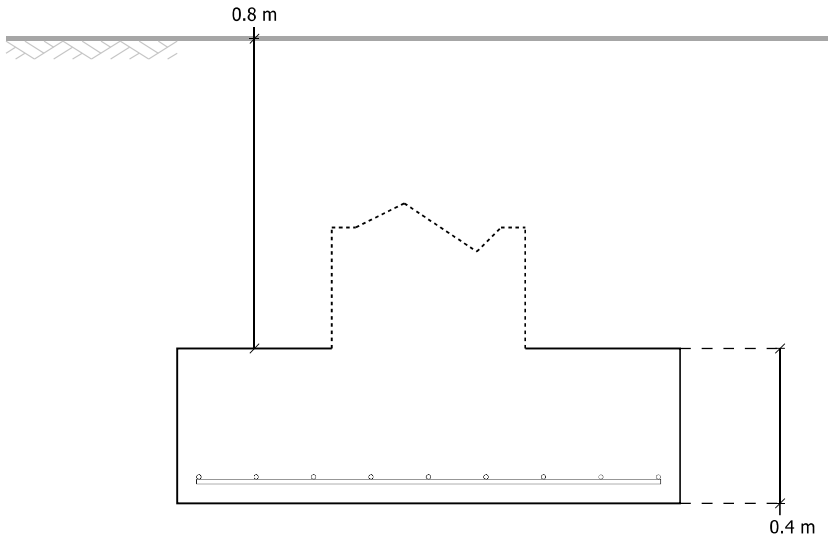
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.113

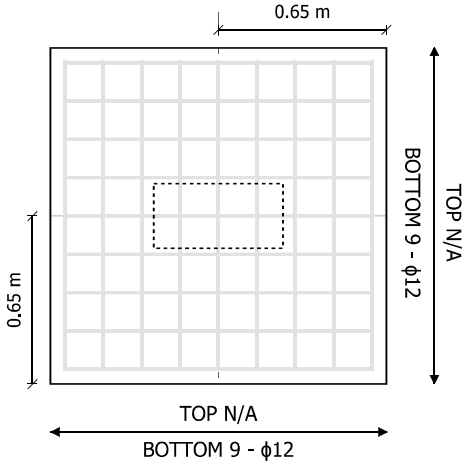
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

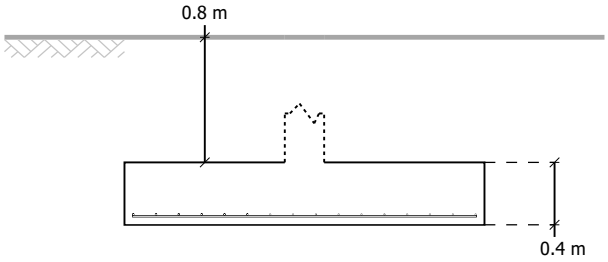
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

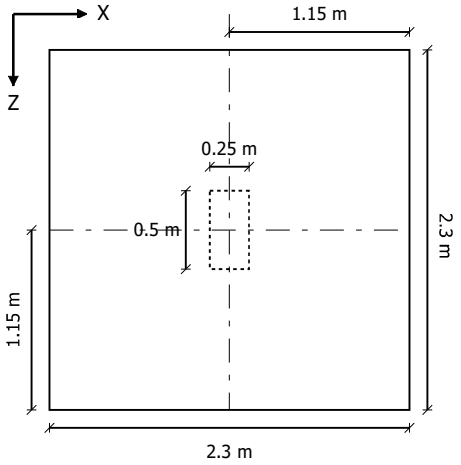
FOUNDATION-F2-GRID-2 & F

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
31	1	2.300m	2.300m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
31	16 - ϕ12	16 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2300.000mm
Footing Width - Z (Fw) : 2300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.300m

Initial Width (W_o) = 2.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	96.438	26.526	33.961	11.428	-16.537
12	94.470	28.856	34.012	11.445	-18.024
13	94.962	28.274	33.999	11.441	-17.652
14	81.435	26.450	38.593	18.596	-16.478
15	105.891	33.427	33.974	11.452	-21.507
16	102.255	33.742	33.752	11.275	-22.474
17	72.714	29.765	42.674	23.728	-18.527
18	67.586	23.207	20.047	6.636	-16.352
19	28.199	17.904	31.942	23.239	-11.089
20	67.316	22.816	20.390	6.881	-14.893
21	42.860	15.839	25.009	14.025	-9.863

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	106.162	33.818	33.631	11.207	-22.966
23	66.775	28.514	45.527	27.811	-17.703
39	87.585	22.594	33.744	11.258	-12.703
40	116.058	29.459	25.367	-0.825	-18.476
41	48.026	8.344	20.037	6.614	-3.323
42	85.991	17.497	8.867	-9.498	-11.020
45	86.602	18.954	33.621	11.185	-9.938
46	124.567	28.107	22.451	-4.927	-17.635

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	135.014	37.137	47.546	15.999	-23.152
25	134.029	38.302	47.571	16.008	-23.895
26	134.029	38.302	47.571	16.008	-23.895
27	120.356	41.393	40.571	13.564	-27.368
28	67.280	36.178	59.284	39.934	-22.454
29	130.299	44.663	40.251	13.369	-30.875
30	67.280	36.178	59.284	39.934	-22.454
31	129.230	41.689	40.772	13.748	-26.945
32	94.293	31.722	47.371	23.954	-19.760
33	102.352	35.540	30.037	9.932	-25.170
34	39.333	27.055	49.070	36.497	-16.749
35	65.362	23.764	37.182	20.526	-14.799
36	100.298	33.732	30.584	10.320	-21.984
47	104.708	29.502	40.563	13.546	-16.945
48	159.747	35.526	22.363	-12.446	-22.345
49	99.003	20.881	40.235	13.333	-10.029
50	159.747	35.526	22.363	-12.446	-22.345
53	71.056	11.758	30.021	9.896	-4.324
54	131.800	26.403	12.150	-15.883	-16.640

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 5.290m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.052m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.300 m Governing Load Case : # 11

Width (W_2) = 2.300 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 5.290 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 50.782 kN

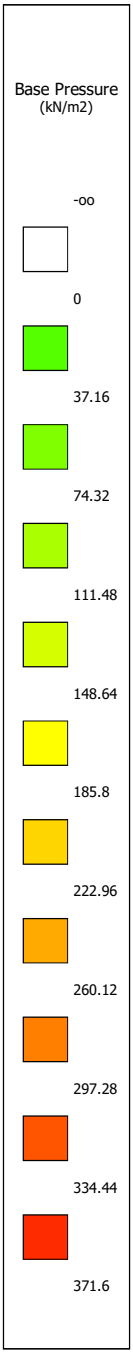
Gross Soil Bearing Capacity = 371.60 kN/m2

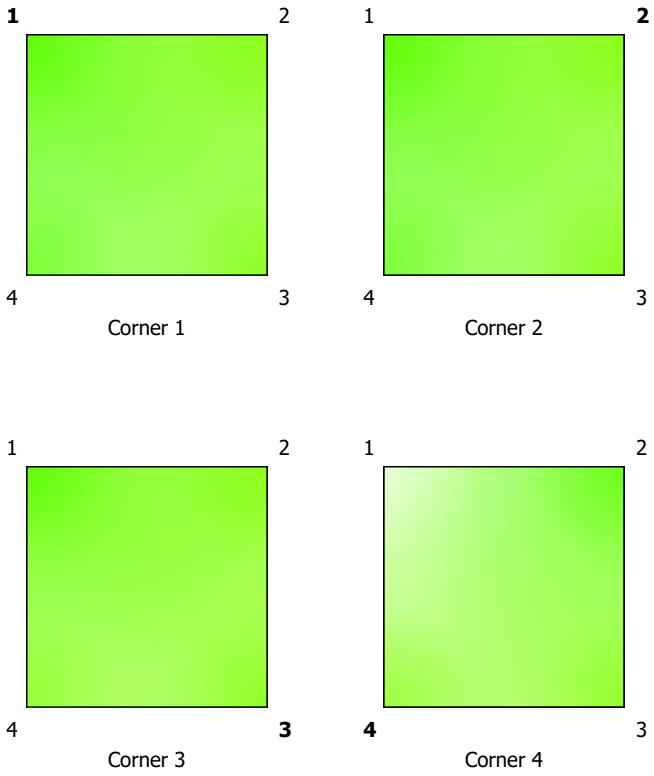
Soil Weight On Top Of Footing = 74.373 kN

Pressures at 4 Corners

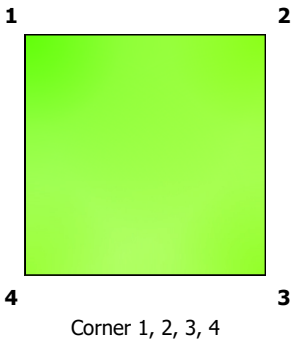
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner





Governing Base Pressure Contour



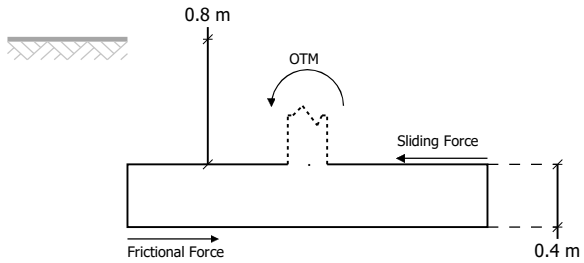
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
42	**33.9634**	51.7351	45.8652	28.0936	0.000
46	30.9667	**59.4484**	63.4462	34.9645	0.000
22	13.5701	49.5631	**73.8844**	37.8914	0.000
23	-0.7682	27.9416	73.3316	**44.6218**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
42	**33.9634**	51.7351	45.8652	28.0936
46	30.9667	**59.4484**	63.4462	34.9645
22	13.5701	49.5631	**73.8844**	37.8914
23	0.0000	27.9416	73.3318	**44.6219**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	4.177	3.262	3.636	1.500	10.188	9.444	1.500
12	3.806	3.229	3.482	1.500	10.083	8.595	1.500
13	3.893	3.237	3.520	1.500	10.109	8.794	1.500
14	3.905	2.677	3.122	1.500	6.981	8.838	1.500
15	3.456	3.400	3.428	1.500	10.610	7.662	1.500
16	3.370	3.369	3.369	1.500	10.555	7.313	1.500
17	3.324	2.318	2.689	1.500	5.578	7.528	1.500
18	4.153	4.807	4.444	1.500	15.125	8.707	1.500
19	4.283	2.401	2.961	1.500	4.897	9.748	1.500
20	4.218	4.720	4.448	1.500	14.719	9.280	1.500
21	5.304	3.359	4.013	1.500	8.041	12.024	1.500
22	3.420	3.439	3.429	1.500	10.787	7.332	1.500
23	3.365	2.108	2.526	1.500	4.796	7.636	1.500
39	4.708	3.152	3.704	1.500	9.882	11.324	1.500
40	4.094	4.754	4.387	1.500	29.759	9.218	1.500
41	10.378	4.322	5.642	1.500	13.614	30.135	1.500
42	6.034	11.906	7.612	1.500	40.799	13.562	1.500
45	5.586	3.149	3.880	1.500	9.886	13.989	1.500
46	4.442	5.561	4.909	1.500	70.848	9.998	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 17

Governing Disturbing Force : 29.765kN

Governing Restoring Force : 98.935kN

Minimum Sliding Ratio for the Critical Load Case : 3.324

Critical Load Case for Overturning about X-Direction : 23

Governing Overturning Moment : 46.021kNm

Governing Resisting Moment : 220.716kNm

Minimum Overturning Ratio for the Critical Load Case : 4.796

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 23

Governing Disturbing Force : 45.527kN

Governing Restoring Force : 95.965kN

Minimum Sliding Ratio for the Critical Load Case : 2.108

Critical Load Case for Overturning about Z-Direction : 16

Governing Overturning Moment : -35.971kNm

Governing Resisting Moment : 263.069kNm

Minimum Overturning Ratio for the Critical Load Case : 7.313

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 23

Governing Disturbing Force : 53.719kN

Governing Restoring Force : 135.715kN

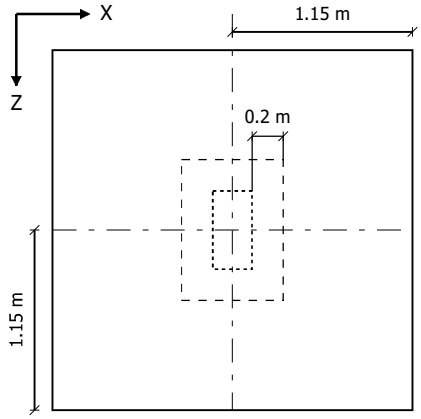
Minimum Sliding Ratio for the Critical Load Case : 2.526

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 257.902kN, Load Case # 50

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

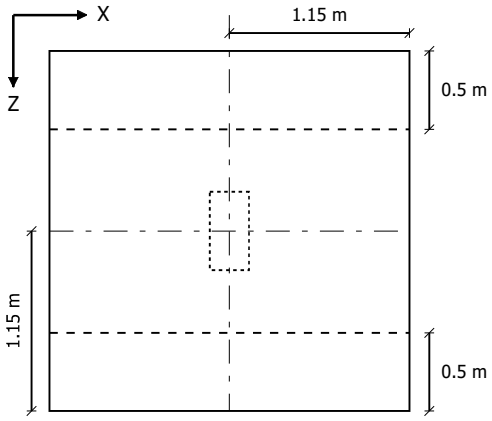
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 777.338kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.556m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

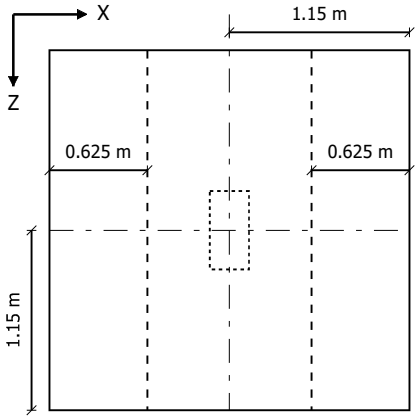
From above calculations, 0.75 X V_c = 583.003 kN

Critical load case for V_{ux} is # 24 $V_{ux} = V_{ux}|_{x=D_x}$ = 79.638 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 777.338 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.681 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

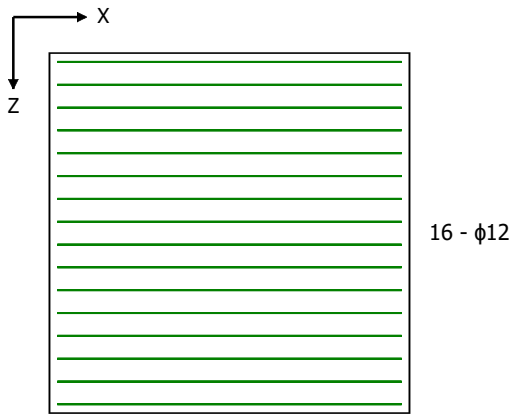
From above calculations, 0.75 X V_c = 583.003 kN

Critical load case for V_{uz} is # 50 $V_{uz} = V_{uz}|_{z=D_z}$ = 104.231 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 50

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00157
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.275 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	80.389 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	89.321 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00079
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00068
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1448.606 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.975	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	16	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1809.545	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00229	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

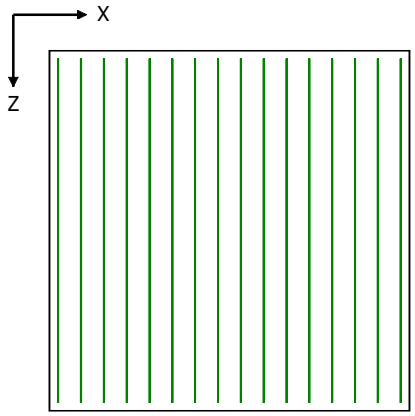
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.249

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



16 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 24

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00153

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.400 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 57.713 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 64.126 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00060

(Based on gross depth) ρ x d_{eff} / Depth = 0.00050

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1405.041 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.850 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 16

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1809.545 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

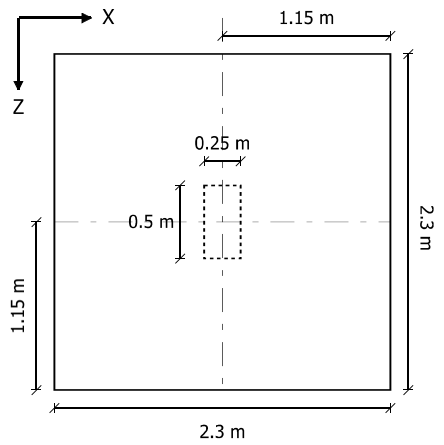
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

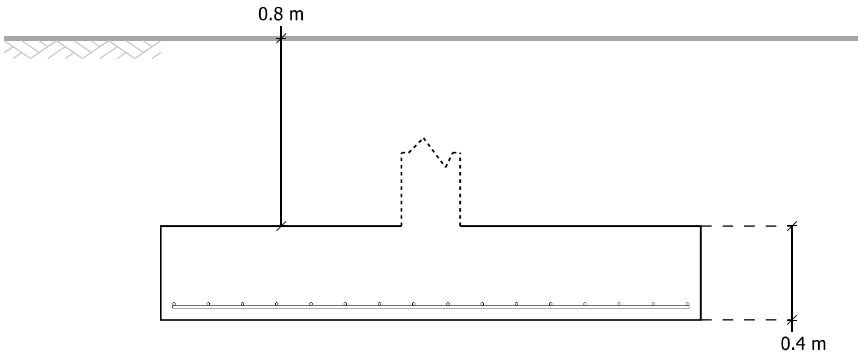
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.288

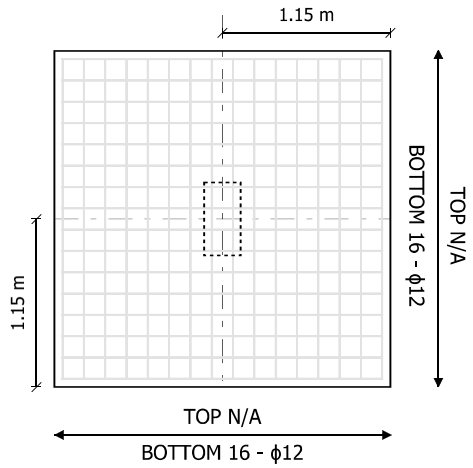
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

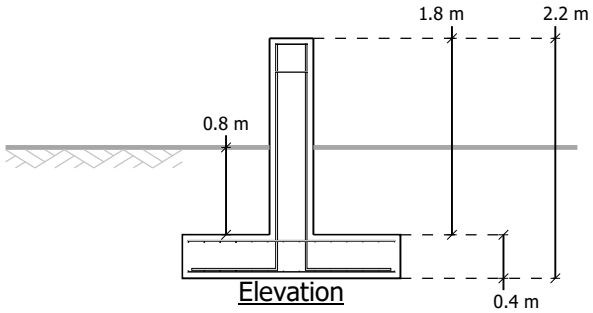
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.3 m	2.3 m	0.4 m	N/A	N/A	16 - ϕ12	16 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

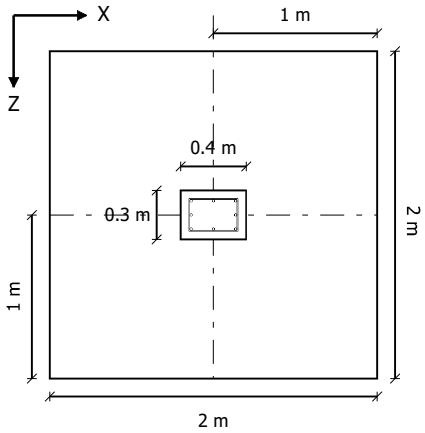
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
96	1	2.000m	2.000m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
96	14 - ϕ12	14 - ϕ12	14 - ϕ12	14 - ϕ12	8 - #16	#6 @ 115 mm

Isolated Footing 96



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2000.000mm
Footing Width - Z (Fw) : 2000.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.240m

Column Width - Z (B_{col}) : 0.120m

Pedestal

Include Pedestal? Yes
Pedestal Shape : Rectangular
Pedestal Height (Ph) : 1.800m
Pedestal Length - X (Pl) : 0.400m
Pedestal Width - Z (Pw) : 0.300m

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Always calculate based on self weight
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.000m

Initial Width (W_o) = 2.000m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	6.221	3.304	-0.002	-0.001	-5.186
12	6.549	3.615	-0.002	-0.001	-5.672
13	6.467	3.537	-0.002	-0.001	-5.551
14	5.909	3.267	0.172	0.259	-5.136
15	9.876	6.567	-0.001	0.000	-10.292
16	10.208	8.249	-4.337	-0.002	-13.104
17	3.114	3.018	6.477	2.994	-4.740
18	8.720	8.265	-5.781	-0.003	-13.183
19	-0.738	1.291	8.637	3.993	-2.030
20	7.387	5.245	-0.001	0.001	-8.217
21	3.421	1.945	0.173	0.259	-3.062

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	11.209	9.587	-5.782	-0.003	-15.257
23	1.750	2.612	8.636	3.992	-4.104
39	2.919	-1.003	-4.338	-0.003	1.738
40	9.147	3.426	-5.690	-2.996	-5.396
41	-0.999	-4.071	-5.782	-0.004	6.607
42	7.306	1.833	-7.585	-3.994	-2.905
45	1.490	-2.750	-5.783	-0.004	4.532
46	9.795	3.155	-7.586	-3.995	-4.980

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	8.710	4.626	-0.003	-0.001	-7.260
25	8.874	4.781	-0.003	-0.001	-7.504
26	8.874	4.781	-0.003	-0.001	-7.504
27	11.980	9.489	-4.626	-0.003	-15.058
28	0.476	3.014	13.819	6.388	-4.736
29	15.610	14.173	-9.250	-0.004	-22.580
30	0.476	3.014	13.819	6.388	-4.736
31	12.687	8.626	-0.001	0.001	-13.517
32	7.020	3.911	0.247	0.370	-6.152
33	13.579	13.026	-9.250	-0.004	-20.781
34	-1.555	1.867	13.820	6.389	-2.937
35	5.153	2.920	0.248	0.370	-4.596
36	10.820	7.635	-0.001	0.001	-11.961
47	4.205	-0.381	-4.627	-0.003	0.773
48	13.347	3.882	-12.136	-6.391	-6.137
49	0.059	-5.566	-9.251	-0.006	9.083
50	13.347	3.882	-12.136	-6.391	-6.137
53	-1.971	-6.712	-9.251	-0.006	10.882
54	11.316	2.735	-12.135	-6.391	-4.338

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.000m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.694m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.000 m Governing Load Case : # 11

Width (W_2) = 2.000 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 48

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.000 m^2

Final Pedestal Height = 1.800 m

Final Soil Height = 0.800 m

Footing Self Weight = 43.582 kN

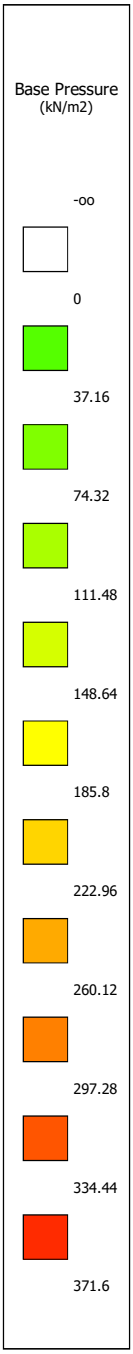
Gross Soil Bearing Capacity = 371.60 kN/m2

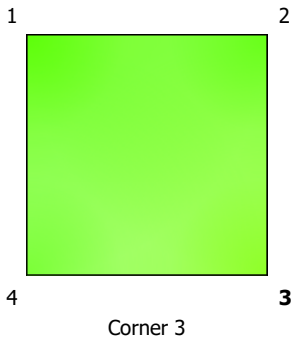
Soil Weight On Top Of Footing = 55.870 kN

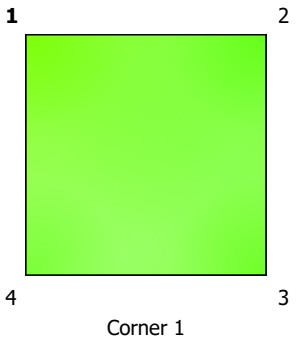
Pressures at 4 Corners

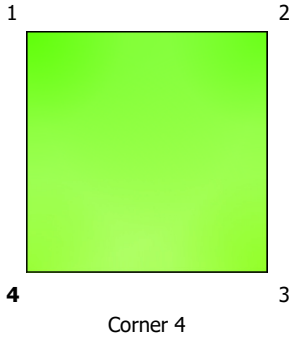
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

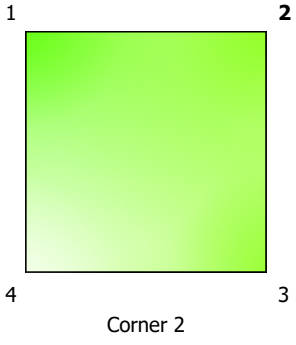
Governing Base Pressure Contour for Each Corner



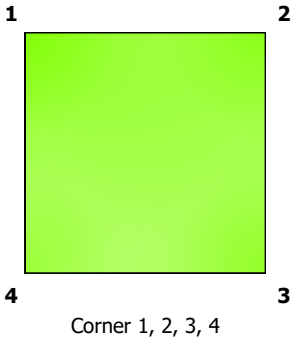








Governing Base Pressure Contour



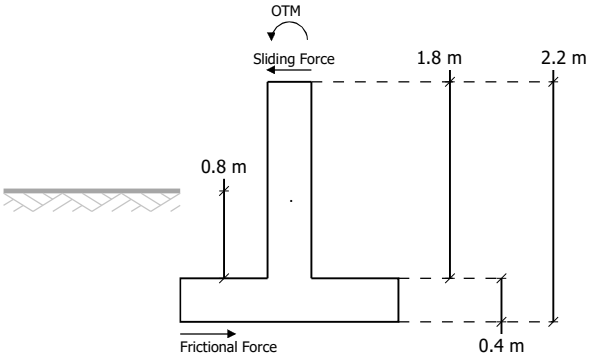
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
41	**45.8291**	22.4837	3.3978	26.7432	0.000
22	9.9466	**64.4691**	45.3841	-9.1383	0.198
23	0.6674	15.4448	**49.9338**	35.1565	0.000
19	3.7800	11.0845	45.5770	**38.2725**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
41	**45.8291**	22.4837	3.3978	26.7432
22	9.3626	**64.8722**	45.0488	0.0000
23	0.6674	15.4448	**49.9338**	35.1565
19	3.7800	11.0845	45.5770	**38.2725**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	15.992	24348.796	22.616	1.500	18501.456	8.531	1.500
12	14.661	25891.941	20.734	1.500	20153.060	7.822	1.500
13	14.972	25485.961	21.173	1.500	19710.295	7.987	1.500
14	16.127	305.421	22.775	1.500	166.756	8.597	1.500
15	8.324	39640.495	11.772	1.500	38975.075	4.443	1.500
16	6.647	12.643	8.320	1.500	11.604	3.527	1.500
17	16.990	7.918	10.149	1.500	6.011	9.063	1.500
18	6.544	9.356	7.584	1.500	8.588	3.467	1.500
19	38.241	5.714	7.993	1.500	4.340	20.390	1.500
20	10.184	104540.092	14.403	1.500	205487.802	5.437	1.500
21	26.446	296.714	37.252	1.500	162.271	14.092	1.500
22	5.772	9.570	6.989	1.500	8.782	3.060	1.500
23	19.370	5.859	7.931	1.500	4.448	10.331	1.500
39	51.032	11.801	16.260	1.500	10.837	26.099	1.500
40	15.851	9.544	11.563	1.500	7.070	8.442	1.500
41	12.091	8.514	9.845	1.500	7.823	6.363	1.500
42	29.114	7.038	9.674	1.500	5.214	15.468	1.500
45	18.355	8.728	11.147	1.500	8.017	9.594	1.500
46	17.313	7.201	9.403	1.500	5.334	9.212	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 22

Governing Disturbing Force : 9.587kN

Governing Restoring Force : 55.331kN

Minimum Sliding Ratio for the Critical Load Case : 5.772

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 22.995kNm

Governing Resisting Moment : 99.792kNm

Minimum Overturning Ratio for the Critical Load Case : 4.340

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 8.637kN

Governing Restoring Force : 49.357kN

Minimum Sliding Ratio for the Critical Load Case : 5.714

Critical Load Case for Overturning about Z-Direction : 22

Governing Overturning Moment : -36.348kNm

Governing Resisting Moment : 111.235kNm

Minimum Overturning Ratio for the Critical Load Case : 3.060

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 22

Governing Disturbing Force : 11.195kN

Governing Restoring Force : 78.249kN

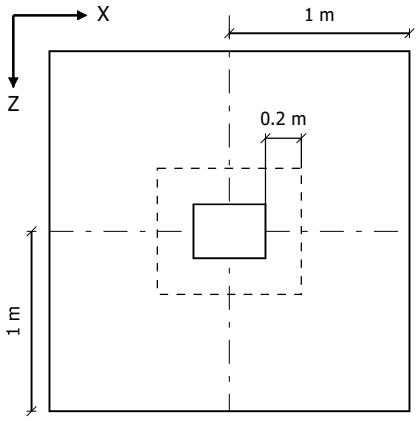
Minimum Sliding Ratio for the Critical Load Case : 6.989

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.333

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 128.649kN, Load Case # 29

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.776m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2345.533kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3263.468kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1876.426kN

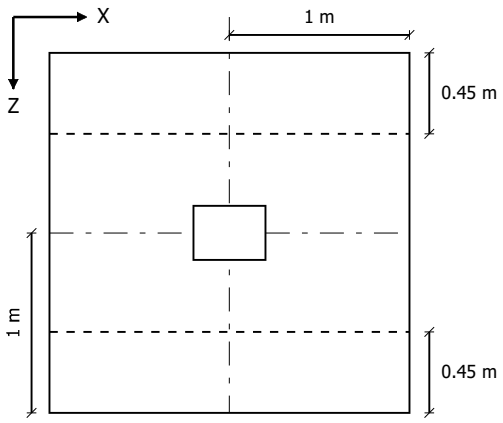
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1407.320kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 675.946kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.506m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

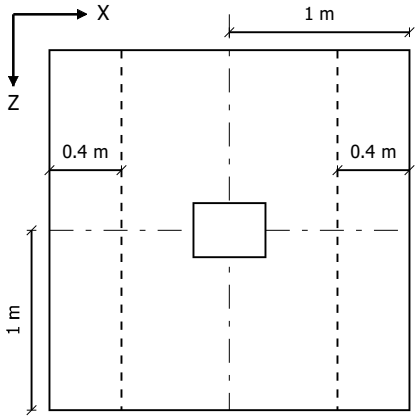
From above calculations, 0.75 X V_c = 506.960 kN

Critical load case for V_{ux} is # 48 $V_{ux} = V_{ux}|_{x=D_x}$ = 47.299 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 675.946 kN

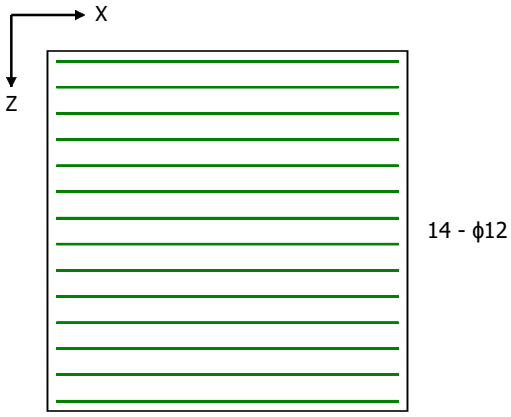
Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.456 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

From above calculations,	0.75 X V_c =	506.960	kN
Critical load case for V_{uz} is # 29	$V_{uz} = V_{uz}\|_{z=D_z}$ =	49.412	kN
	0.75 X V_c > V_{uz} hence, OK		

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00160
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200	m
Ultimate moment,	$M_u\|_{z=D_z}$ =	42.739	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	47.488	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00048	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00041	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1279.167	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.750	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	14	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1583.352	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00230	

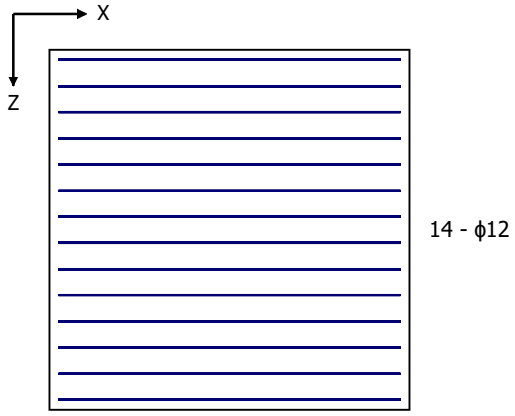
From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.238

Check to see if width is sufficient to accomodate bars

Design For Top Reinforcement Parallel to X Axis



Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_z. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00000	

From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)} =$	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd} =$	1.200	m
Ultimate moment,	$M_u\big\vert_{x=D_x} =$	15.359	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi} =$	17.066	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right] =$	0.000172	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.000148	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff} =$	118.264	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1583.363	mm2

Provided Steel Area / Required Steel Area = 13.388

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

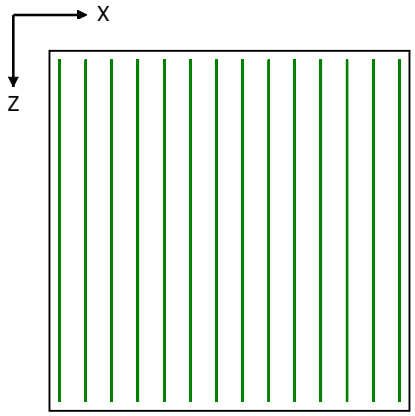
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



14 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =	0.332	m

Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl.7.12.2, ρ_{min} =		0.00163
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.150 m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	33.369 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	37.076 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00040
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00033
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1307.167 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410 m
Available development length for bars, D_L =	$0.5 \times \left(L - D_{col}\right) - C_{cover}$ =	0.800 m
Try bar size # 12	Area of one bar =	113.097 mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	14

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1583.352 mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.332 m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ =	0.00238

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

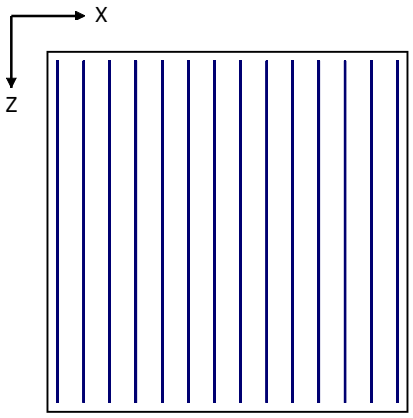
Provided Steel Area / Required Steel Area = 1.211

Check to see if width is sufficient to accomodate bars

Bending moment for uplift cases will be calculated based solely on selfweight, soil depth and surcharge loading.

As the footing size has already been determined based on all servicebility load cases, and design moment calculation is based on selfweight, soil depth and surcharge only, top reinforcement value for all pure uplift load cases will be the same.

Design For Top Reinforcement Parallel to Z Axis



14 - ϕ12

Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_x. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.850	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	17.339	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	19.266	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00021	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00017	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	138.370	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1583.363	mm2

Provided Steel Area / Required Steel Area = 11.443

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

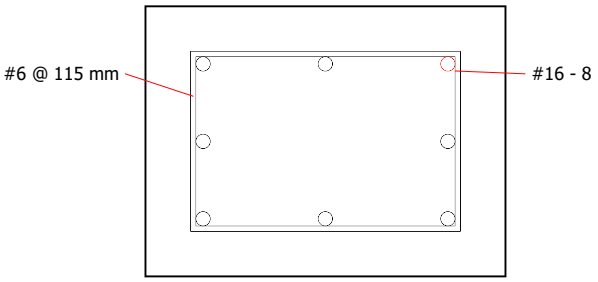
Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Pedestal Design

Pedestal Reinforcement Arrangement : 4 Faces

Calculation Of Shear Capacity



Design Passed in Shear

Critical Load Case For Shear Along X = 29

Critical Load Case For Shear Along Z = 34

Shear along X = 14.173kN

Shear along Z = 13.820kN

(ϕ = 0.75, ref ACI 318 Clause No 9.3.2.3)

Shear Capacities

$$\phi VcX = \phi 2\left[1+\frac{N_u}{2000A_g}\right]\sqrt{f'_c}\,b_w d = 70.434\text{kN}$$

(ϕVcX ref eqn ACI 318-08 Eqn 11.4 and Clause no 11.2.1.2)

$$\phi VsX = \phi A_v f_{yt}\frac{d}{s} = 35.626\text{kN}$$

(ϕVsX ref eqn ACI 318-08 Eqn 11.15 and Clause No 11.4.7.2)

$$\phi VcZ = \phi 2\left[1+\frac{N_u}{2000A_g}\right]\sqrt{f'_c}\,b_w d = 74.438\text{kN}$$

(ϕVcZ ref eqn ACI 318-08 Eqn 11.4 Clause no 11.2.1.2)

$$\phi VsZ = \phi A_v f_{yt}\frac{d}{s} = 50.721\text{kN}$$

(ϕVsZ ref eqn ACI 318-08 Eqn 11.15 and Clause No 11.4.7.2)

Tie bar for shear X = 6

Tie bar for shear Z = 6

Tie bar provided = 6

Legs for resisting FX = 2

Legs for resisting FZ = 2

(Spacing Calculation ref Clause no 11.4.5.1 and 11.4.6.3 and Eqn 11.13)

Tie spacing required to resist FX = 0.118m

Tie spacing required to resist FZ = 0.168m

Spacing provided = 0.118m

Note- Actual stirrup provided is dependent upon design requirement and bar binding and positioning. So Stirrup provided in detail drawing may vary from designed stirrup requirements.

Critical Load Case Based on Main Reinforcement Design : 29

Check for Pedestal Strength Against Axial Load

Factored Axial Load acting on Pedestal = -31.990kN

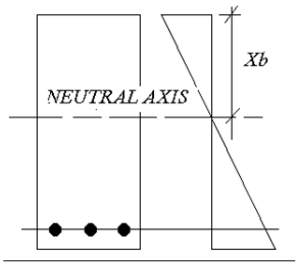
Gross Section Area = 119999.994sq.mm

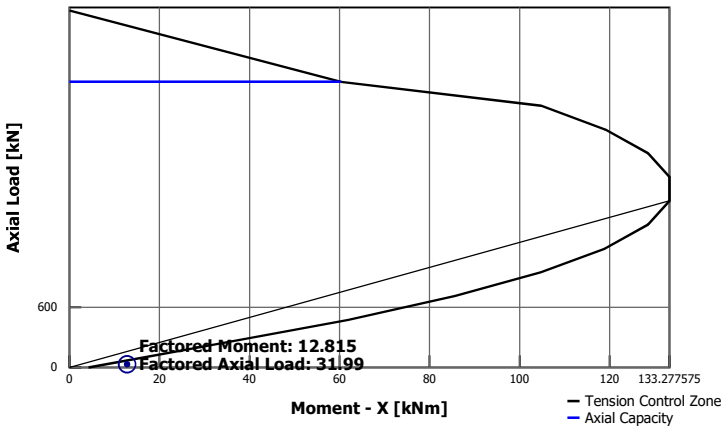
Required Area of steel to resist Axial Force = 0.000sq.mm

Provided Steel Area = 1608.513sq.mm

Hence Safe

Strength and Moment Along Reinforcement in X direction





Note: Factored pedestal design loads (inclusive of Φ = 0.650 based on ACI 318 Chapter 9)

Factored Moment = 12.815kNm

Factored Axial Load = -31.990kN

Bar size = 16 mm

Number of Bars = 8

Required Steel Area = 1199.9999sq.mm

Provided Steel Area = 1608.5128sq.mm

Balanced Neutral Axis Depth= $\frac{\text{deff x } 0.003}{0.003 + \text{fy/Es}}$ = 0.2012m

Actual Neutral Axis Depth (X_b)= = 0.0546m

Strength and Moment from Concrete

C_c = $0.85 \times X_b \times b \times \beta \times f_c$ = 516.875kN

M_c = $C_c \times \left[\frac{d}{2} - \left(0.85 \times \frac{X_b}{2}\right)\right]$ = 66.305kNm

Calculate strength and moment from one bar.

Distance between extreme fiber and bar (d_b) = 0.064m

Strain in bar, ε_s= $\varepsilon_c\left(1 - \frac{d_b}{X_b}\right)$ = -0.0005

Maximum Strain, ε_y= $\frac{F_y}{E_s}$ = 0.0021

as $|\varepsilon_s| < \varepsilon_y$

-103864.843kN/m2

$f_s' = F_{y.main} =$

$\varepsilon_o = \frac{2 \times 0.85 \times f_c}{57 \times \sqrt{0.001 \times f_c}} =$ 0.0021

as $\varepsilon_s < 0$

$f_c' =$ 0.000kN/m2

Tension Resisted by Bars = $C_{s1} = A_{st} \times \left(f_s' - f_c'\right) =$ -20.883kN

Moment Resisted by Bars = $M_{s1} = C_{s1} \times \left(\frac{d}{2} - \frac{d_b}{2}\right) =$ -1.796kNm

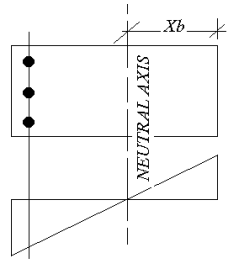
Total Bar Capacity,C_s= Σ(Bar Stress-Concrete Stress)X Bar Area = -484.885kN

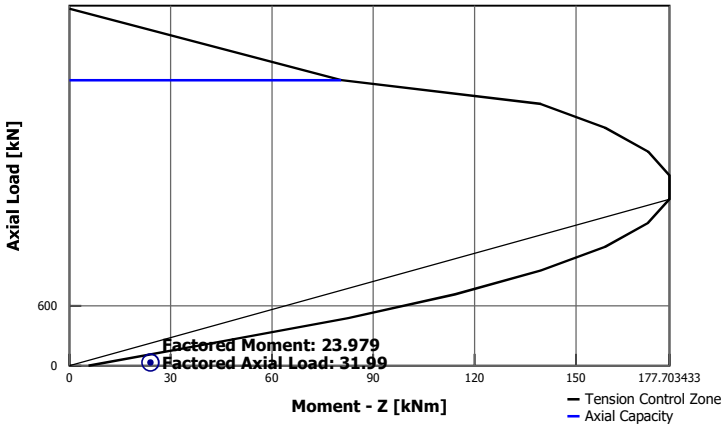
Capacity of Column = Capacity of effective concrete+ Capacity of steel = $C_c + C_s =$ 31.990kN

Total Moment Capacity provided by bars,M_s = Σ(Individual Bar Capacity X Moment Arm) = 16.399kNm

Total Moment, Mox = $M_c + M_s =$ 82.704kNm

Strength and Moment Along Reinforcement in Z direction





Note: Factored pedestal design loads (inclusive of Φ = 0.650 based on ACI 318 Chapter 9)

Factored Moment = 23.979kNm

Factored Axial Load = -31.990kN

Bar size = 16 mm

Number of Bars = 8

Required Steel Area = 1199.9999sq.mm

Provided Steel Area = 1608.5128sq.mm

Balanced Neutral Axis Depth= $\frac{\text{deff x } 0.003}{0.003 + \text{fy/Es}}$ = 0.1423m

Actual Neutral Axis Depth (X_b)= = 0.0640m

Strength and Moment from Concrete

$C_c =$ $0.85 \times X_b \times b \times \beta \times f_c =$ 454.471kN

M_c = $C_c \times \left[\frac{d}{2} - \left(0.85 \times \frac{X_b}{2}\right)\right]$ = 79.322kNm

Calculate strength and moment from one bar.

Distance between extreme fiber and bar, d_b = 0.064m

Strain in bar, ε_s = $\varepsilon_c\left(1 - \frac{d_b}{X_b}\right)$ = -0.0000

Maximum Strain, ε_y = $\frac{F_y}{E_s}$ = = 0.0021

as $|\varepsilon_s| < \varepsilon_y$

$f_s' = F_{y.main}$ = -408.624kN/m2

$\varepsilon_o = \frac{2 \times 0.85 \times f_c}{57 \times \sqrt{0.001 \times f_c}}$ = 0.0021

as $\varepsilon_s < 0$

$f_c' = 0$ = kN/m2

Tension Resisted by Bars, $C_{s1} = A_{st} \times \left(f_s' - f_c'\right)$ = -0.082kN

Moment Resisted by Bars, $M_{s1} = C_{s1} \times \left(\frac{d}{2} - \frac{d_b}{2}\right)$ = -0.011kNm

Total Bar Capacity, C_s = Σ(Bar Stress-Concrete Stress)X Bar Area = -422.481kN

Capacity of Column = Capacity of effective concrete+ capacity of steel = $C_c + C_s$ = 31.990kN

Total Moment capacity provided by Bars,M_s = Σ(Individual Bar Capacity X Moment Arm) = 34.420kNm

Total Moment, Moz= $M_c + M_s$ = 113.742kNm

Check for bi-axial bending, $\left(\frac{M_{nx}}{M_{0x}}\right)^{\alpha} + \left(\frac{M_{nz}}{M_{0z}}\right)^{\alpha}$ = 0.697

Design Moment M_{nx} = 25.622kNm

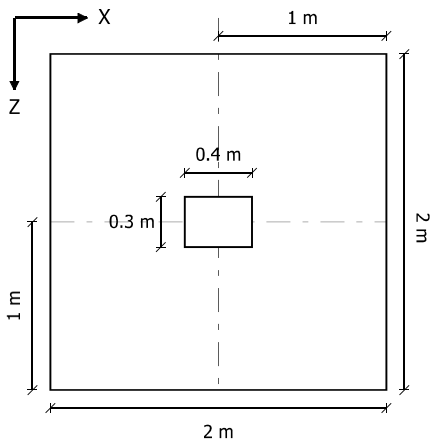
Design Moment M_{nz} = 73.986kNm

Total Moment M_{ox} = 82.704kNm

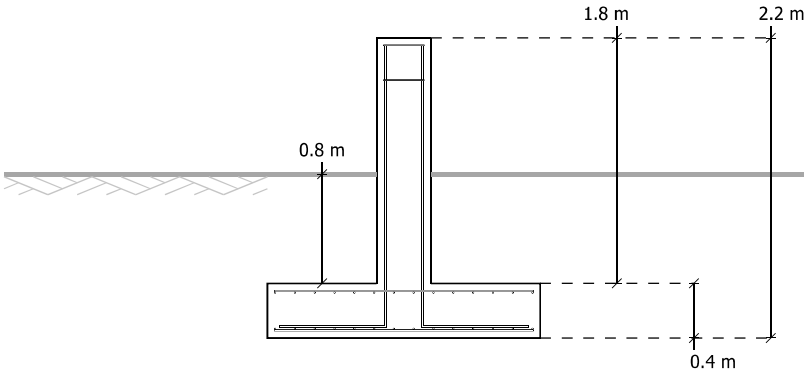
Total Moment M_{oz} = 113.742kNm

if M_{nx} or M_{nz} = 0, then α = 1.0

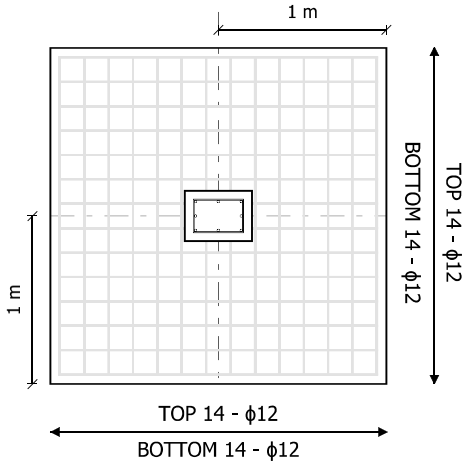
otherwise, α = 1.5



Plan



Elevation



Plan

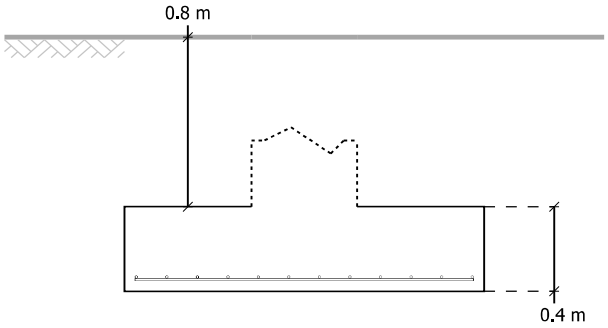
	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	2 m	2 m	0.4 m	14 - ϕ12	14 - ϕ12	14 - ϕ12	14 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	0.4 m	0.3 m	1.8 m	#16 - 8		#6 @ 115 mm	

Print Calculation Sheet

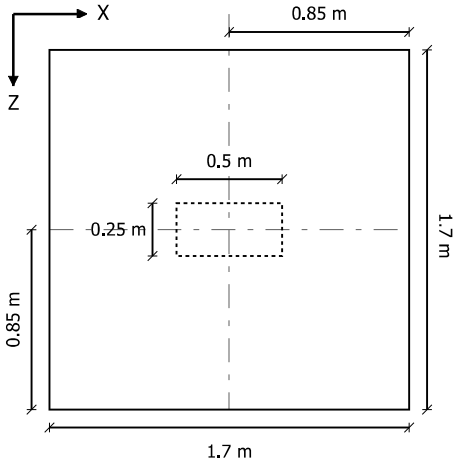
FOUNDATION-F4-GRID-1 & G

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
71	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
71	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	180.008	-43.464	-21.445	-7.982	11.351
12	186.494	-43.596	-20.569	-7.697	11.230
13	184.872	-43.563	-20.788	-7.768	11.260
14	178.965	-43.462	-20.132	-6.728	11.351
15	178.343	-40.659	-21.445	-7.981	5.608
16	180.950	-34.834	-21.203	-8.073	-3.872
17	181.408	-42.338	-17.421	-4.731	8.147
18	102.774	-14.440	-13.421	-5.196	-13.366
19	103.385	-24.446	-8.377	-0.739	2.659
20	106.340	-23.273	-12.867	-4.788	1.068
21	106.962	-26.077	-11.554	-3.535	6.811

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	174.778	-31.826	-21.999	-8.389	-8.825
23	175.388	-41.831	-16.955	-3.932	7.199
39	189.331	-52.758	-21.206	-8.079	27.556
40	185.559	-42.107	-24.070	-10.741	7.611
41	113.949	-38.339	-13.424	-5.204	28.538
42	108.920	-24.137	-17.243	-8.753	1.945
45	185.952	-55.725	-22.002	-8.397	33.078
46	180.923	-41.523	-25.821	-11.946	6.485

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	252.011	-60.850	-30.023	-11.175	15.892
25	255.254	-60.915	-29.585	-11.033	15.831
26	255.254	-60.915	-29.585	-11.033	15.831
27	222.203	-43.057	-24.776	-9.448	-2.713
28	211.861	-49.610	-18.113	-2.956	6.918
29	210.884	-33.601	-26.182	-10.087	-18.721
30	211.861	-49.610	-18.113	-2.956	6.918
31	213.631	-48.150	-25.734	-9.577	5.417
32	214.519	-52.154	-23.858	-7.786	13.622
33	153.639	-20.496	-20.186	-7.834	-22.066
34	154.616	-36.505	-12.117	-0.704	3.573
35	160.517	-39.115	-17.424	-5.392	10.216
36	159.629	-35.111	-19.300	-7.183	2.012
47	231.143	-62.176	-24.779	-9.454	30.809
48	220.717	-49.116	-32.297	-15.778	5.776
49	228.764	-71.839	-26.188	-10.099	48.324
50	220.717	-49.116	-32.297	-15.778	5.776
53	171.518	-58.734	-20.192	-7.847	44.979
54	163.471	-36.011	-26.302	-13.526	2.431

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.371 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 11

Width (W_2) = 1.700 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

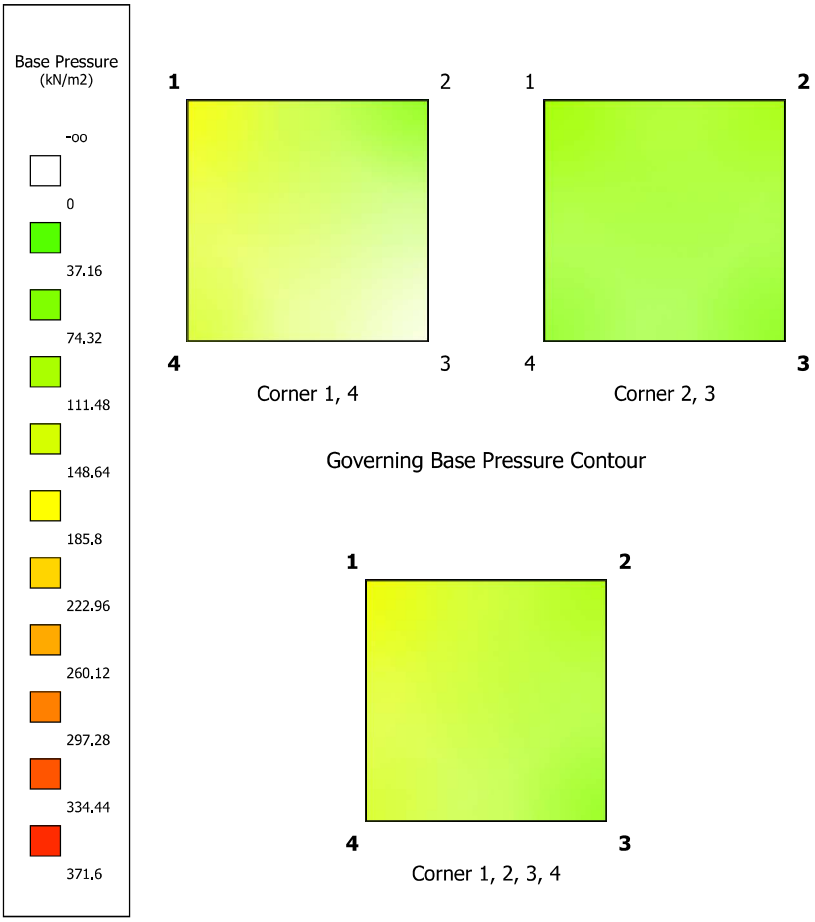
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 39.815 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



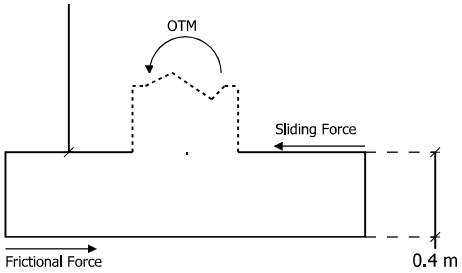
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**176.3410**	41.1035	-0.9017	134.3358	0.000
22	109.6132	**100.0754**	58.0928	67.6306	0.000
22	109.6132	100.0754	**58.0928**	67.6306	0.000
45	176.3410	41.1035	-0.9017	**134.3358**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**176.3413**	41.1034	0.0000	134.3360
22	109.6132	**100.0754**	58.0928	67.6306
22	109.6132	100.0754	**58.0928**	67.6306
45	176.3413	41.1034	0.0000	**134.3360**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	2.848	5.772	3.612	1.500	12.776	7.323	1.500
12	2.914	6.176	3.727	1.500	13.632	7.532	1.500
13	2.897	6.071	3.698	1.500	13.412	7.480	1.500
14	2.836	6.123	3.639	1.500	14.255	7.292	1.500
15	3.024	5.733	3.783	1.500	12.691	9.556	1.500
16	3.567	5.860	4.309	1.500	12.829	20.994	1.500
17	2.940	7.146	3.845	1.500	18.186	8.437	1.500
18	5.898	6.346	6.110	1.500	13.814	19.076	1.500
19	3.496	10.203	4.678	1.500	35.804	11.683	1.500
20	3.736	6.758	4.624	1.500	14.993	14.244	1.500
21	3.346	7.552	4.327	1.500	18.328	8.604	1.500
22	3.807	5.508	4.429	1.500	12.051	52.750	1.500
23	2.904	7.164	3.806	1.500	19.381	8.629	1.500
39	2.435	6.057	3.195	1.500	13.254	4.487	1.500
40	3.006	5.258	3.690	1.500	10.619	8.798	1.500
41	2.367	6.760	3.160	1.500	14.700	3.516	1.500
42	3.656	5.117	4.207	1.500	9.658	12.932	1.500
45	2.275	5.761	2.992	1.500	12.597	3.892	1.500
46	2.992	4.812	3.593	1.500	9.534	9.145	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 45
Governing Disturbing Force : -55.725kN
Governing Restoring Force : 126.755kN
Minimum Sliding Ratio for the Critical Load Case : 2.275
Critical Load Case for Overturning about X-Direction : 46
Governing Overturning Moment : -22.274kNm
Governing Resisting Moment : 212.352kNm
Minimum Overturning Ratio for the Critical Load Case : 9.534

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 46
Governing Disturbing Force : -25.821kN
Governing Restoring Force : 124.240kN
Minimum Sliding Ratio for the Critical Load Case : 4.812

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 43.873kNm

Governing Resisting Moment : 154.278kNm

Minimum Overturning Ratio for the Critical Load Case : 3.516

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 45

Governing Disturbing Force : 59.911kN

Governing Restoring Force : 179.259kN

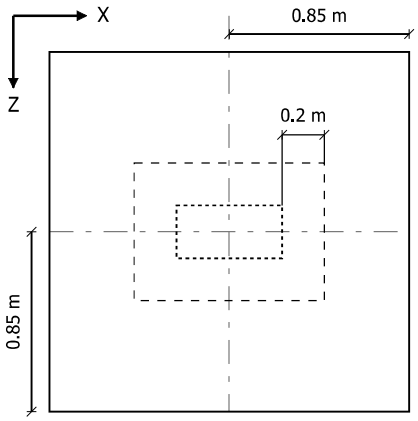
Minimum Sliding Ratio for the Critical Load Case : 2.992

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 266.812kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1944.021kN

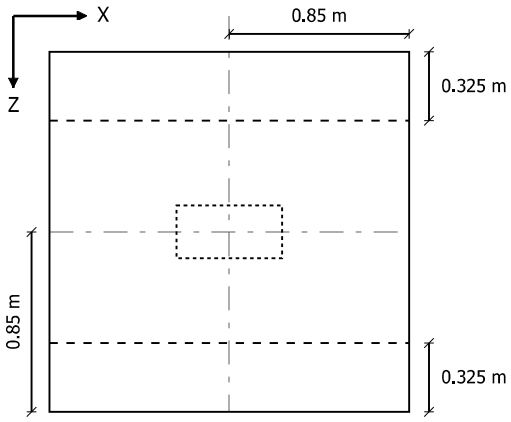
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

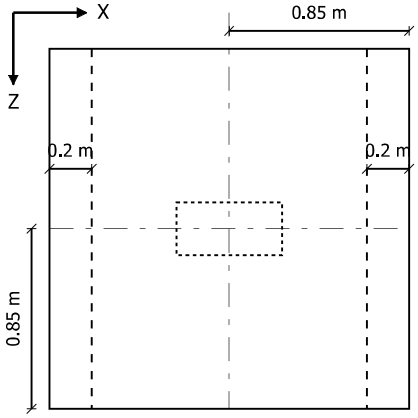
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 25 $V_{ux} = V_{ux}|_{x=D_x}$ = 86.382 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

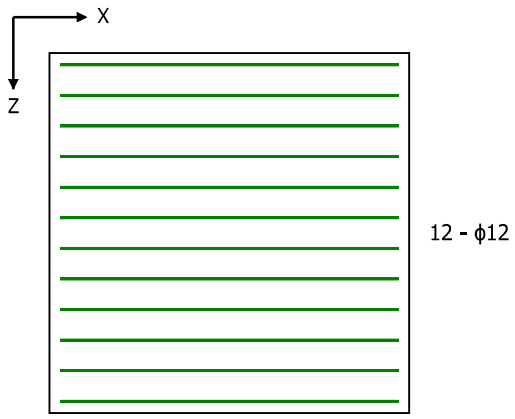
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 79.412 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00165
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.600 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	53.391 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	59.323 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00071
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00061
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1124.583 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}} =$ 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

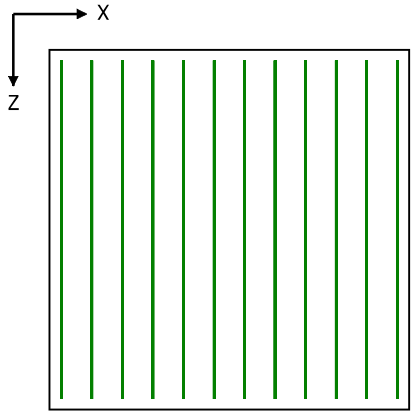
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.207

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.725 m

Ultimate moment, $M_u|_{x=D_x}$ = 58.831 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 65.368 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00084

(Based on gross depth) ρ x d_{eff} / Depth = 0.00069

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1160.523 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

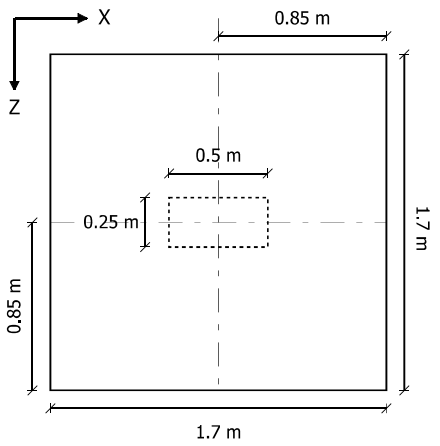
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

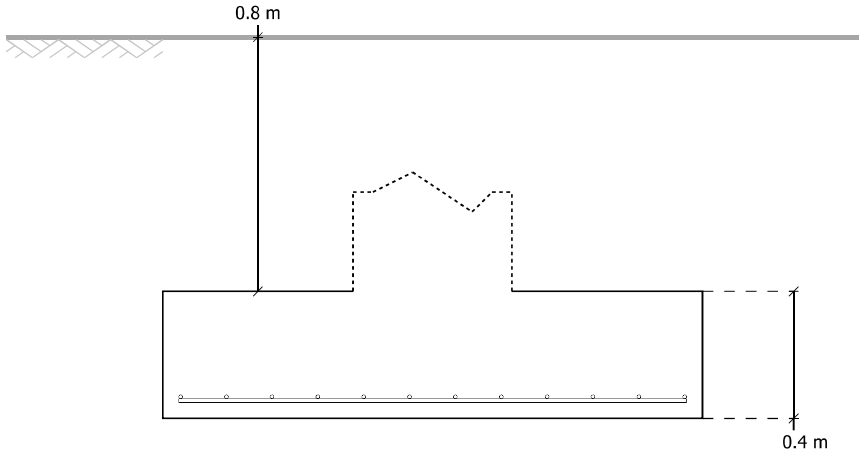
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.169

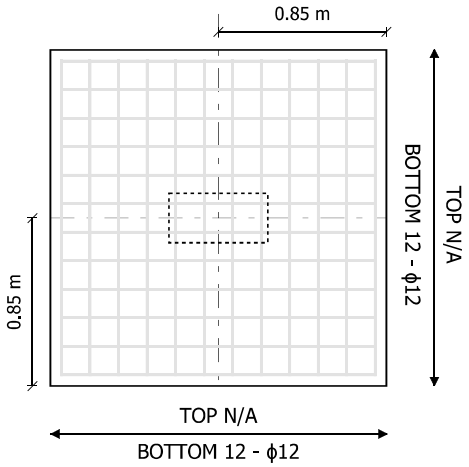
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

COMBINED FOUNDATION DESIGN (ACI 318-08)

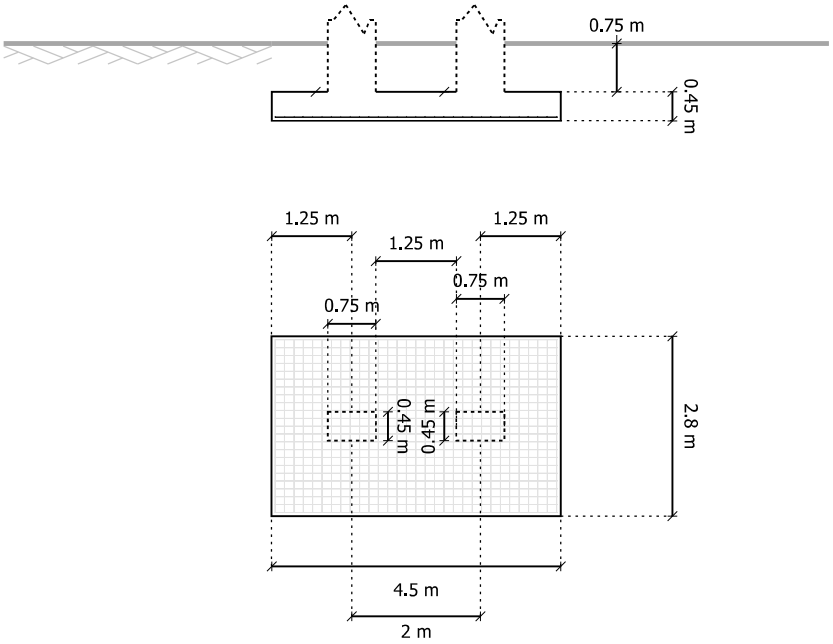
Result Summary

Footing No.	Left Overhang (m)	Right Overhang (m)	Length (m)	Width (m)	Thickness (m)
1	1.250	1.250	4.500	2.800	0.450

Footing No.	Footing Reinforcement			
-	Main Steel Top	Main Steel Bottom	Secondary Steel Top	Secondary Steel Bottom
1	-	23 - ϕ12	-	31 - ϕ12

FOUNDATION-CF1

Input Data



Geometry of Footing

For Column 29

Column Dimensions

Column Shape : Rectangular

Column Length - X (D_{col}) : 0.750m

Column Width - Z (B_{col}) : 0.450m

Pedestal

Include Pedestal? No

Pedestal Shape : N/A

Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Eccentricity

Column Offset in Transverse Direction :	0.000m

For Column 32

Column Dimensions

Column Shape :	Rectangular
Column Length - X (D_{col}) :	0.750m
Column Width - Z (B_{col}) :	0.450m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Eccentricity

Column Offset in Transverse Direction :	0.000m

Length of left overhang :	1.250m
Length of right overhang :	1.250m
Is the length of left overhang fixed?	No
Is the length of right overhang fixed?	No
Maximum length of Footing (Lo) :	450.000mm
Width of Footing (Wo) :	2.800m
Thickness of Footing (Do) :	450.000mm

Cover and Soil Properties

Pedestal Clear Cover :	50.000mm
Footing Clear Cover :	50.000mm
Unit Weight of soil :	18.000kN/m3
Soil Bearing Capacity :	350.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000m
Depth of Soil above Footing :	0.750m
Type of Depth :	Fixed Top
Depth of Water Table :	24.000m

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Compressive Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	415.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Minimum Pedestal Bar Size :	#25
Maximum Pedestal Bar Size :	#25
Minimum Bar Spacing :	125.000mm

Maximum Bar Spacing : 150.000mm

Design Calculations

Footing Size Calculations

Gross Soil Bearing Capacity = 371.60kN/m^2
Reduction of force due to buoyancy = 0.000kN

Area from initial length and width, A_o = L_o X W_o = 12.600m^2
Min. area required from bearing pressure, A_{min} = P / q_{max} = 2.857m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final footing dimensions are:

Length of footing, L : 4.500m
Width of footing, W : 2.800m
Depth of footing, Do : 0.450m
Area, A : 12.600m^2
Length of left overhang, $L_{left_overhang}$: 1.250m
Length of right overhang, $L_{right_overhang}$: 1.250m
Footing self weight : 136.075kN
Soi weight on top of footing : 160.982kN

Load Combinations

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1,4D+1,4T+0,5LR)	1,00	1,00	1,00
26	ULS: (1,4D+1,4T+0,5LR)	1,00	1,00	1,00
27	ULS: (1,2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

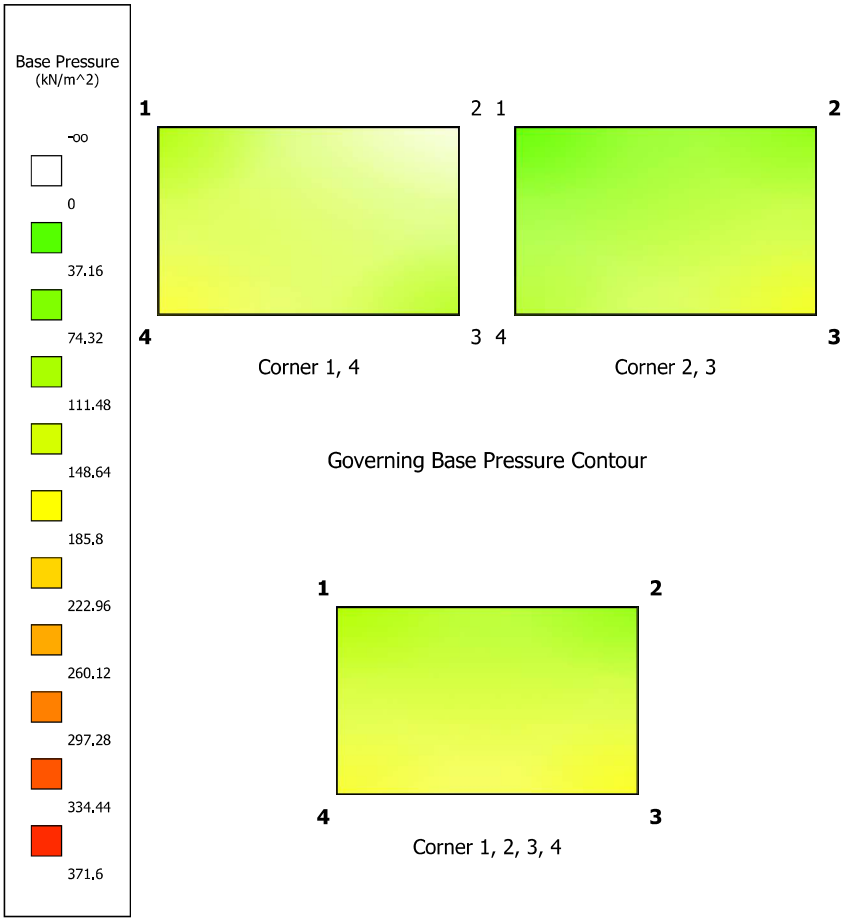
Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 29					
11	366.170	-49.409	34.874	7.941	74.174
12	383.997	-54.216	34.973	7.819	80.723
13	379.540	-53.015	34.948	7.849	79.086
14	328.263	-50.180	51.385	53.991	73.253
15	374.931	-38.554	33.438	7.195	47.270
16	389.413	-31.957	31.744	6.023	42.375
17	323.526	-68.935	64.062	77.265	95.375
18	232.866	-1.568	16.653	2.330	-4.443
19	145.017	-50.872	59.743	97.318	66.224
20	228.463	-18.790	19.489	4.018	17.601
21	181.795	-30.416	37.435	50.815	43.583
22	379.334	-21.332	30.602	5.506	25.227
23	291.485	-70.636	73.693	100.494	95.894
39	371.369	-69.662	34.845	7.309	110.274
40	430.161	-63.059	7.210	-61.021	94.373
41	208.808	-51.843	20.787	4.044	86.088
42	287.196	-43.038	-16.060	-87.062	64.888
45	355.276	-71.606	34.736	7.221	115.758
46	433.664	-62.802	-2.111	-83.886	94.557
-					
Column Number : 32					
11	353.259	-48.852	-35.462	-10.054	73.181
12	369.791	-53.607	-35.608	-10.146	79.635
13	365.658	-52.418	-35.571	-10.123	78.022
14	391.268	-48.089	-19.039	35.944	74.189
15	361.292	-37.993	-34.109	-9.598	46.264
16	375.057	-31.350	-32.393	-8.423	41.290
17	416.458	-62.470	-7.931	58.736	93.424
18	224.487	-1.220	-17.039	-3.766	-5.067
19	279.689	-42.713	15.577	85.779	64.445
20	219.989	-18.452	-19.924	-5.577	16.991
21	249.965	-28.548	-4.854	39.966	44.917
22	365.790	-20.761	-31.224	-7.788	24.206
23	420.993	-62.254	1.392	81.757	93.718
39	357.943	-69.077	-35.434	-9.464	109.230
40	309.533	-68.324	-64.547	-79.407	94.189
41	201.668	-51.523	-21.094	-5.154	85.521
42	137.122	-50.519	-59.911	-98.411	65.466
45	342.972	-71.064	-35.279	-9.176	114.793
46	278.426	-70.060	-74.096	-102.432	94.738

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 29					
24	512.638	-69.173	48.824	11.117	103.843
25	521.551	-71.577	48.873	11.056	107.118
26	521.551	-71.577	48.873	11.056	107.118
27	478.458	-44.521	38.590	7.386	60.330
28	328.821	-95.658	104.008	157.554	127.035
29	469.380	-16.771	35.063	5.572	13.968
30	328.821	-95.658	104.008	157.554	127.035
31	451.919	-43.783	39.798	8.463	50.575
32	385.251	-60.392	65.436	75.315	87.693
33	350.616	0.456	24.552	3.251	-11.559
34	210.057	-78.431	93.497	155.233	101.508
35	275.400	-45.569	54.973	72.933	65.441
36	342.068	-28.961	29.335	6.081	28.323
47	459.211	-84.740	41.897	8.758	132.754
48	556.308	-83.123	-17.277	-137.455	124.897
49	430.887	-97.210	41.678	8.316	158.818
50	556.308	-83.123	-17.277	-137.455	124.897
53	312.122	-79.983	31.166	5.994	133.291
54	437.544	-65.896	-27.789	-139.776	99.370
-					
Column Number : 32					
24	494.563	-68.393	-49.647	-14.075	102.454
25	502.829	-70.770	-49.720	-14.121	105.681
26	502.829	-70.770	-49.720	-14.121	105.681
27	460.387	-43.757	-39.397	-10.399	58.963
28	540.551	-82.443	16.339	134.787	123.903
29	452.227	-16.055	-35.846	-8.485	12.684
30	540.551	-82.443	16.339	134.787	123.903
31	435.387	-43.109	-40.621	-11.414	49.364
32	478.210	-57.532	-19.093	53.647	89.257
33	337.983	0.978	-25.135	-5.423	-12.497
34	426.307	-65.410	27.051	137.849	98.722
35	372.232	-42.877	-8.454	56.663	67.303
36	329.409	-28.453	-29.983	-8.397	27.410
47	442.132	-83.999	-42.641	-11.510	131.433
48	312.444	-94.933	-104.442	-159.916	125.536
49	415.717	-96.539	-42.334	-10.706	157.623
50	312.444	-94.933	-104.442	-159.916	125.536
53	301.473	-79.506	-31.622	-7.644	132.442
54	198.200	-77.900	-93.730	-156.854	100.355

Pressures at 4 Corners

Governing Base Pressure Contour for Each Corner



Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)	Area of footing in uplift (A_u) (sq. m)
46	**77.5020**	-2.0435	82.6801	162.2255	0.000
23	1.1380	**74.2705**	159.1056	85.9732	0.000
23	1.1380	74.2705	**159.1056**	85.9732	0.000
46	77.5020	-2.0435	82.6801	**162.2255**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)
46	**77.5021**	0.0000	82.6803	162.2261
23	1.1380	**74.2705**	159.1056	85.9732
23	1.1380	74.2705	**159.1056**	85.9732
46	77.5021	0.0000	82.6803	**162.2261**

Stability Check

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
11	-98.261	-0.588	98.263	538.738	5.483	916.540	5.483	1.500
12	-107.823	-0.635	107.825	556.948	5.165	877.306	5.165	1.500
13	-105.433	-0.623	105.434	552.395	5.239	886.566	5.239	1.500
14	-98.268	32.346	103.455	538.792	5.483	16.657	5.208	1.500
15	-76.547	-0.670	76.549	547.638	7.154	816.831	7.154	1.500
16	-63.306	-0.648	63.310	562.609	8.887	867.597	8.887	1.500
17	-131.404	56.131	142.891	549.632	4.183	9.792	3.847	1.500
18	-2.788	-0.387	2.815	399.837	143.392	1034.421	142.034	1.500
19	-93.586	75.320	120.131	382.534	4.088	5.079	3.184	1.500
20	-37.242	-0.435	37.245	395.119	10.609	907.636	10.609	1.500
21	-58.964	32.581	67.367	386.273	6.551	11.856	5.734	1.500
22	-42.093	-0.622	42.098	552.356	13.122	888.530	13.121	1.500
23	-132.890	75.085	152.635	535.053	4.026	7.126	3.505	1.500
39	-138.739	-0.589	138.740	543.976	3.921	923.698	3.921	1.500
40	-131.383	-57.337	143.350	549.478	4.182	9.583	3.833	1.500
41	-103.366	-0.307	103.366	374.992	3.628	1221.007	3.628	1.500
42	-93.557	-75.972	120.518	382.329	4.087	5.033	3.172	1.500
45	-142.670	-0.542	142.671	527.511	3.697	972.851	3.697	1.500
46	-132.862	-76.207	153.166	534.848	4.026	7.018	3.492	1.500

Check for stability against overturning (Moments printed against Local axis)

Load Case	Moment X (kNm)	Moment Z (kNm)	Resisting Moment X (kNm)	Resisting Moment Z (kNm)	Ratio X	Ratio Z	Required FOS
11	191.575	15.288	1423.054	2287.052	7.428	149.596	1.500
12	208.881	16.818	1471.156	2364.358	7.043	140.586	1.500
13	204.555	16.435	1459.131	2345.031	7.133	142.681	1.500
14	191.665	-167.495	1423.198	2287.283	7.426	13.656	1.500
15	127.982	16.344	1446.565	2324.837	11.303	142.248	1.500
16	112.155	17.048	1486.110	2388.392	13.251	140.097	1.500
17	247.935	-254.189	1451.832	2333.301	5.856	9.179	1.500
18	-8.254	9.990	1056.155	1697.391	127.952	169.911	1.500
19	172.785	-351.660	1010.450	1623.937	5.848	4.618	1.500
20	51.352	10.228	1043.693	1677.363	20.325	163.994	1.500
21	115.035	-173.610	1020.325	1639.809	8.870	9.445	1.500
22	68.376	16.105	1459.027	2344.865	21.339	145.597	1.500
23	249.415	-345.545	1413.323	2271.412	5.667	6.573	1.500
39	281.941	15.847	1436.890	2309.288	5.097	145.726	1.500
40	247.689	286.853	1451.425	2332.647	5.860	8.132	1.500
41	218.127	8.388	990.528	1591.920	4.541	189.787	1.500
42	172.457	369.730	1009.907	1623.065	5.856	4.390	1.500
45	294.757	14.503	1393.401	2239.394	4.727	154.407	1.500
46	249.087	375.846	1412.780	2270.539	5.672	6.041	1.500

Calculations of Footing Thickness

Footing thickness is calculated based on the ultimate load cases

Punching Shear Check

For Column 29

Critical Load case for Punching Shear Check : 50

Total Footing Depth, D_o = 0.450m

Calculated Effective Depth, d_{eff} = 0.394m

For rectangular column, β_c = B_{col} / D_{col} : 1.667

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 469.534kN

From ACI Cl.11.11.2, b_c for column 3.976m

Equation 11-31, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 3386.015kN

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o}+2\right)\times \lambda \times \sqrt{f_c} \times b_o \times d$ = 4589.422kN

Equation 11-33, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 3078.195kN

Punching shear strength, V_c= 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 2308.647kN

$0.75 * V_c > V_u$ hence, OK

For Column 32

Critical Load case for Punching Shear Check : 28

Total Footing Depth, D_o = 0.450m

Calculated Effective Depth, d_{eff} = 0.394m

For rectangular column, β_c = B_{col} / D_{col} : 1.667

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 442.832kN

From ACI Cl.11.11.2, b_c for column 3.976m

Equation 11-31, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 3386.015kN

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o}+2\right)\times \lambda \times \sqrt{f_c} \times b_o \times d$ = 4589.422kN

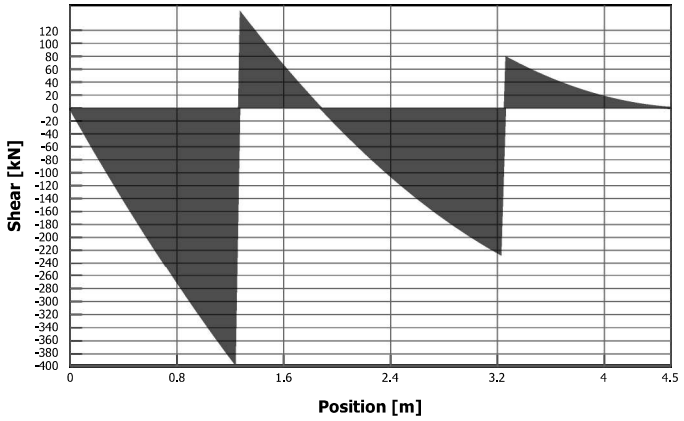
Equation 11-33, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 3078.195kN

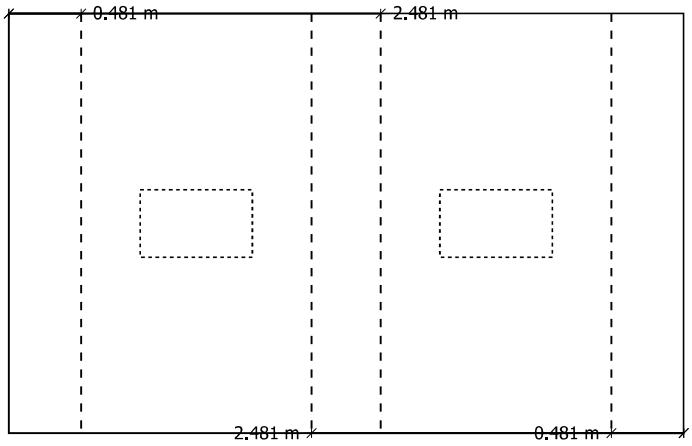
Punching shear strength, V_c= 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 2308.647kN

$0.75 * V_c > V_u$ hence, OK

One-Way Shear

Shear Diagram





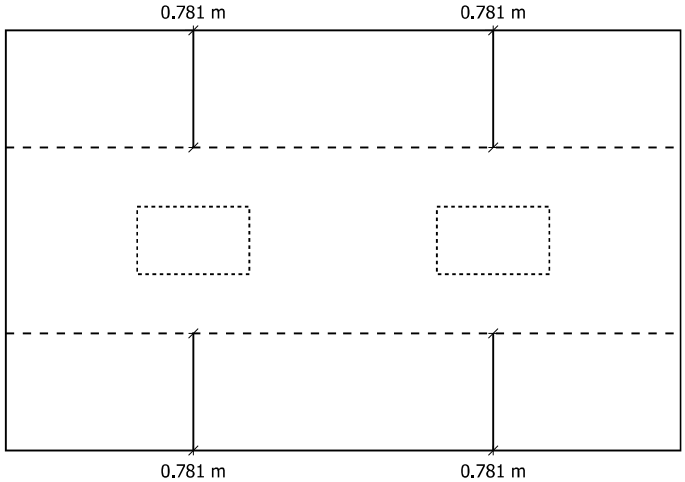
Shear Plane Parallel to Foundation Width

Critical load case for maximum shear force along the length of footing : 50

Critical Shear force, V_u	For the critical load case:	167.749kN
Point of occurance of V_u	Critical one-way shear position:	0.481m
From ACI Cl.11.2.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	1083.872kN

0.75 x V_c = 812.904kN

Since 0.75 * V_c > V_u hence, OK



Shear Plane Parallel to Foundation Length

Critical load case for maximum shear force along the width of footing : 49

Critical Shear force,V_u	For the critical load case:	407.561kN
Point of occurance of V_u	Critical one-way shear position:	2.019m
From ACI Cl.11.2.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	1741.937kN
	0.75 x V_c =	1306.452kN
	Since 0.75 * V_c > V_u	hence, OK

Design for Flexure

Design of flexure

Moment Diagram

Bottom Reinforcement

Critical load case :	28	
Required Effective Depth :	0.394m	
β_1, from ACI Cl.10.2.7.3	max[0.85-0.05x(Fc-4),0.65]=	0.7962
From ACI Cl. 10.3.2, ρ_{bal}	$0.85 \times \beta_1 \times F_c' \times \frac{87}{[f_y \times (87 + F_y)]}$ =	0.03374
From ACI Cl. 10.3.3, ρ_{max}	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min}	$\max\left(0.0018 \cdot \frac{60ksi}{F_y}, 0.0014\right)$ =	0.00200
Modular Ratio,m	$\frac{F_y}{(0.85 \times F_c')}$ =	13.9496

Ultimate Moment (Mx) :	171.387kNm
Point of occurrence of the ultimate moment along the length of footing :	1.625m
Nominal Moment Capacity :	190.430kNm
Required ρ (based on effective depth) :	0.0023
ρ x d_{eff} / Depth (based on gross depth) :	0.0020
Area of main steel required, A_s = ρ * W * deff :	2520.000mm2

Distribution Reinforcement

Critical load case :	49		
Critical Moment for distribution steel (Mz) :		365.5316	kNm
Nominal moment Capacity :	406.1463		kNm
Point of occurance of the critical moment along length:	1.6250		m
Required ρ (based on effective depth):	0.0024		
ρ x d_{eff} / Depth (based on gross depth) :	0.0020		
Area of distribution steel required, A_s = ρ * L * deff :	2585.279		mm2

Provided Reinforcement

Based on spacing reinforcement increment; provided reinforcement is

Main bar no. for bottom Reinforcement:	#12	
Spacing of bottom reinforcement bar :	125.000	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 125mm o.c.

Distribution bar no. (Bottom):	#12	
Spacing of distribution bars (Bottom):	146.667	mm

Based on spacing reinforcement increment; provided reinforcement is

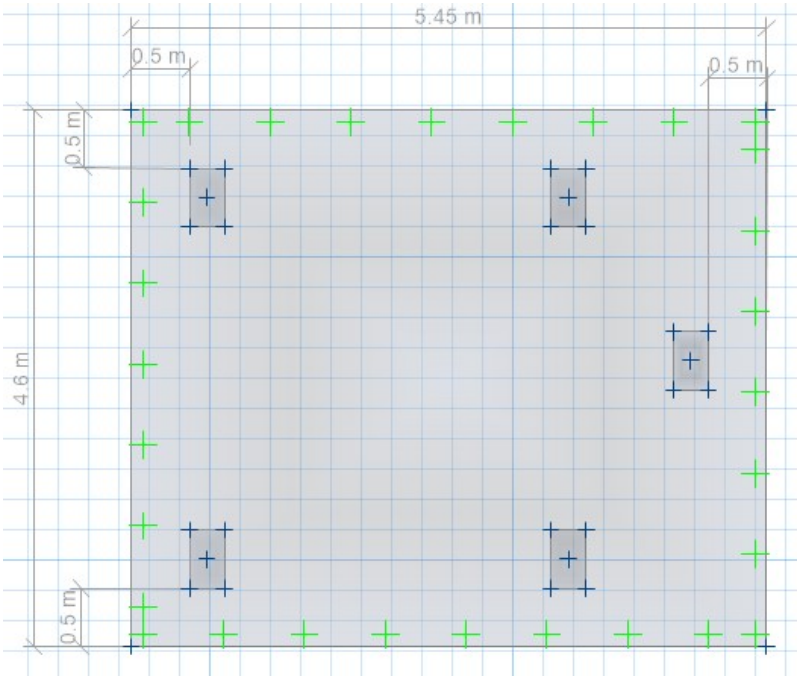
#12 @ 145mm o.c.

Based on spacing reinforcement increment; provided reinforcement is

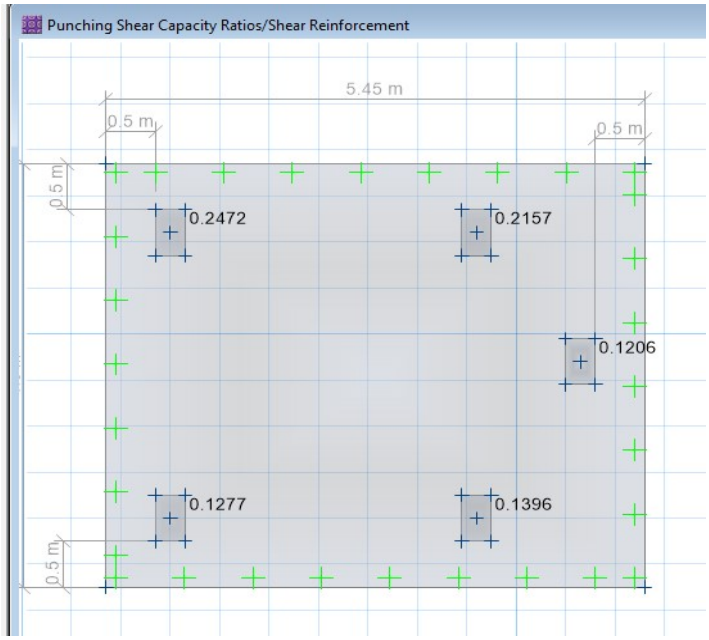
Pedestal Design

Print Calculation Sheet

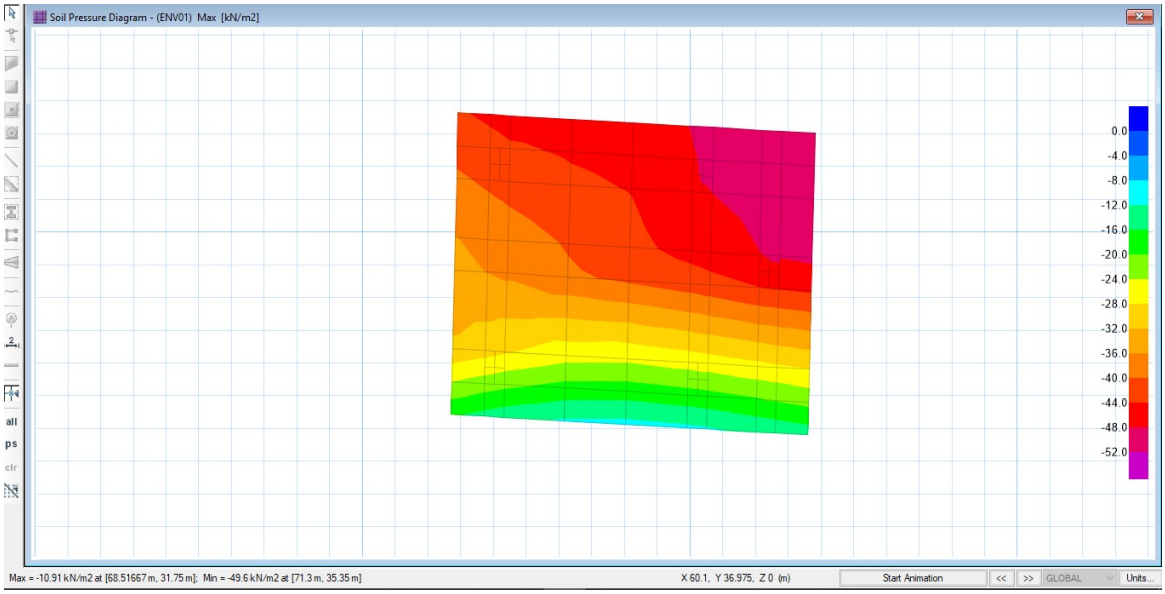
FOUNDATION CF2



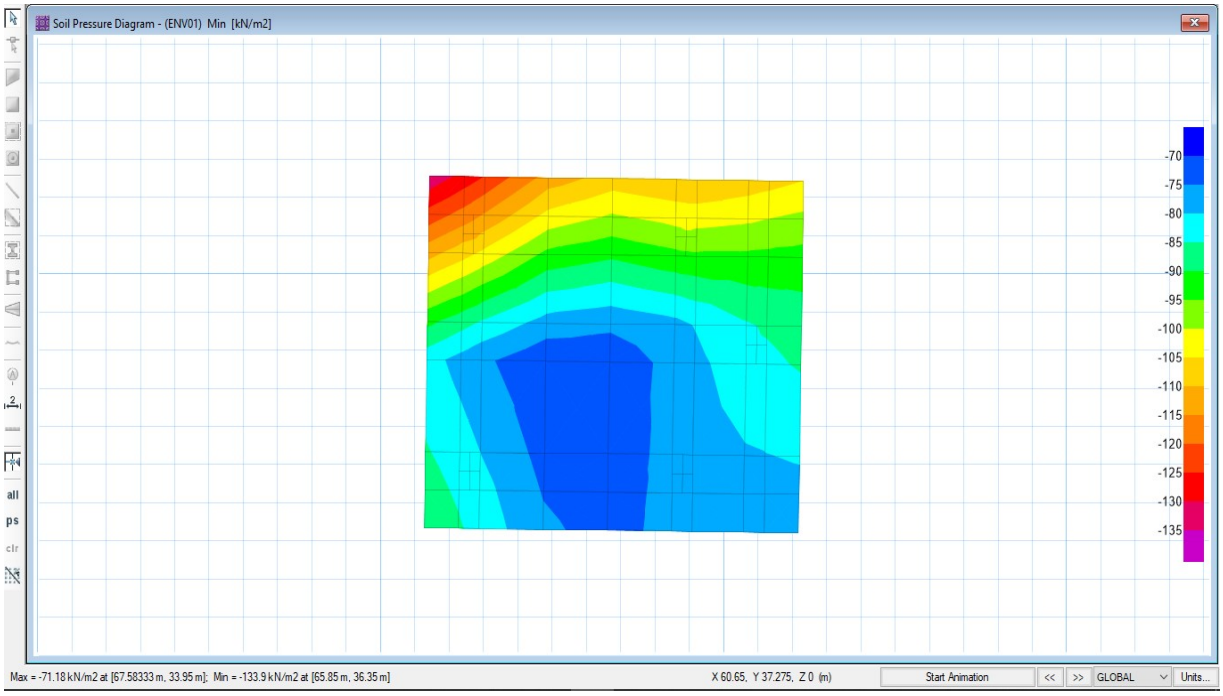
FOOTING GEOMETRICAL DIMENSIONS



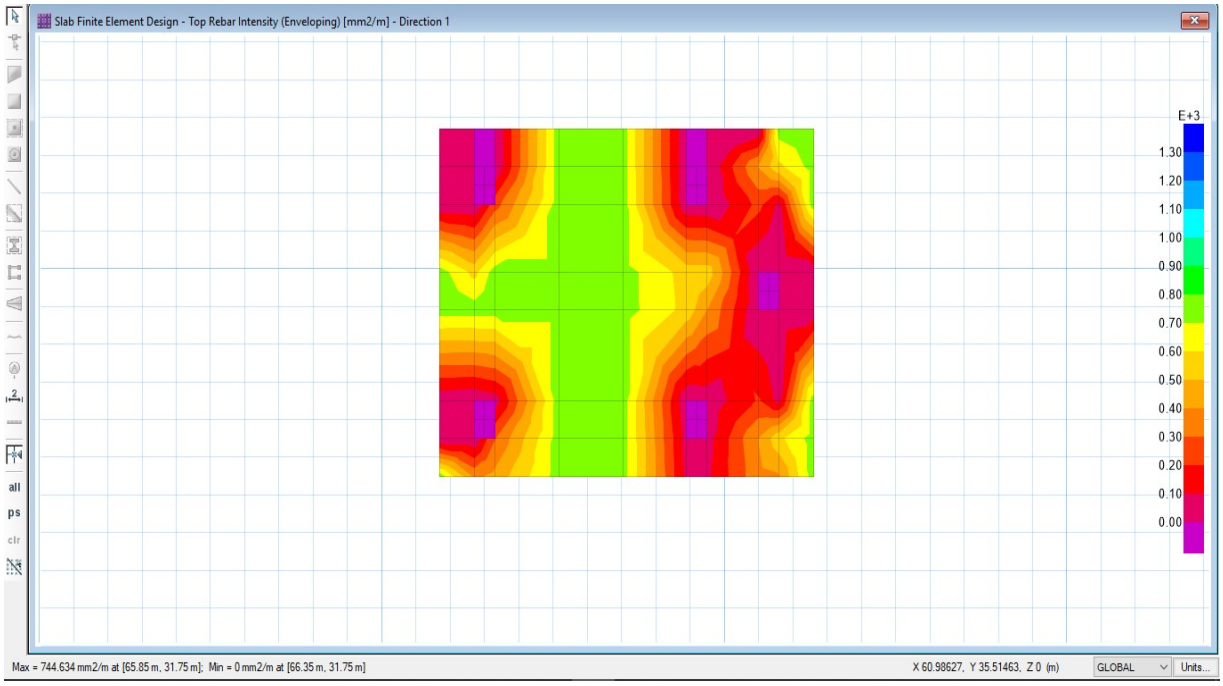
PUNCHING SHEAR RATIO LESS THAN 1, HENCE SAFE FOR TWO WAY SHEAR CHECK.



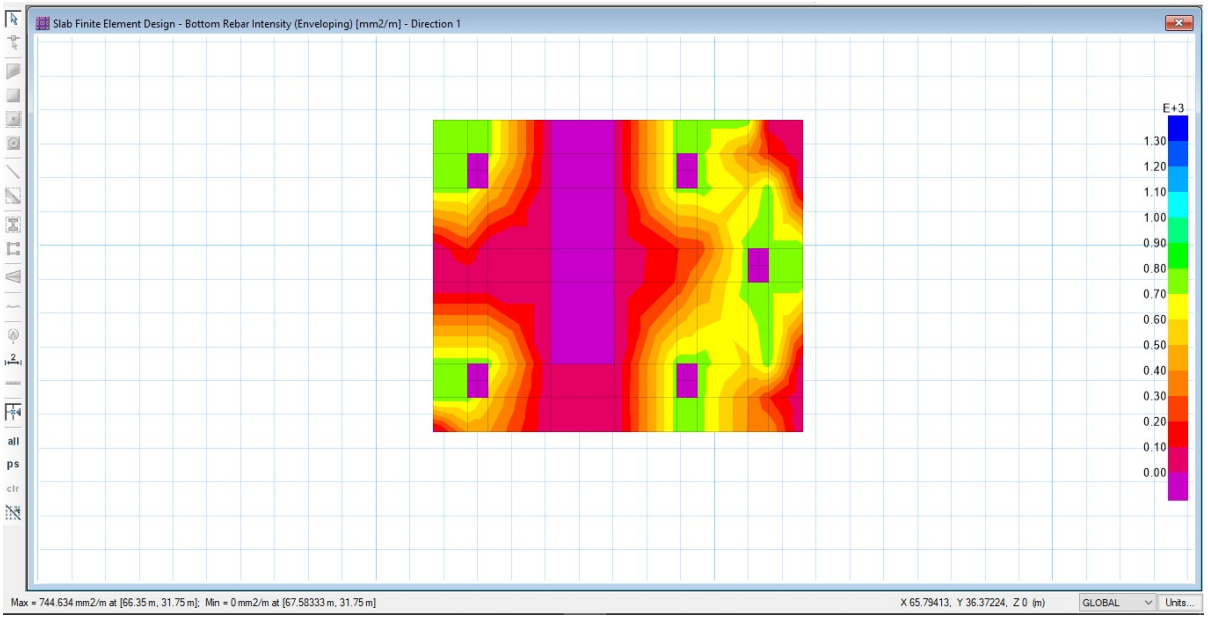
MINIMUM BASE PRESSURE OBTAINED FOR SERVICE ENVELOPE IS 10 KPA>>>>0, NO TENSION AVAILABLE IN FOOTING



MAXIMUM BASE PRESSURE OBTAINED FOR SERVICE ENVELOPE IS 133KPA<<<<350KPA, HENCE SAFE FOR BASE PRESSURE CHECK.

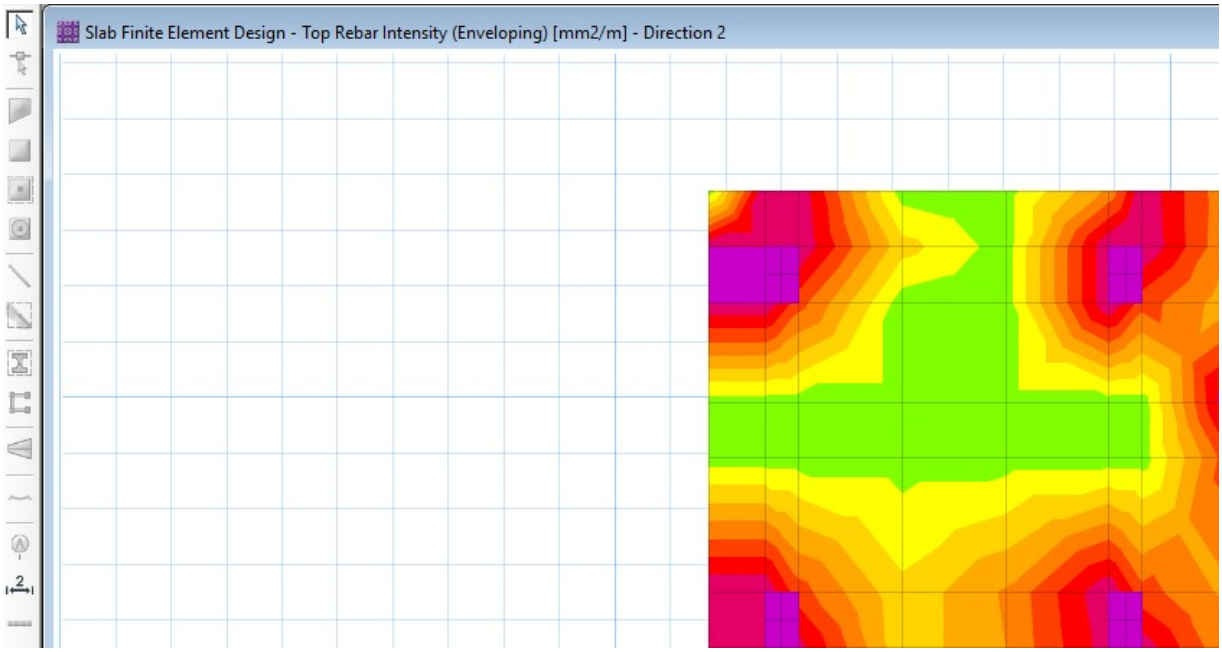


DIRECTION-1: TOP REINFORCEMENT

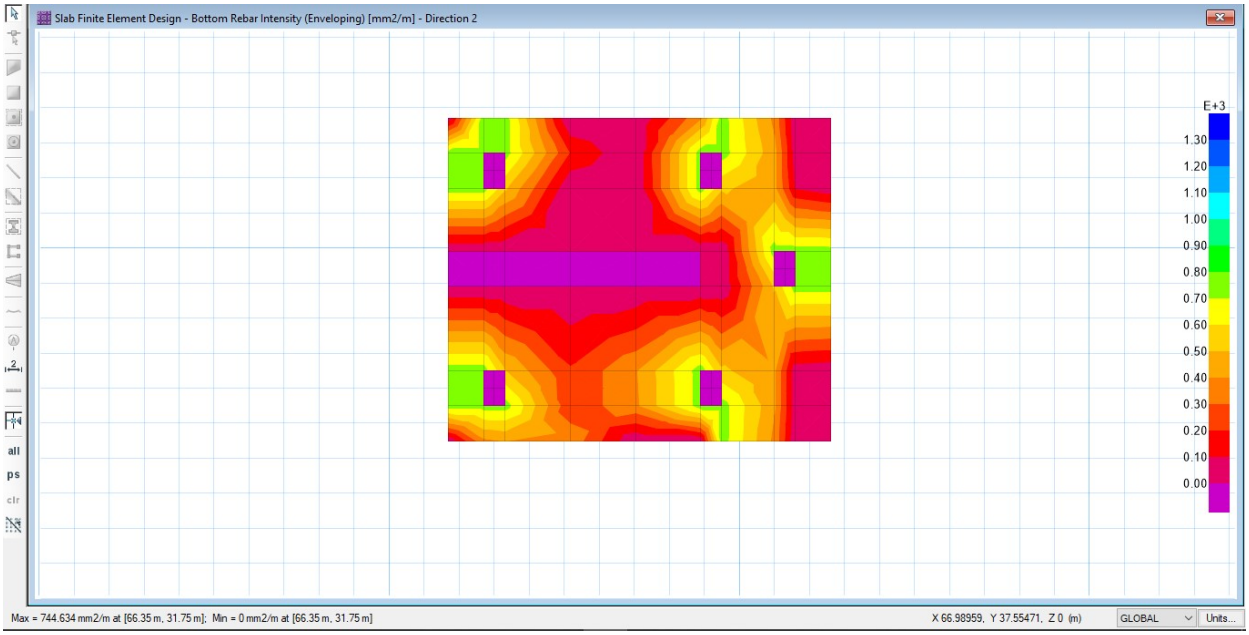


DIRECTION-1: BOTTOM REINFORCEMENT

REQUIRED REINFORCEMENT IN X-DIRECTION T&B IS 744 mm2/M.HENCE PROVIDE DIA12@150mm2/M T&B.



DIRECTION-2: TOP REINFORCEMENT



DIRECTION-2: BOTTOM REINFORCEMENT

REQUIRED REINFORCEMENT IN Y-DIRECTION T&B IS 744 mm2/M.HENCE PROVIDE DIA12@150mm2/M T&B.